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LINE</td><td>A</td></tr><tr><td>13</td><td>3041-PID-0310-1</td><td>PEM SKID N2 HEADERS</td><td>A</td></tr><tr><td>14</td><td>3041-PID-0310-2</td><td>PEM SKID CW HEADERS</td><td>A</td></tr><tr><td>15</td><td>3041-PID-0301-1</td><td>TRC SG LINE</td><td>A</td></tr><tr><td>16</td><td>3041-PID-0400-1</td><td>DRAIN POWER SUPPLIES</td><td>A</td></tr><tr><td>17</td><td>3041-PID-0400-2</td><td>AC POWER SUPPLY</td><td>A</td></tr><tr><td>18</td><td>3041-PID-0400-3</td><td>DC POWER SUPPLY</td><td>A</td></tr><tr><td>19</td><td>3041-PID-0500-1</td><td>AC ELECTRODE HOLDERS</td><td>A</td></tr><tr><td>20</td><td>3041-PID-0500-2</td><td>DC ELECTRODE FEEDERS</td><td>A</td></tr><tr><td>21</td><td>3041-PID-0500-3</td><td>DC RETURN ELECTRODE HOLDERS</td><td>A</td></tr><tr><td>22</td><td>3041-PID-0600-1</td><td>TRC</td><td>A</td></tr><tr><td>23</td><td>3041-PID-0600-2</td><td>TRC GAD INJECTION</td><td>A</td></tr><tr><td>24</td><td>3041-PID-0601-1</td><td>TRC GAD FUEL TRAIN</td><td>A</td></tr><tr><td>25</td><td>3041-PID-0900-1</td><td>UTILITY SKID N2 HEADERS</td><td>A</td></tr><tr><td>26</td><td>3041-PID-0900-2</td><td>UTILITY SKID O2 HEADER</td><td>A</td></tr><tr><td>27</td><td>3041-PID-0900-3</td><td>UTILITY SKID CW HEADERS</td><td>A</td></tr><tr><td>28</td><td>3041-PID-0900-4</td><td>UTILITY SKID COOLING PUMPS</td><td>A</td></tr><tr><td>29</td><td>3041-PID-0900-5</td><td>UTILITY SKID LS HEADER</td><td>A</td></tr><tr><td>30</td><td>3041-PID-0900-6</td><td>UTILITY SKID PREHEATERS</td><td>A</td></tr><tr><td>31</td><td>3041-PID-0900-7</td><td>UTILITY SKID IA HEADER</td><td>A</td></tr><tr><td>32</td><td>3041-PID-0900-8</td><td>GAS ANALYZER</td><td>A</td></tr><tr><td>33</td><td>3041-PID-1000-1</td><td>FEED HANDLING 1 & SHREDDING</td><td>A</td></tr><tr><td>34</td><td>3041-PID-1000-2</td><td>FEED SHREDDER VENTILATION SYSTEM</td><td>A</td></tr><tr><td>35</td><td>3041-PID-1000-3</td><td>FEED HANDING 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TRACE</td><td>A</td></tr><tr><td>47</td><td>3041-PID-3200-1</td><td>SCRUBBER</td><td>A</td></tr><tr><td>48</td><td>3041-PID-3200-2</td><td>SCRUBBER PUMP AND PIPING</td><td>A</td></tr><tr><td>49</td><td>3041-PID-3200-3</td><td>CAUSTIC SUPPLY</td><td>A</td></tr><tr><td>50</td><td>3041-PID-3300-1</td><td>SYNGAS BLOWER 1</td><td>A</td></tr><tr><td>51</td><td>3041-PID-3300-2</td><td>SYNGAS BLOWER 2</td><td>A</td></tr><tr><td>52</td><td>3041-PID-3500-1</td><td>SULFUR ADSORBER SYSTEM</td><td>A</td></tr><tr><td>53</td><td>3041-PID-3800-1</td><td>HG REMOVAL & FILTRATION</td><td>A</td></tr><tr><td>54</td><td>3041-PID-3950-1</td><td>GAS ANALYZER</td><td>A</td></tr><tr><td>55</td><td>3041-PID-4100-1</td><td>FLARE</td><td>A</td></tr><tr><td>56</td><td>3041-PID-5000-1</td><td>SYNGAS COMPRESSOR</td><td>A</td></tr></table>																#	DRAWING NUMBER	DRAWING TITLE	REVISION	1	3041-PID-0000-1	DRAWING LIST	A	2	3041-PID-0001-1	LEGEND SHEET	A	3	3041-PID-0002-1	LEGEND 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PUMPS	A	29	3041-PID-0900-5	UTILITY SKID LS HEADER	A	30	3041-PID-0900-6	UTILITY SKID PREHEATERS	A	31	3041-PID-0900-7	UTILITY SKID IA HEADER	A	32	3041-PID-0900-8	GAS ANALYZER	A	33	3041-PID-1000-1	FEED HANDLING 1 & SHREDDING	A	34	3041-PID-1000-2	FEED SHREDDER VENTILATION SYSTEM	A	35	3041-PID-1000-3	FEED HANDING 2	A	36	3041-PID-2000-1	GLASS HANDLING SYSTEM	A	37	3041-PID-2100-1	METAL HANDLING SYSTEM	A	38	3041-PID-2200-1	GLASS ADDITIVES CONVEYOR	A	39	3041-PID-2200-2	GLASS ADDITIVES FEEDER	A	40	3041-PID-2300-1	WASTEWATER COLLECTION	A	41	3041-PID-2300-2	WASTEWATER TREATMENT	A	42	3041-PID-3000-1	PRIMARY QUENCH	A	43	3041-PID-3000-2	PRIMARY QUENCH TANK & PUMPS	A	44	3041-PID-3100-1	BAGHOUSE	A	45	3041-PID-3100-2	BAGHOUSE	A	46	3041-PID-3100-3	BAGHOUSE HEAT TRACE	A	47	3041-PID-3200-1	SCRUBBER	A	48	3041-PID-3200-2	SCRUBBER PUMP AND PIPING	A	49	3041-PID-3200-3	CAUSTIC SUPPLY	A	50	3041-PID-3300-1	SYNGAS BLOWER 1	A	51	3041-PID-3300-2	SYNGAS BLOWER 2	A	52	3041-PID-3500-1	SULFUR ADSORBER 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NOTES: 1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.			
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NOTES:
1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.

PLOT DATE:
3/21/2025

DRAWN: A BENNETT	
DATE:	
CHECKED:	
DATE:	
ENGR:	
DATE:	
APVD:	
DATE:	

REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

3041 P&ID DRAWING LIST	
SCALE	NONE
DWG NO	3041-PID-0000-1
CADFILE	3041-PID-0000-1.DWG
SHEET NO	1
SHEETS	1
REV	B
3/21/2025	

MISC. PIPING SYMBOLS

ABBREVIATIONS

DRAWING IDENTIFICATION

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G100P-PID-0620-1
(1) (2) (3) (4) (5)

AREA

- 1 JOB NUMBER
2 DISCIPLINE (SEE BELOW)
3 AREA (2 DIGITS)
4 SUB AREA (2 DIGITS)
5 SHEET NUMBER

DISCIPLINE

- A ARCHITECTURAL
C CIVIL
CS CONTROL SYSTEM
E ELECTRICAL
GA GENERAL ARRANGEMENT
M MECHANICAL
P PIPING
PEM MELTER SHELL
PFD PROCESS FLOW DIAGRAM
PID PIPING AND INSTRUMENTATION DIAGRAM
R REFRACTORY
S STRUCTURAL
SCH SCHEDULE

- 02 PREGASIFIER
03 PEM
04 INDUCTION, AC, & DC POWER
05 AC & DC ELECTRODE SYSTEMS
06 TRC
09 PEM UTILITIES
10 SOLIDS FEED SYSTEM
20 GLASS DRAIN
21 METAL DRAIN
22 GLASS ADDITIVES SYSTEM
23 WASTEWATER COLLECTION
30 PRIMARY QUENCH
31 BAGHOUSE
32 SCRUBBER & CAUSTIC SUPPLY
33 SYNGAS BLOWER
35 SULFUR ADSORBER
38 MERCURY ADSORBER & FILTRATION
39 SYNGAS ANALYSIS
41 E-FLARE
50 SYNGAS COMPRESSION
52 PSA
53 H2 COMPRESSION
54 H2 FILLING STATION
57 THERMAL OXIDIZER
84 VENTILATION
90 FUEL GAS UTILITIES & DISTRIBUTION
91 PROCESS WATER UTILITES & DISTRIBUTION
92 COOLING WATER SYSTEMS & DISTRIBUTION
93 N2 & O2 UTILITES & DISTRIBUTION
95 IA SYSTEMS & DISTRIBUTION
98 BFW & STEAM SYSTEMS & DISTRIBUTION

PIPING IDENTIFICATION

PIPE SPEC.NO.	CLASS RATING (#)	DESCRIPTION
1AA1	150#	304L STAINLESS STEEL PIPE
1AA2	150#	304L STAINLESS STEEL PIPE
UAB1	N/A	304L STAINLESS STEEL TUBING
1AC1	150#	316L STAINLESS STEEL PIPE
1AC2	150#	316L STAINLESS STEEL PIPE
UAD1	N/A	316L STAINLESS STEEL TUBING
1AE1	150#	HASTELLOY C-276 PIPE
UAF1	N/A	HASTELLOY C-276 TUBING
1AG1	150#	AL6XN PIPE
UAH1	N/A	AL6XN TUBING
1CA1	150#	CARBON STEEL PIPE
1CA2	150#	CARBON STEEL PIPE
1CA3	150#	CARBON STEEL PIPE
1CB1	150#	GALVANIZED CARBON STEEL PIPE
1CC1	150#	1-1/4 CHROME-1/2 MOLY PIPE
UKA1	N/A	COPPER PIPE
1LA1	150#	TEFLON LINED CARBON STEEL PIPE
UPA1	N/A	PVC PIPE
UPB1	N/A	PVC TUBING
UPC1	N/A	CPVC PIPE
UPD1	N/A	PP PIPE
UPE1	N/A	PP TUBING
UPF1	N/A	LDPE TUBING
UPG1	N/A	PVDF PIPE
UPH1	N/A	PVDF TUBING
UP1	N/A	PTFE PIPE
UPJ1	N/A	PTFE TUBING
UPK1	N/A	EPDM RUBBER HOSE
UPL1	N/A	PVC HOSE
UPM1	N/A	SPRINKLER HOSE
UPN1	N/A	NONCONDUCTIVE HOSE
UPO1	250#	STEAM HOSE

INSULATION TYPE

- NONE
H HEAT CONSERVATION
IC COLD
IH HOT
PP PERSONNEL PROTECTION

HEAT TRACE TYPE

- NONE
E ELECTRIC
ET ELECTRIC TRACE
S STEAM
SJ STEAM JACKET

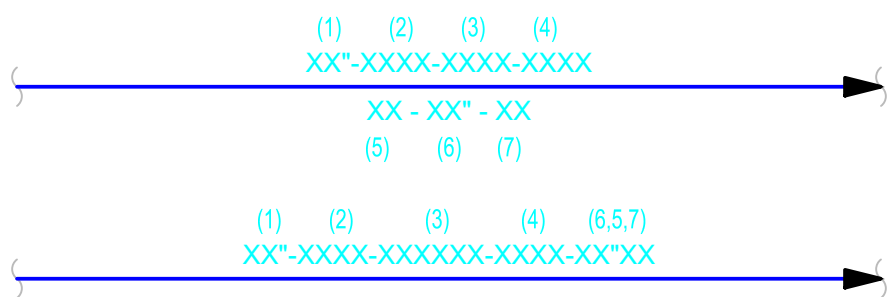
EQUIPMENT ID.

LINE IDENTIFICATION

FA-0730
(1) (2) (3)

1. EQUIPMENT TYPE (SEE BELOW)
2. AREA (2 DIGITS)
3. SEQUENCE (2 DIGITS)

- | | |
|-------------------------|-----------------------------|
| A MIXING EQUIPMENT | R REACTORS |
| B BLOWERS | T TOWERS / COLUMNS |
| C COMPRESSORS | TK TANKS |
| D MECHANICAL DRIVERS | U MISCELLANEOUS |
| E HEAT EXCHANGERS | V EQUIPMENT |
| EE ELECTRICAL EQUIPMENT | Y VESSELS |
| F FURNACES | |
| P PUMPS | ELECTRODE HOLDERS / FEEDERS |



- 1 LINE SIZE
2 LINE SERVICE CODE
3 LINE NUMBER
4 PIPE SPECIFICATION
5 INSULATION TYPE
6 INSULATION THICKNESS
7 HEAT TRACING (HT)

SERVICE IDENTIFICATION

- | | | |
|---------------------------------|--------------------------------|---------------------------|
| AV ATMOSPHERIC VENT | H HYDROGEN | SG SYNTHESIS GAS |
| BA BREATHING AIR | HS HIGH PRESSURE STEAM | SO SEAL OIL |
| BD BLOWDOWN | IA INSTRUMENT AIR | SS SANITARY SEWER |
| BFW BOILER FEED WATER | IS INTERMEDIATE PRESSURE STEAM | ST STEAM |
| BRR BRINE RETURN | LNG LIQUID NATURAL GAS | STS STORM SEWER |
| BRS BRINE SUPPLY | LO LUBE OIL | SW SERVICE WATER |
| CC CONTAMINABLE CONDENSATE | LPG LIQUIFIED PETROLEUM GAS | SWR SEA WATER RETURN |
| CE COMBUSTION EXHAUST | LS LOW PRESSURE STEAM | SWS SEA WATER SUPPLY |
| CF COLD FLARE | ME METHANOL | TWR TEMPERED WATER RETURN |
| CHS CHEMICAL SEWER | MR MIXED REFRIGERANT | TWS TEMPERED WATER SUPPLY |
| CHWR CHILLED WATER RETURN | MS MEDIUM PRESSURE STEAM | UR UREA |
| CHWS CHILLED WATER SUPPLY | MTL METAL | VC VACUUM CONDENSATE |
| CO2 CARBON DIOXIDE | N NITROGEN | VE VACUUM EXHAUST |
| CTWR COOLING TOWER WATER RETURN | NA CAUSTIC | WF WARM FLARE OIL |
| CTWS COOLING TOWER WATER SUPPLY | NAS CAUSTIC SEWER | WO WASH OIL |
| CV COLD VENT | NG NATURAL GAS | WW WASTEWATER |
| CWR COOLING WATER RETURN | NH AMMONIA | |
| CWS COOLING WATER SUPPLY | OWS OILY WATER SEWER | |
| DMV DEMINERALIZED WATER | OX OXYGEN | |
| DR DRAIN | P GENERAL PROCESS | |
| DRS DILUTION STEAM | PA PLANT AIR | |
| DW DRINKING WATER | PC PROCESS CONDENSATE | |
| ER ETHYLENE REFRIGERANT | PR PROPYLENE REFRIGERANT | |
| FF FLUSHING FLUID | PW PROCESS WATER | |
| FO FUEL GAS | QO QUENCH OIL | |
| FW FUEL OIL | RS RECOVERED SLOPS | |
| GLR GLASS | RV RELIEF VENT | |
| GLS GLYCOL RETURN | RW RAW WATER | |
| | SB SCRUB WATER | |
| | SC STEAM CONDENSATE | |

REVISIONS

REV NO	DCR/DESCRIPTION
	INITIAL ISSUE

3041
P&ID
LEGEND SHEET

SCALE	NONE	SHEET NO	1	SHEETS	1
DWG NO	3041-PID-0001-1	REV			A
CADFILE	3041-PID-0001-1.DWG				4/20/2023

EQUIPMENT

PUMPS & BLOWERS

CENTRIFUGAL PUMP

POSITIVE DISPLACEMENT PUMP

PROGRESSIVE CAVITY PUMP

CENTRIFUGAL BLOWER

POSITIVE DISPLACEMENT BLOWER

CENTRIFUGAL COMPRESSOR

SCREW COMPRESSOR

EXCHANGERS

PLATE EXCHANGER (PFD)

PLATE EXCHANGER (P&ID)

SHELL & TUBE EXCHANGER (PFD)

SHELL & TUBE EXCHANGER (P&ID)

AIR COOLER

MISCELLANEOUS

FILTER

PACKED BED

FLAME ARRESTOR

DETONATION ARRESTOR

FLARE

TANKS & VESSELS

VERTICAL VESSEL

TANK

VALVES

GATE

DIAPHRAGM

KNIFE GATE

PINCH

GLOBE

CHECK

BUTTERFLY

BALL

PLUG

NEEDLE

3-WAY

4-WAY

ANGLE

PRESSURE RELIEF VALVE (PSV)

VACUUM RELIEF VALVE

PRESSURE & VACUUM RELIEF VALVE

PILOT OPERATED RELIEF VALVE

RUPTURE DISC FOR PRESSURE RELIEF

RUPTURE DISC FOR VACUUM RELIEF

VALVE ACTUATORS

DIAPHRAGM

DIAPHRAGM W/ MANUAL OVERRIDE

DIAPHRAGM W/ POSITIONER

SELF-CONTAINED PRESSURE REGULATOR

PISTON, SINGLE-ACTION

PISTON, DOUBLE-ACTION

MOTOR

TEMPERATURE REGULATOR (FILLED SYSTEM)

SOLENOID VALVES

SOLENOID

SOLENOID W/ AUTOMATIC RESET

SOLENOID W/ MANUAL RESET

FOUR-WAY SOLENOID

THREE-WAY SOLENOID FLOW DIRECTION IN DE-ENERGIZED CONDITION

SIGNAL CONDITIONING DEVICE SYMBOLS

IP CURRENT-TO-PRESSURE

PI PRESSURE-TO-CURRENT

EL VOLTAGE-TO-CURRENT

√ SQUARE ROOT

Σ SUMMING

× MULTIPLYING

÷ DIVIDING

+ BIAS

Δ DIFFERENCE

< HIGH SELECTING

> LOW SELECTING

∞ HIGH LIMITING

∞ LOW LIMITING

∫ INTEGRAL

K PROPORTIONAL

1/s DERIVATIVE

C CONSTANT

OR LOGICAL OR

AND LOGICAL AND

Z RELAY OR GATE

SIGNAL CONDITIONING DEVICE SYMBOL TO BE IN BOX LOCATED IN APPROX. POSITION AS SHOWN TO INSTRUMENT BALLOON.

INSTRUMENT IDENTIFICATION

FIRST LETTER		SUCCEEDING LETTERS			
MEASURED OR INITIATING VARIABLE		MODIFIER	READOUT OR PASSIVE FUNCTION	OUTPUT FUNCTION	MODIFIER
A	ANALYSIS, ANALYZER	DIFFERENTIAL	ALARM	USERS CHOICE	USERS CHOICE
B	BURNER, FLAME, COMBUSTION, BELT		USER'S CHOICE		USERS CHOICE
C	CONDUCTIVITY		SENSOR (PRIMARY ELEMENT)	CONTROL	CLOSE
D	DENSITY				DIVERT
E	VOLTAGE (EMF)		GAUGE, GLASS	INDICATE	HIGH
F	FLOW RATE	RATIO (FRACTION)			
G	USER'S CHOICE	SCAN	CONTROL STATION	LOW	
H	HAND				MIDDLE OR INTERMEDIATE
I	CURRENT (ELECTRICAL)	TIME RATE OF CHANGE	USERS CHOICE	USERS CHOICE	
J	POWER			OPEN	
K	TIME, TIME SCHEDULE	MOMENTARY	ORIFICE, RESTRICTION POINT (TEST) CONNECTION	REGULATING SWITCH	THROUGH
L	LEVEL				
M	USER'S CHOICE	SAFETY	RECORD	MULTIFUNCTION	MULTIFUNCTION
N	USER'S CHOICE				
O	USER'S CHOICE	X AXIS	UNCLASSIFIED	UNCLASSIFIED	UNCLASSIFIED
P	PRESSURE, VACUUM				
Q	QUANTITY OR HEAT DUTY	Y AXIS	WELL	UNCLASSIFIED	UNCLASSIFIED
R	RADIATION				
S	SPEED, FREQUENCY	Z AXIS	UNCLASSIFIED	UNCLASSIFIED	UNCLASSIFIED
T	TEMPERATURE				
U	MULTIVARIABLE	Z AXIS	UNCLASSIFIED	UNCLASSIFIED	UNCLASSIFIED
V	VIBRATION, MECHANICAL ANALYSIS				
W	WEIGHT OR FORCE	Z AXIS	UNCLASSIFIED	UNCLASSIFIED	UNCLASSIFIED
X	UNCLASSIFIED				
Y	EVENT, STATE, OR PRESENCE	Z AXIS	UNCLASSIFIED	UNCLASSIFIED	UNCLASSIFIED
Z	POSITION, DIMENSION				

EXAMPLES:

LEVEL

↑↑↑

LAL

↑↑↑

LOW

ALARM

PRESSURE

↑↑↑

PDAH

↑↑↑

HIGH

DIFFERENTIAL

ALARM

POSITION

↑↑↑

ZSC

↑↑↑

CLOSED

SWITCH

EXAMPLES:

LEVEL LAL LOW ALARM

PRESSURE POAH HIGH ALARM

POSITION ZSC SWITCH

INSTRUMENTATION

LOCAL

MOUNTED ON MAIN PANEL

MOUNTED BEHIND MAIN PANEL

MOUNTED ON LOCAL PANEL

MOUNTED BEHIND LOCAL PANEL

ROTAMETER WITH INTEGRAL VALVE

FLOW ELEMENT ROTAMETER

FLOW ELEMENT VENTURI

FLOW ELEMENT TURBINE

FLOW ELEMENT PITOT

FLOW ELEMENT THERMAL MASS

DIAPHRAGM SEAL

N = STREAM NUMBER ON PFD

PLC FUNCTION OPERATOR ACCESSIBLE

PLC FUNCTION OPERATOR INACCESSIBLE

DCS FUNCTION OPERATOR ACCESSIBLE

DCS FUNCTION OPERATOR INACCESSIBLE

FLOW ELEMENT MAGMETER

FLOW ELEMENT ULTRASONIC

FLOW ELEMENT VORTEX

FLOW ELEMENT ORIFICE

FLOW ELEMENT TARGET

INTERLOCK/LOGIC FUNCTION

X = LOGIC NOTE NUMBER (CAUSE = NUMERIC / EFFECT = ALPHA)

ELECTRIC SIGNAL

PNEUMATIC SIGNAL

SOFTWARE LINK

LEADER

HYDRAULIC SIGNAL

ELECTROMAGNETIC SIGNAL

MECHANICAL LINK

CAPILLARY TUBE

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DRAWN: A BENNETT
DATE:
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APVD:
DATE:

REVISIONS	
REV NO	DCR/DESCRIPTION

3041 P&ID LEGEND SHEET	
SCALE NONE	SHEET NO 1 SHEETS 1
DWG NO 3041-PID-0002-1	REV A
CADFILE 3041-PID-0002-1.DWG	4/20/2023

PA-0210 F-0210 V-0200
PREGASIFIER PREHEATER PACKAGE PREGASIFIER PREHEATER PREGASIFIER VESSEL

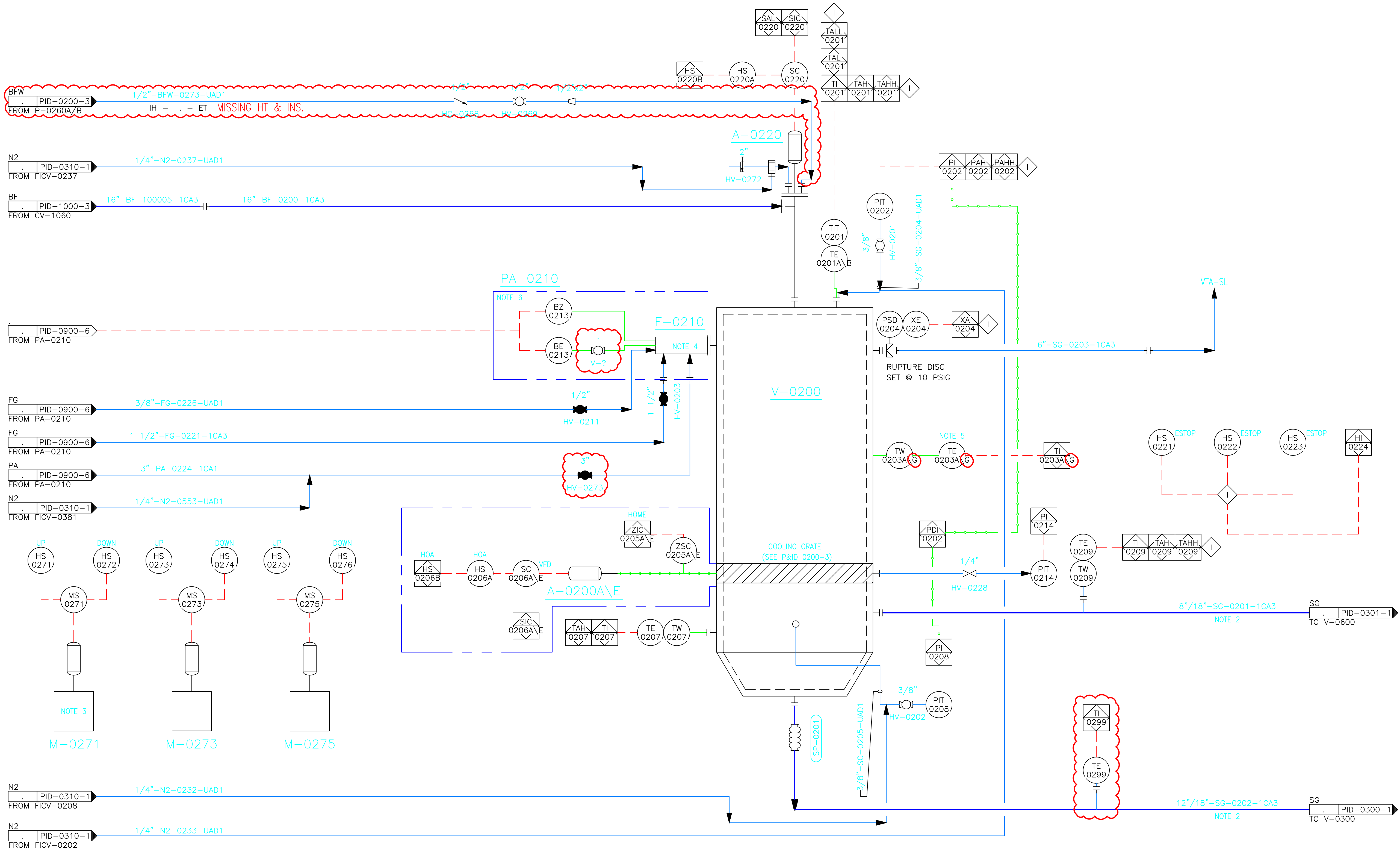
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NOTES:

1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
2. PIPE SIZE SHOWN IS APPROXIMATE INSIDE / OUTSIDE DIAMETER REQUIRED. THE DIFFERENCE INDICATES THE ALLOWANCE FOR REFRACTORY.
3. PREGASIFIER LIFTING MECHANISM FOR GRATE INSTALLATION AND REMOVAL.
4. PREHEATER MAYBE REMOVED BEFORE PROCESSING AND REQUIRES SUPPORT TO REMOVE. AIR AND GAS LINES NEED TO BE EASILY DISCONNECTED FROM PREHEATER. ALSO THE PREHEATER REQUIRES A MOUNTING PAD FOR THE IGNITER TRANSFORMER BOX NEAR THE PREHEATER.
5. AN ARRAY OF TE THERMOCOUPLES DOWN THE SIDE OF THE PREGASIFIER.
6. PREGASIFIER PREHEATER BLOWER, VALVES, AND CONTROLS ON 3004-PID-0900-6
7. TEMPORARY.



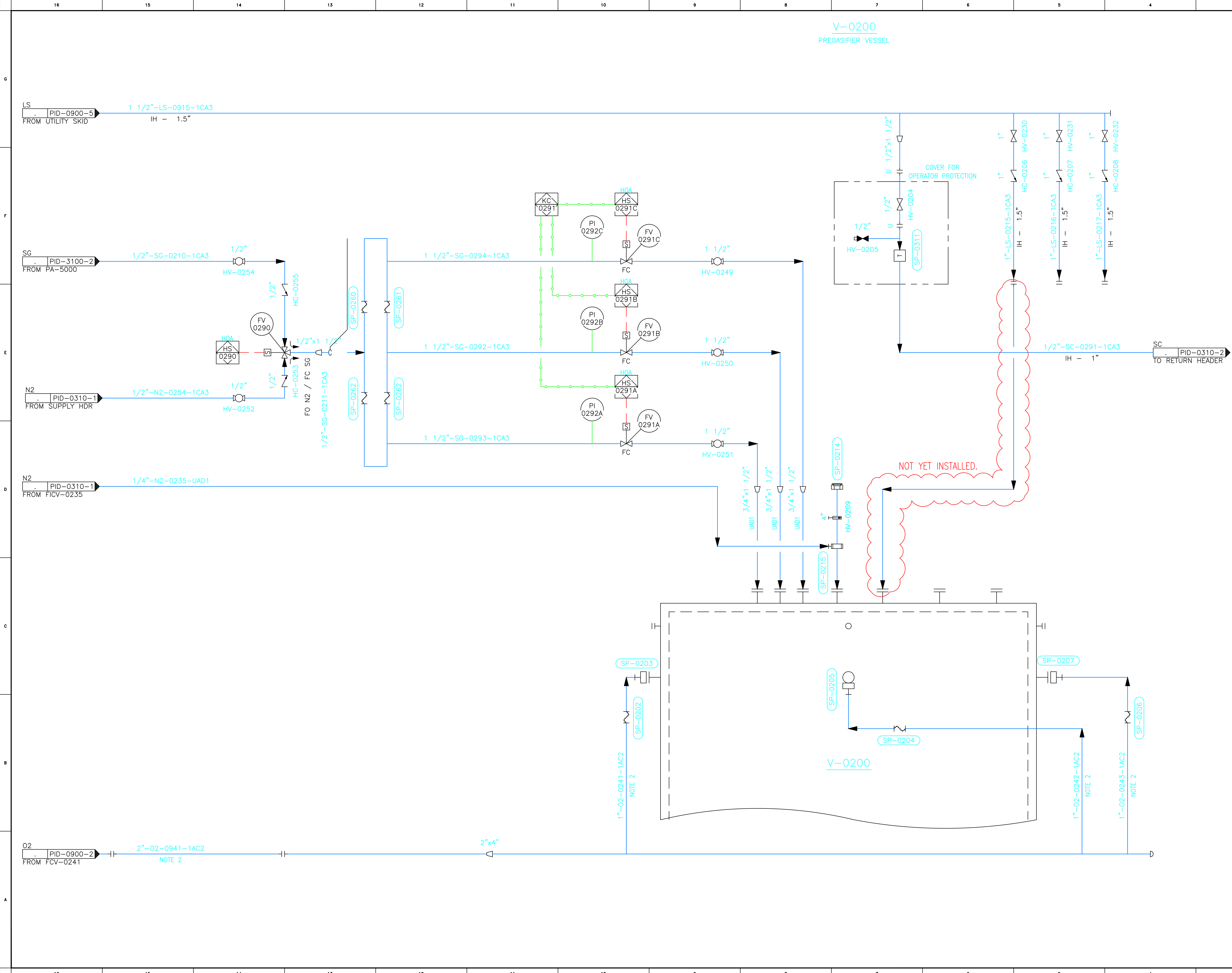
DRAWN: A BENNETT
DATE:
CHECKED:
DATE:
ENGR:
DATE:
APVD:
DATE:

REVISIONS

REV NO	DCR/DESCRIPTION

3041
P&ID
PREGASIFIER

SCALE	NONE	SHEET NO	1	SHEETS	1
DWG NO	3041-PID-0200-1	REV	A		
CADFILE	3041-PID-0200-1.DWG				4/20/2023



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- NOTES:
1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
 2. ALL O2 SERVICE VALVES AND INSTRUMENTATION TO BE OXYGEN CLEANED AND COMPATIBLE.
 3. A MOUNTING PAD MAY BE REQUIRED NEAR EACH GAD FOR THE IGNITER TRANSFORMER.
 4. TEMPORARY

DRAWN: A BENNETT DATE:	
CHECKED: P. PEREZ DATE: 6/28/2020	
ENGR: DATE:	
APVD: DATE:	
REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE
B	REVISED AND REDRAWN
C	REVISED AND REDRAWN
3041 P&ID PREGASIFIER O2, LS, N2	
SCALE	NONE
DWG NO	3041-PID-0200-2
CADFILE	3041-PID-0200-2.DWG
SHEET NO	1
SHEETS	1
REV	C
DATE	9/27/2023

V-0260
ACTIVE GRATE STEAM DRUM

E-0200
PREGASIFER FEED SPOOL HEAT EXCHANGER

A-0200
GRATE SYSTEM

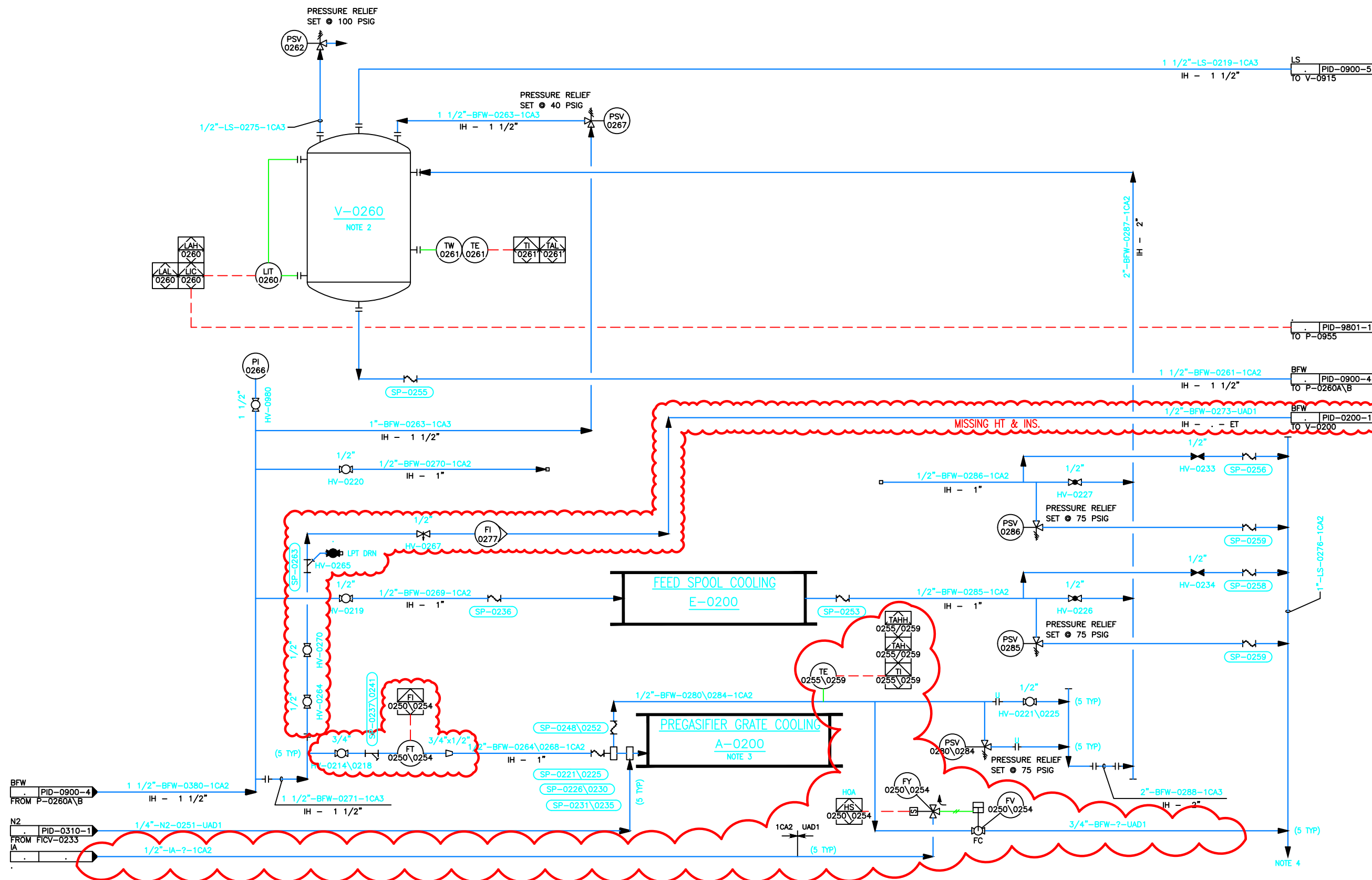
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NOTES:

1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
2. STEAM DRUM MUST BE LOCATED ABOVE THE T-SPINNER AND THE STEAM DRUM HAS 3" OF INSULATION.
3. THERE ARE FIVE GRATE BARS TO BE COOLED.
4. VENT AT GROUND LEVEL TO ALLOW WATER TO DRAIN OUT FROM LINE.



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APVD:
DATE:

REVISIONS

REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE
B	REVISED AND REDRAWN
C	REVISED AND REDRAWN

3041
P&ID
PREGASIFIER COOLING

SCALE NONE
DWG NO 3041-PID-0200-3
CADFILE 3041-PID-0200-3.DWG
SHEET NO 1
SHEETS 1
REV C
9/27/2023

V-0200?
DDG VESSEL

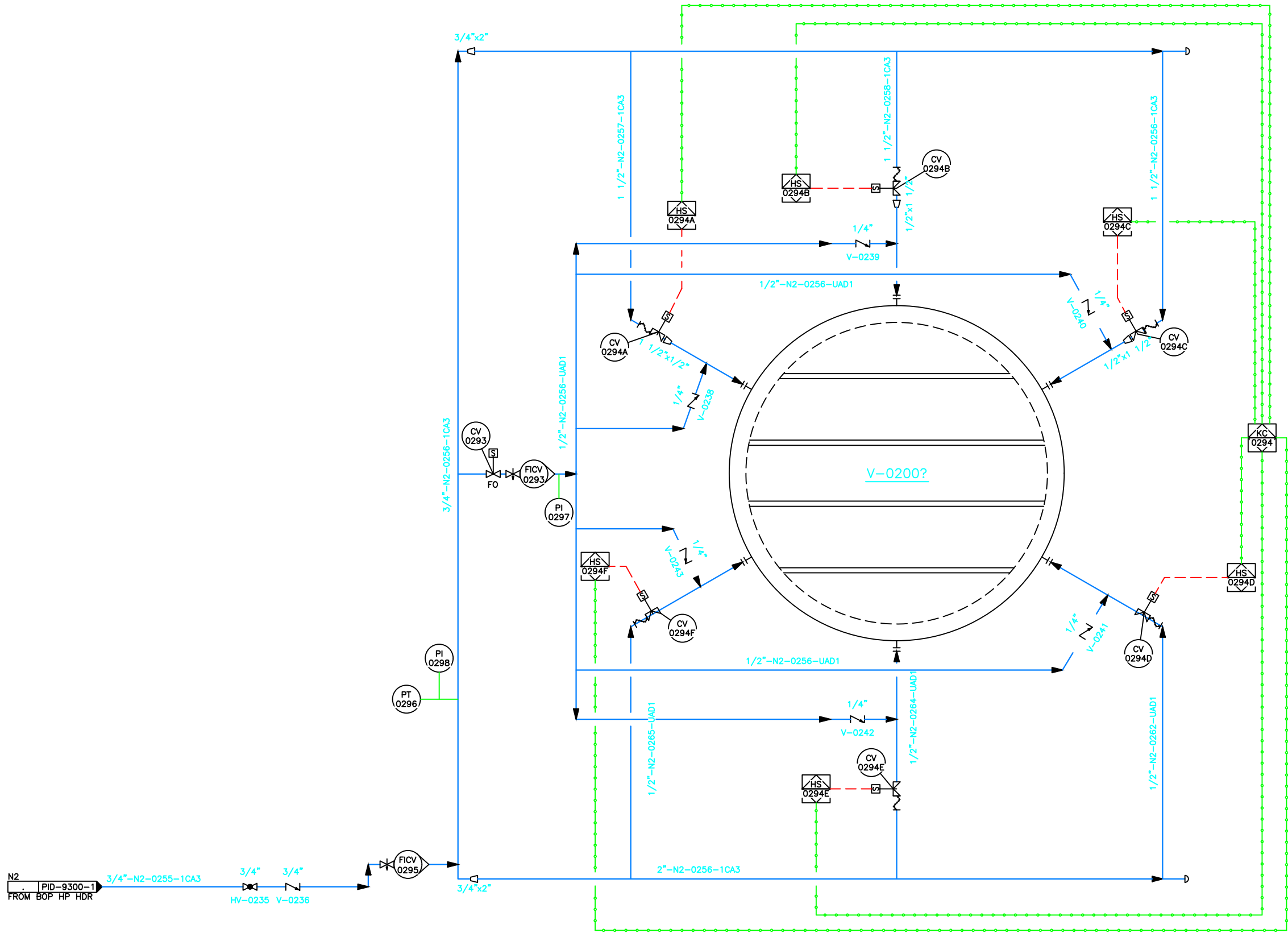
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NOTES:

1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
2. VIEWPORT LOCATED ON 14 IN FLANGE LOOKING AT DRAIN.
3. ACCESS PORT FOR WORKING ON DRAIN.



DRAWN: P SALAS
DATE: 9/11/23

CHECKED:
DATE:

ENGR:
DATE:

APVD:
DATE:

REVISIONS

REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE
B	ADDED FILL LINE ROTAMETER AND PRESSURE GAUGES

3041
P&ID
DDG GRATE PULSERS

SCALE NONE

DWG NO 3041-PID-0200-4

CADFILE 3041-PID-0200-4.DWG

SHEET NO 1

REV B

3/21/2025

V-0200?
DDG VESSEL

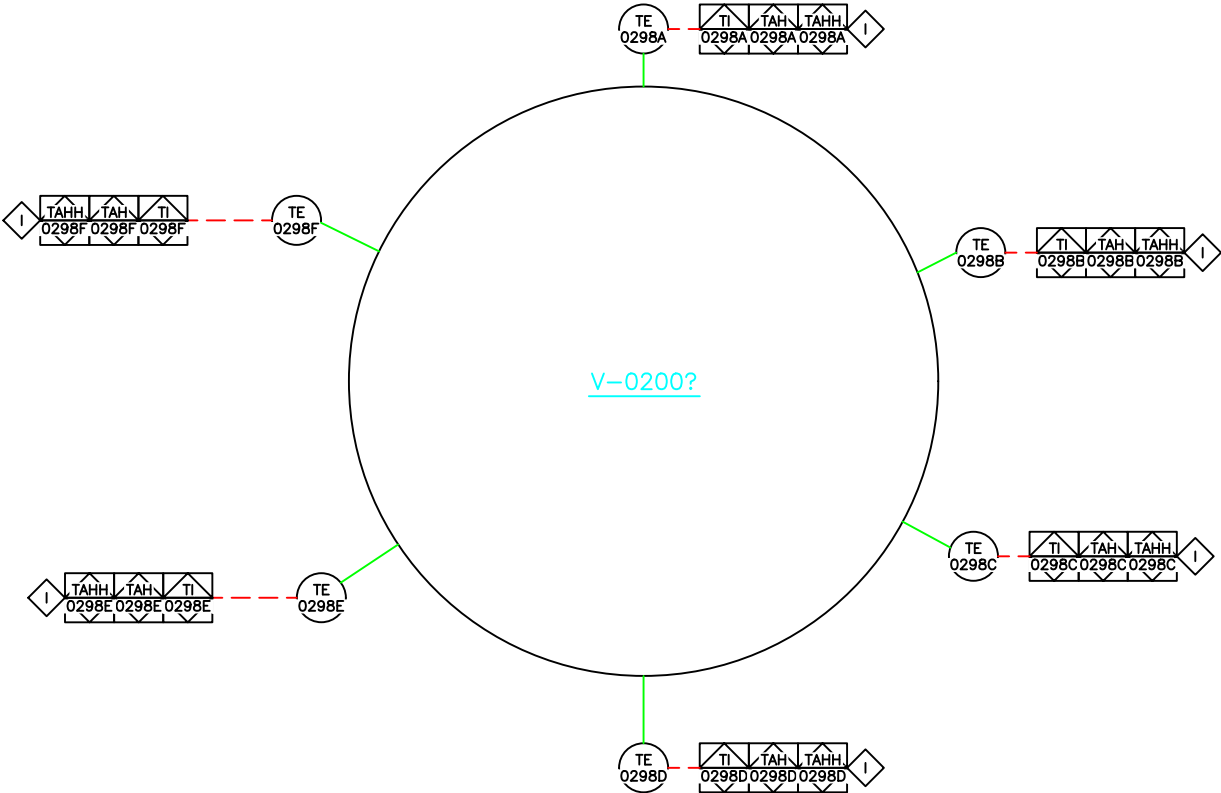
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- 2. VIEWPORT LOCATED ON 14 IN FLANGE LOOKING AT DRAIN.
- 3. ACCESS PORT FOR WORKING ON DRAIN.



FROM BOP HP HDR

DRAWN: P SALAS
DATE: 9/11/23

CHECKED:
DATE:

ENGR:
DATE:

APVD:
DATE:

REVISIONS

REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

3041
P&ID
T-CUP TEMPERATURE INDICATION

SCALE NONE

DWG NO 3041-PID-0200-5

CADFILE 3041-PID-0200-5.DWG

SHEET NO 1

SHEETS 1

REV A

3/20/2024

V-0300
PEM VESSEL

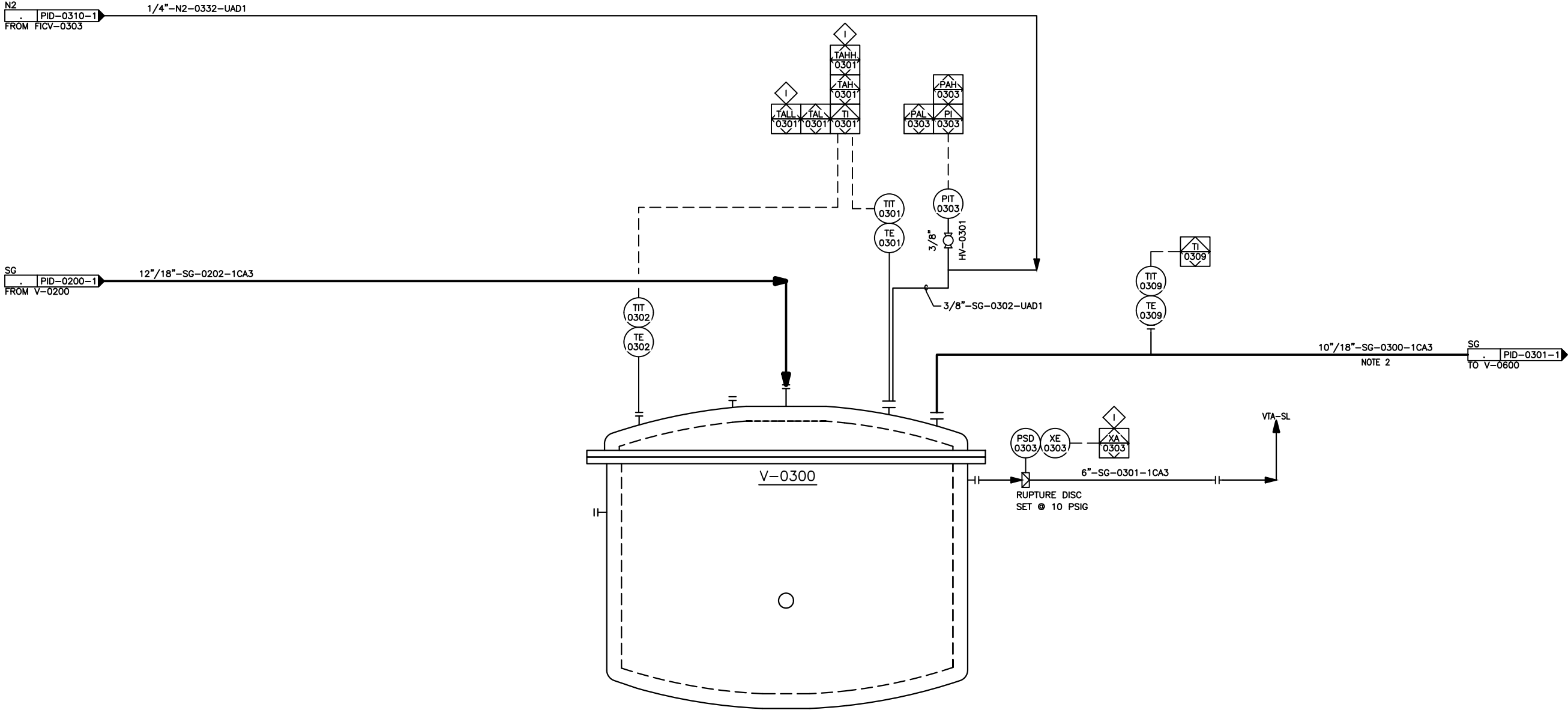
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REVISIONS

REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE
B	REVISED AND REDRAWN
C	REVISED AND REDRAWN

3041
P&ID
PEM

SCALE NONE

DWG NO 3041-PID-0300-1

CADFILE 3041-PID-0300-1.DWG

SHEET NO 1

SHEETS 1

REV C

9/27/2023

V-0300
PEM VESSEL

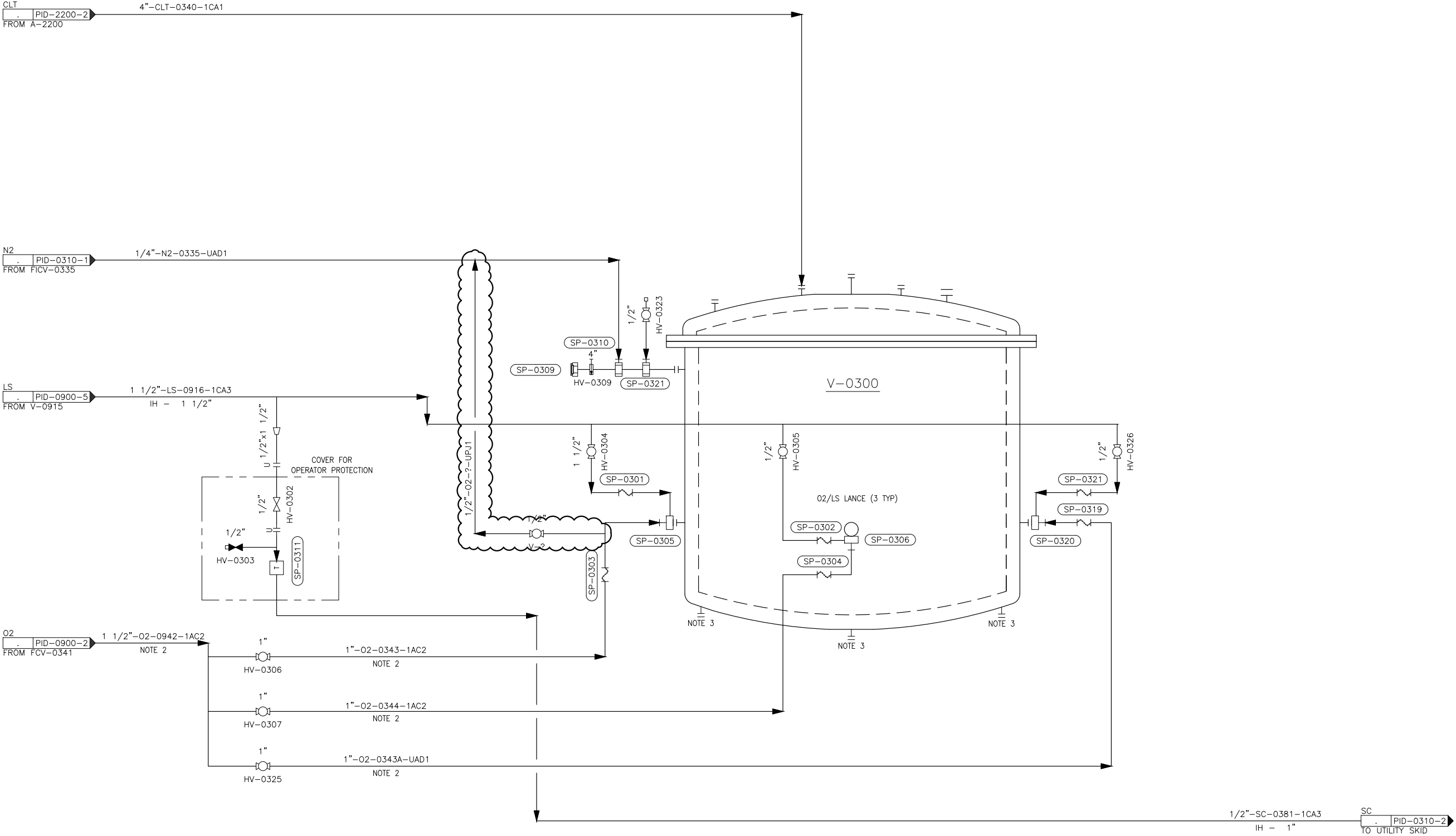
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NOTES:

1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
2. ALL O2 SERVICE VALVES AND INSTRUMENTATION TO BE OXYGEN CLEANED AND COMPATIBLE.
3. DRAIN PORT FOR GLASS AND METAL REMOVAL.



DRAWN: A BENNETT
DATE:

CHECKED: P PEREZ
DATE: 6/28/2020

ENGR:
DATE:

APVD:
DATE:

REVISIONS

REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE
B	REVISED AND REDRAWN
C	REVISED AND REDRAWN

3041
P&ID
PEM LS, O2, & N2

SCALE	NONE	SHEET NO	1	SHEETS	1
DWG NO	3041-PID-0300-2	REV	D		
CADFILE	3041-PID-0300-2.DWG				3/21/2024

V-0300
PEM VESSEL

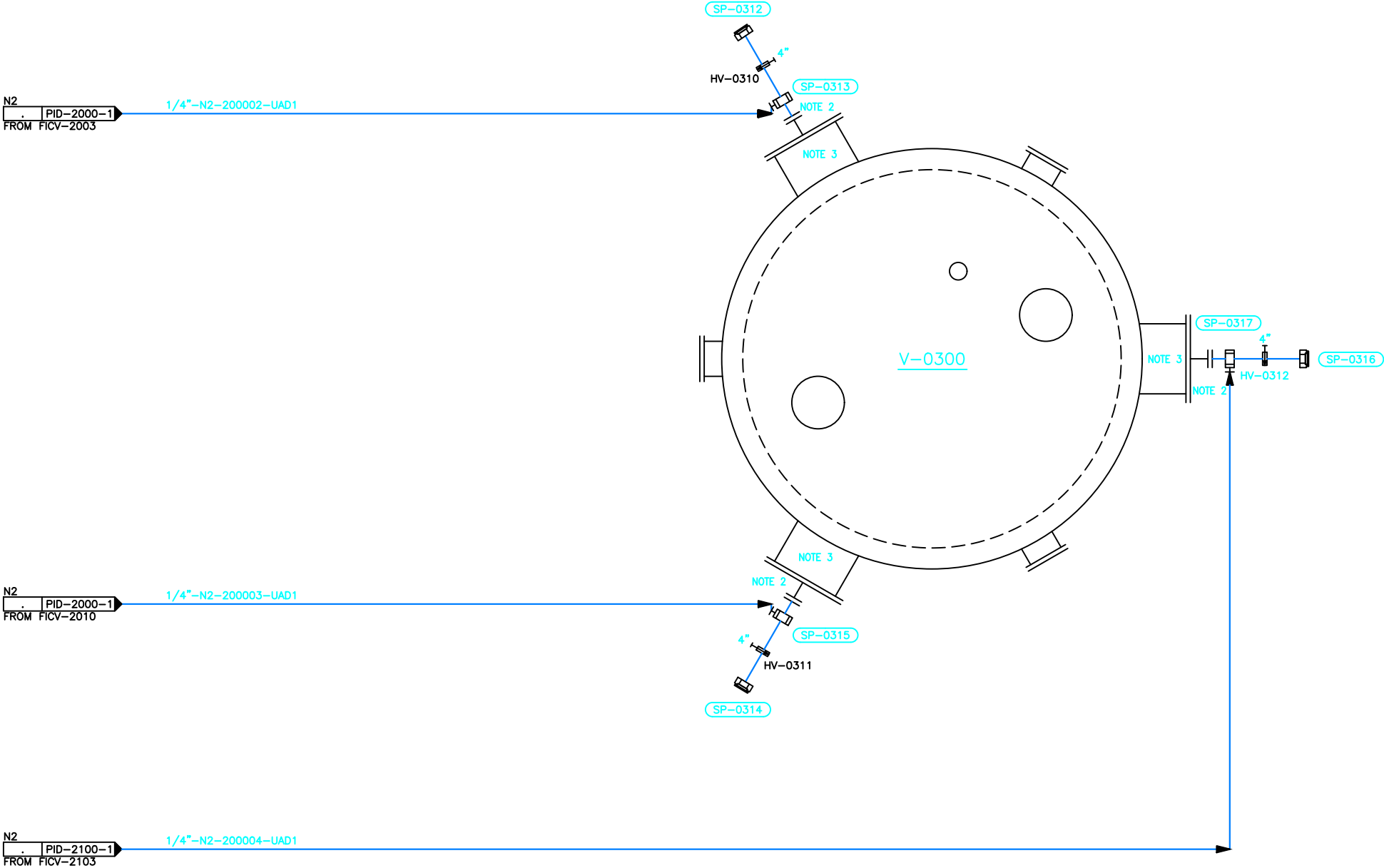
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NOTES:

1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
2. VIEWPORT LOCATED ON 14 IN FLANGE LOOKING AT DRAIN.
3. ACCESS PORT FOR WORKING ON DRAIN.



DRAWN: A BENNETT
DATE:

CHECKED: P PEREZ
DATE: 6/28/2020

ENGR:
DATE:

APVD:
DATE:

REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE
B	REVISED AND REDRAWN
C	REVISED AND REDRAWN

3041
P&ID
PEM DRAINS

SCALE NONE

DWG NO 3041-PID-0300-3

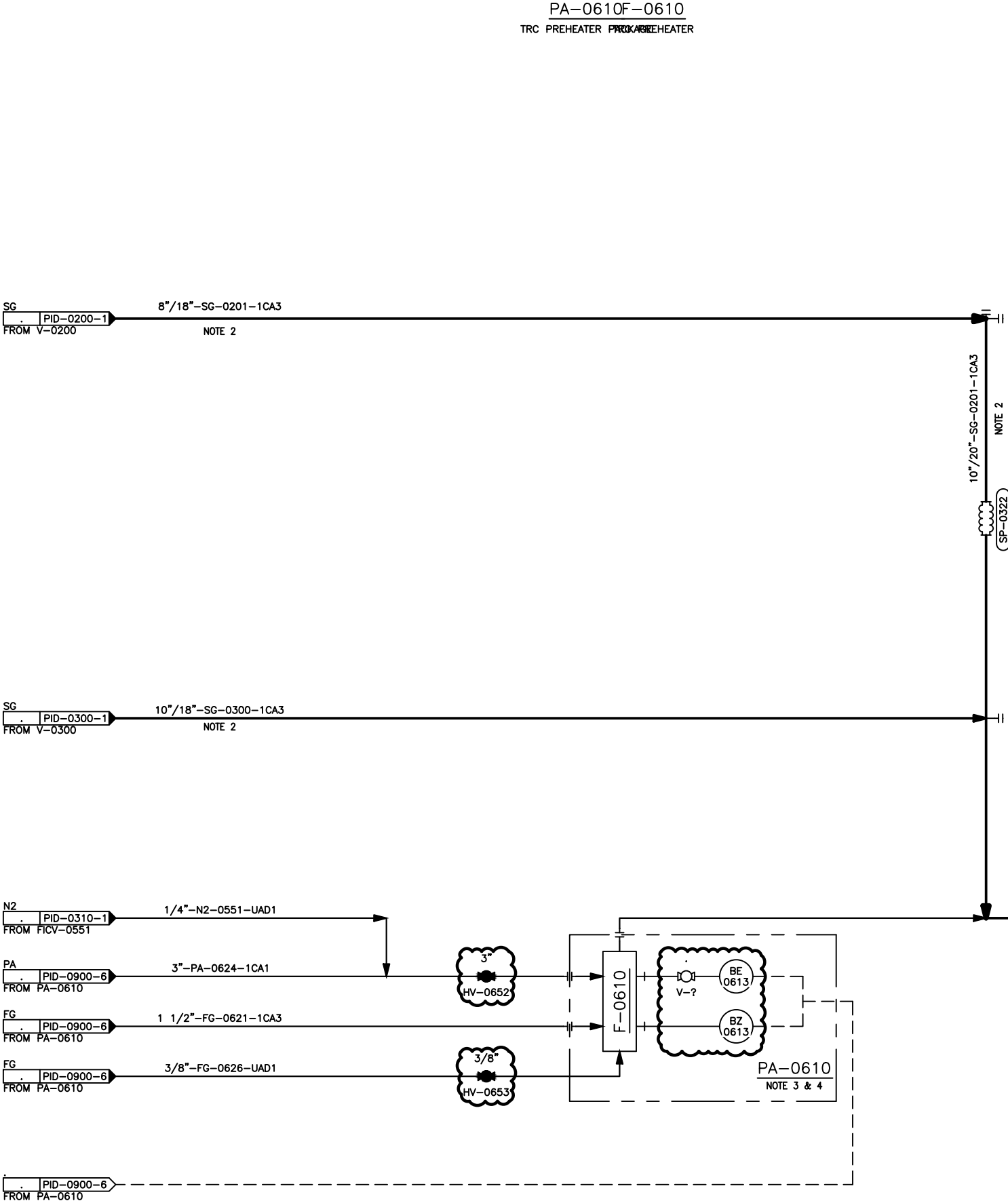
CADFILE 3041-PID-0300-3.DWG

SHEET NO 1

SHEETS 1

REV C

9/11/2023



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NOTES:

1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
2. PIPE SIZE SHOWN IS APPROXIMATE INSIDE / OUTSIDE DIAMETER REQUIRED. THE DIFFERENCE INDICATES THE ALLOWANCE FOR REFRACTORY.
3. PREHEATER MAYBE REMOVED BEFORE PROCESSING AND REQUIRES SUPPORT TO REMOVE. AIR AND GAS LINES NEED TO BE EASILY DISCONNECTED FROM PREHEATER. ALSO THE PREHEATER REQUIRES A MOUNTING PAD FOR THE IGNITER TRANSFORMER BOX NEAR THE PREHEATER.
4. TRC PREHEATER BLOWER, VALVES, AND CONTROLS ON 3041-PID-0900-6.

DRAWN: A BENNETT
DATE:
CHECKED: P PEREZ
DATE: 6/28/2020

ENGR:
DATE:

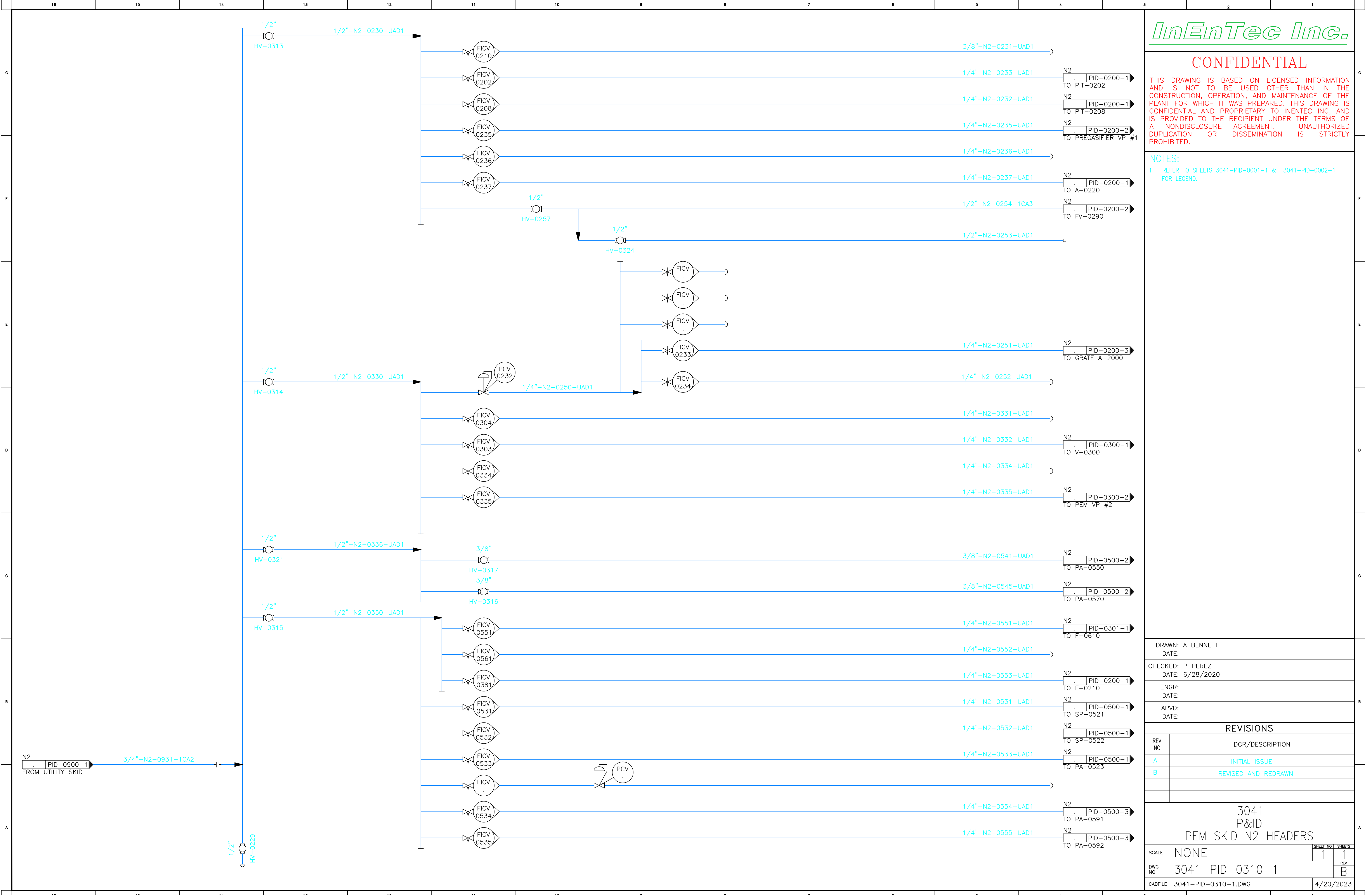
APVD:
DATE:

REVISIONS

REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

3041
P&ID
PEM-TRC SG DUCT

SCALE	NONE	SHEET NO	1	SHEETS	1
DWG NO	3041-PID-0301-1	REV	A		
CADFILE	3041-PID-0301-1.DWG				



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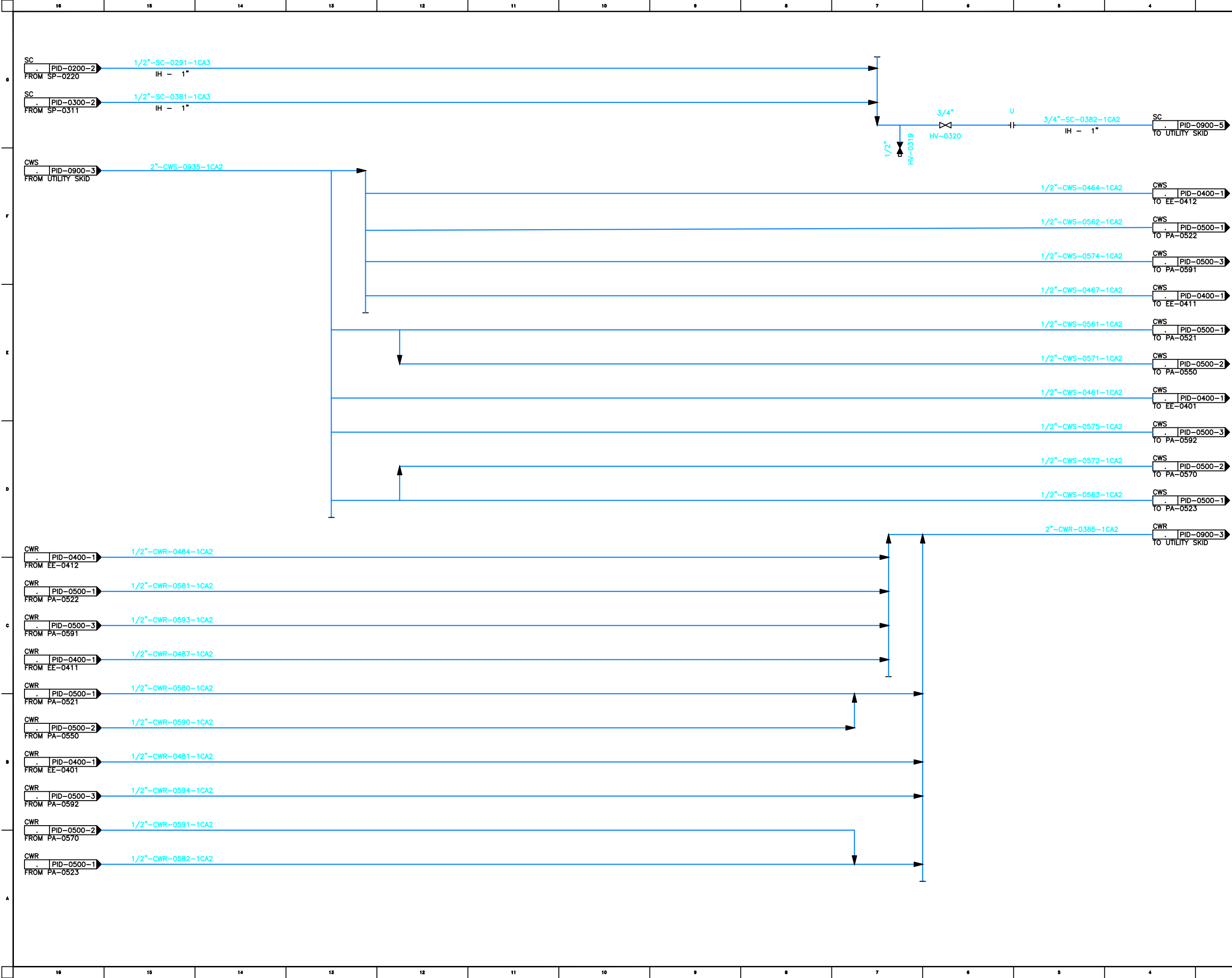
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- NOTES:
- 1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.

DRAWN: A BENNETT	
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DATE: 6/28/2020	
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APVD:	
DATE:	

REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE
B	REVISED AND REDRAWN

3041	
P&ID	
PEM SKID N2 HEADERS	
SCALE	NONE
DWG NO	3041-PID-0310-1
CADFILE	3041-PID-0310-1.DWG
SHEET NO	1
SHEETS	1
REV	B
DATE	4/20/2023



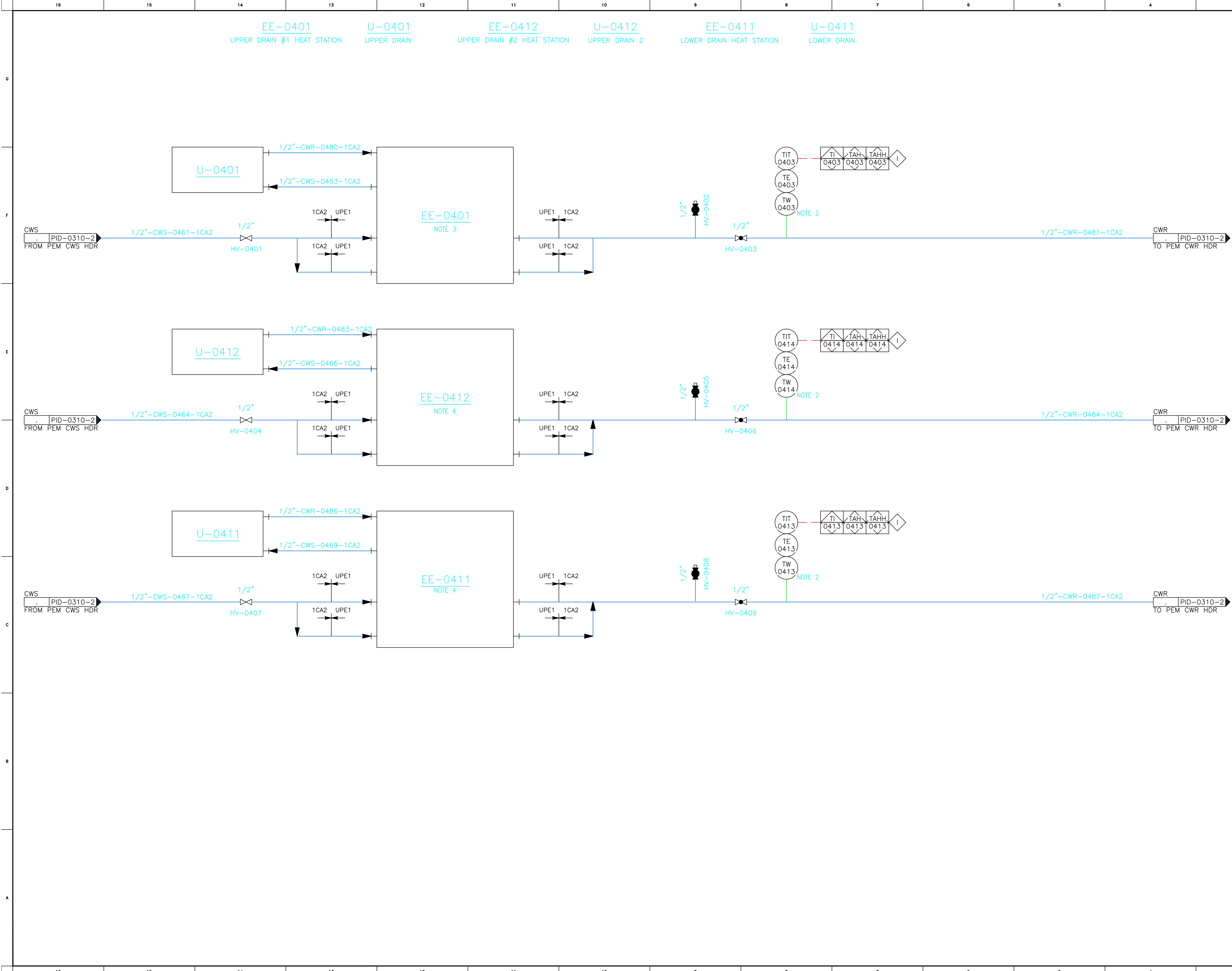
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- NOTES:
- 1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.

DRAWN: A BENNETT	
DATE:	
CHECKED: P PEREZ	
DATE: 6/528/2020	
ENGR:	
DATE:	
APVD:	
DATE:	
REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE
B	REVISED AND REDRAWN
3041 P&ID PEM SKID CW HEADERS	
SCALE	NONE
DWG NO	3041-PID-0310-2
CADFILE	3041-PID-0310-2.DWG
SHEET NO	1
SHEETS	1
REV	B
4/20/2023	



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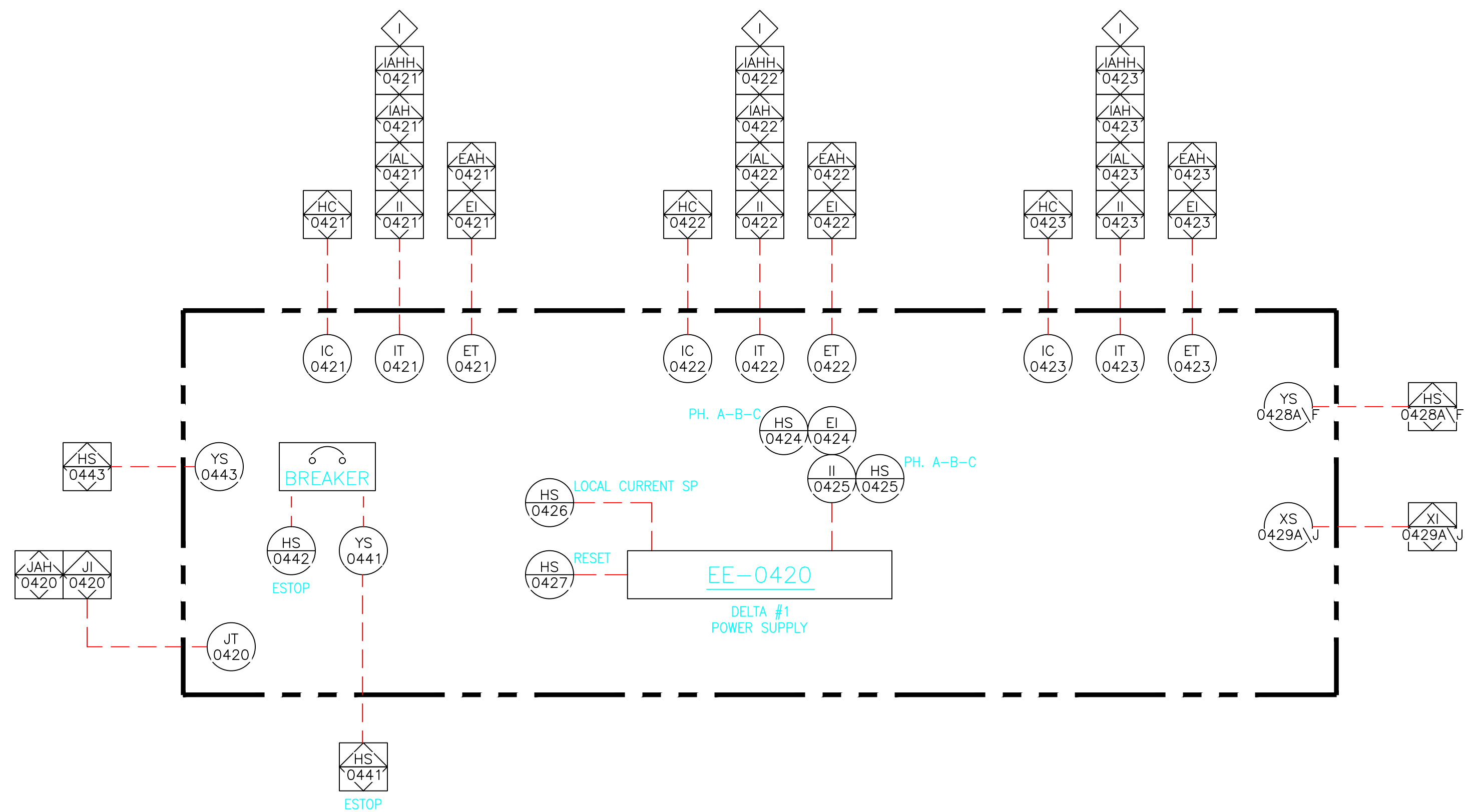
THIS DRAWING IS BASED ON LICENSED INFORMATION AND IS NOT TO BE USED OTHER THAN IN THE CONSTRUCTION, OPERATION, AND MAINTENANCE OF THE PLANT FOR WHICH IT WAS PREPARED. THIS DRAWING IS CONFIDENTIAL AND PROPRIETARY TO INENTEC INC, AND IS PROVIDED TO THE RECIPIENT UNDER THE TERMS OF A NONDISCLOSURE AGREEMENT. UNAUTHORIZED DUPLICATION OR DISSEMINATION IS STRICTLY PROHIBITED.

- NOTES:
- REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
 - THERMOWELL ASSEMBLY FOR SMALL PIPE.
 - HEAT STATION EE-0401 IS CONNECTED TO DRAIN POWER SUPPLY EE-0400 ON 3041-PID-0900-3.
 - HEAT STATIONS EE-0411 AND EE-0412 ARE CONNECTED TO DRAIN POWER SUPPLY EE-0410 AND SWITCH EE-0415 ON 3041-PID-0900-3. ONLY ONE OF THESE HEAT STATIONS CAN BE OPERATED AT A TIME BY USING THE SWITCH TO SELECT IT FROM THE HMI.

DRAWN: A BENNETT
DATE:
CHECKED: P PEREZ
DATE: 6/28/2020
ENGR:
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APVD:
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REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

3041 P&ID DRAIN POWER SUPPLIES	
SCALE NONE	SHEET NO 1 SHEETS 1
DWG NO 3041-PID-0400-1	REV A
CADFILE 3041-PID-0400-1.DWG	4/20/2023



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NOTES:

1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.

DRAWN: A BENNETT			
DATE:			
CHECKED: P PEREZ			
DATE: 6/28/2020			
ENGR:			
DATE:			
APVD:			
DATE:			
REVISIONS			
REV NO	DCR/DESCRIPTION		
A	INITIAL ISSUE		
3041 P&ID AC POWER SUPPLY			
SCALE	NONE	SHEET NO	SHEETS
		1	1
DWG NO	3041-PID-0400-2	REV	
		A	
CADFILE	3041-PID-0400-2.DWG	4/20/2023	



EE-0470
DC POWER SUPPLY #2

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- REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.

DRAWN: A BENNETT
DATE:

CHECKED: P PEREZ
DATE: 6/28/2020

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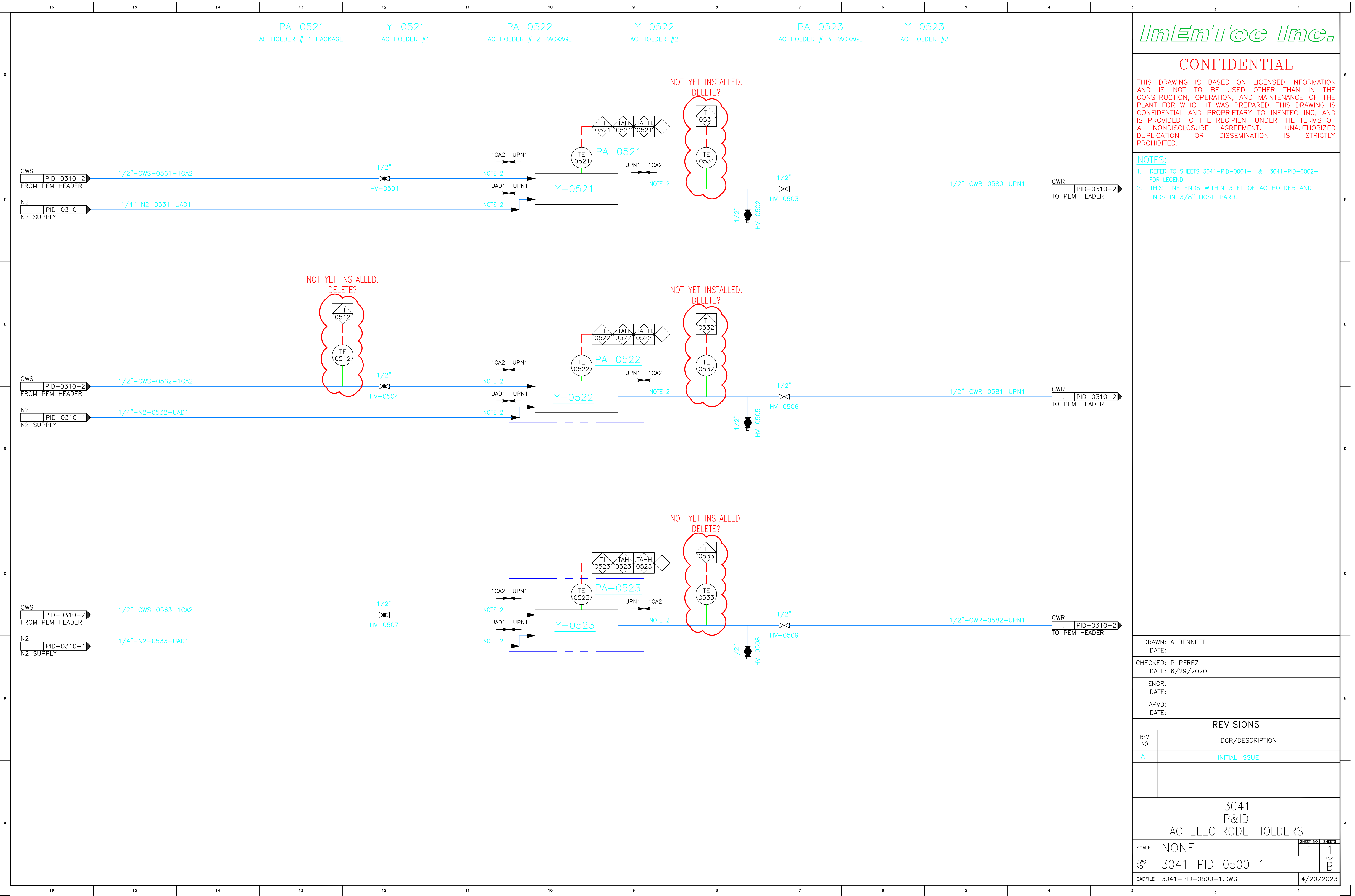
REVISIONS

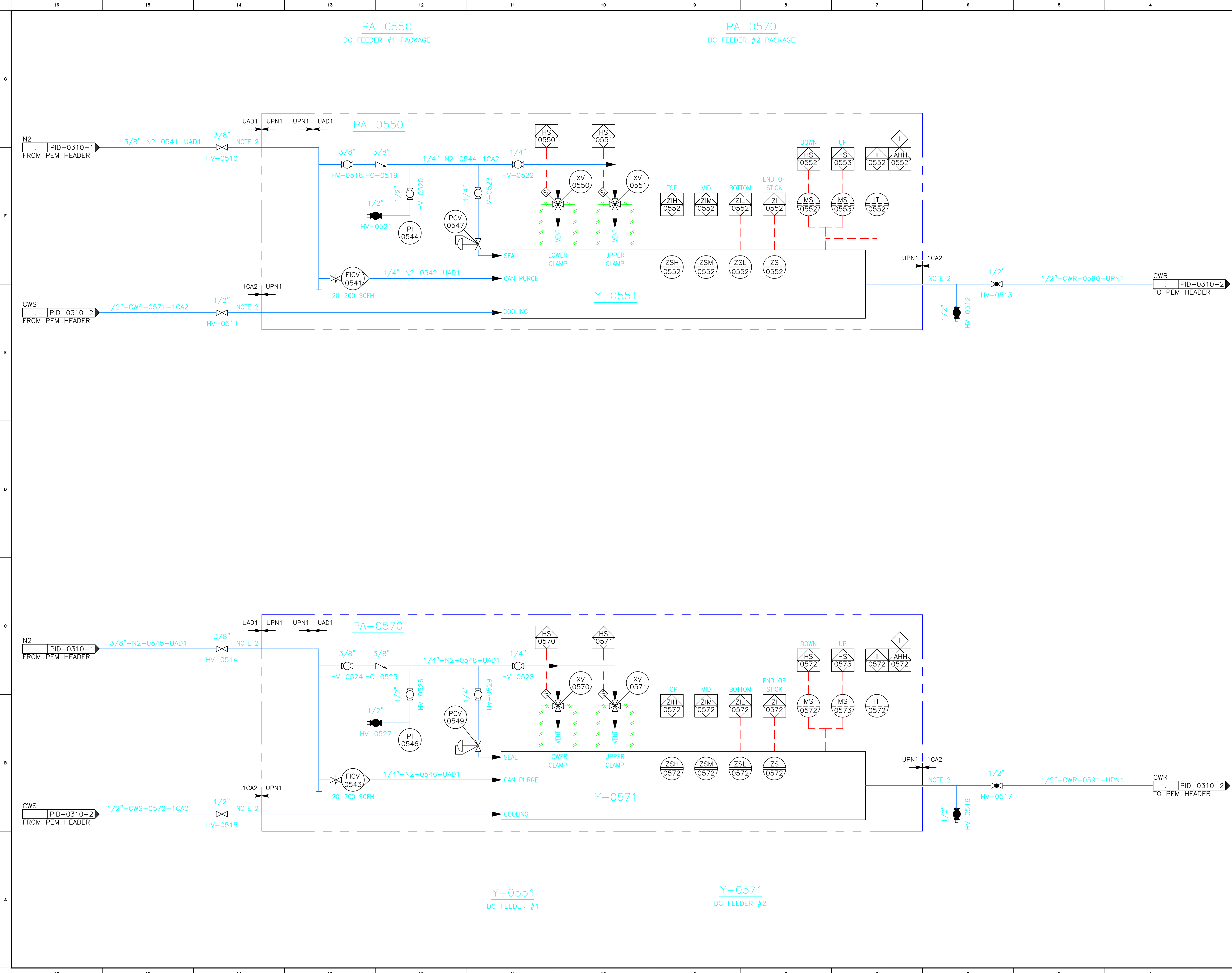
3041
P&ID
DC POWER SUPPLY

SCALE	NONE	SHEET NO	SHEET
		1	1

3041-PID-0400-3

CADFILE 3041-PID-0400-3.DWG 4/20/202





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NOTES:

- 1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
- 2. THIS LINE ENDS WITHIN 4 FT OF THE DC FEEDER AND ENDS IN A FEMALE NPT FITTING MATCHING THE SIZE OF THE LINE.

DRAWN: A BENNETT
DATE:

CHECKED:
DATE: P PEREZ

ENGR: 6/29/2020
DATE:

APVD:
DATE:

REVISIONS

REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

3041
P&ID
DC ELECTRODE FEEDERS

SCALE
NONE

DWG NO
3041-PID-0500-2

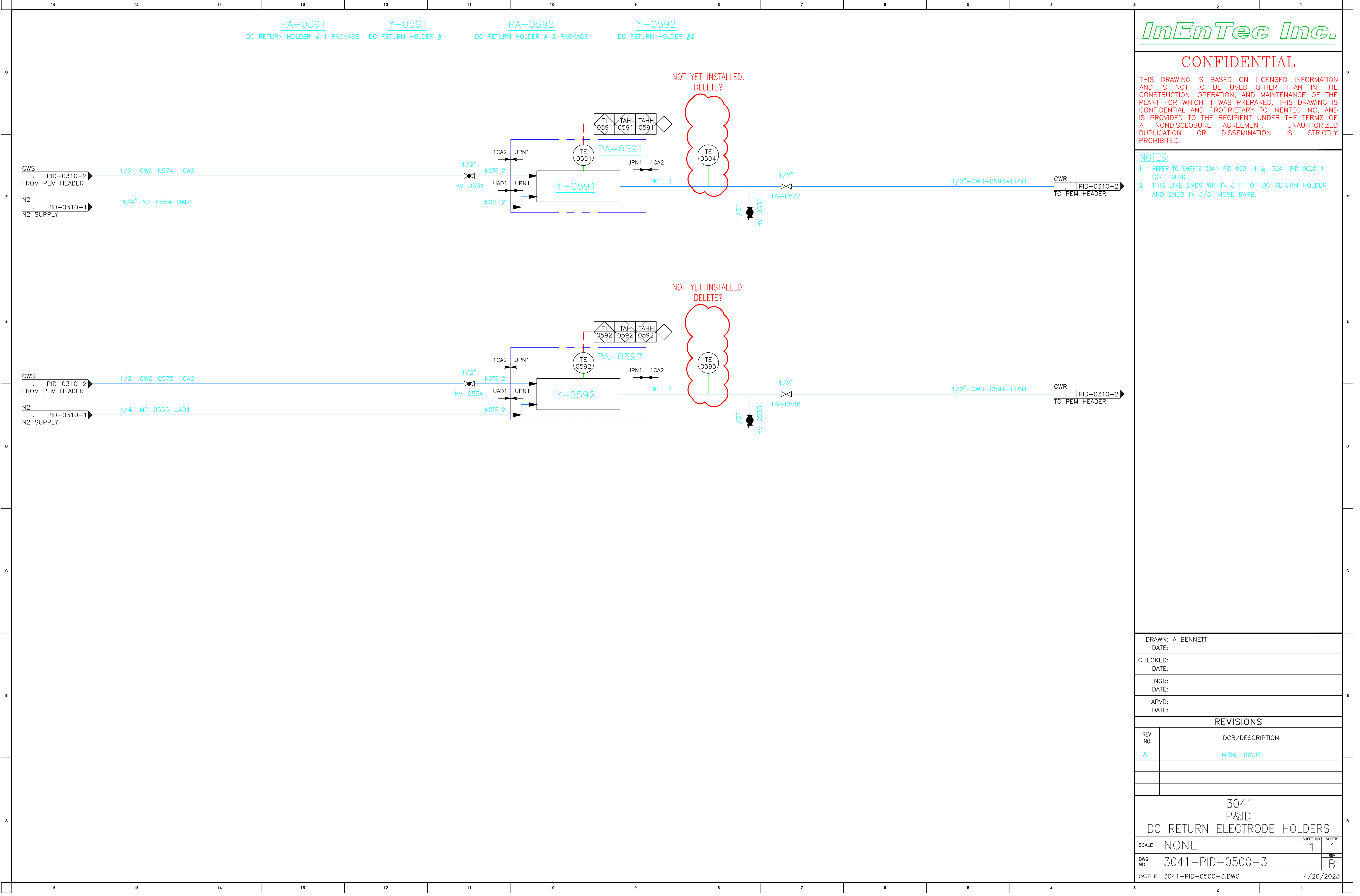
CADFILE
3041-PID-0500-2.DWG

SHEET NO
1

REV
A

SHEETS
1

4/20/2023



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NOTES:

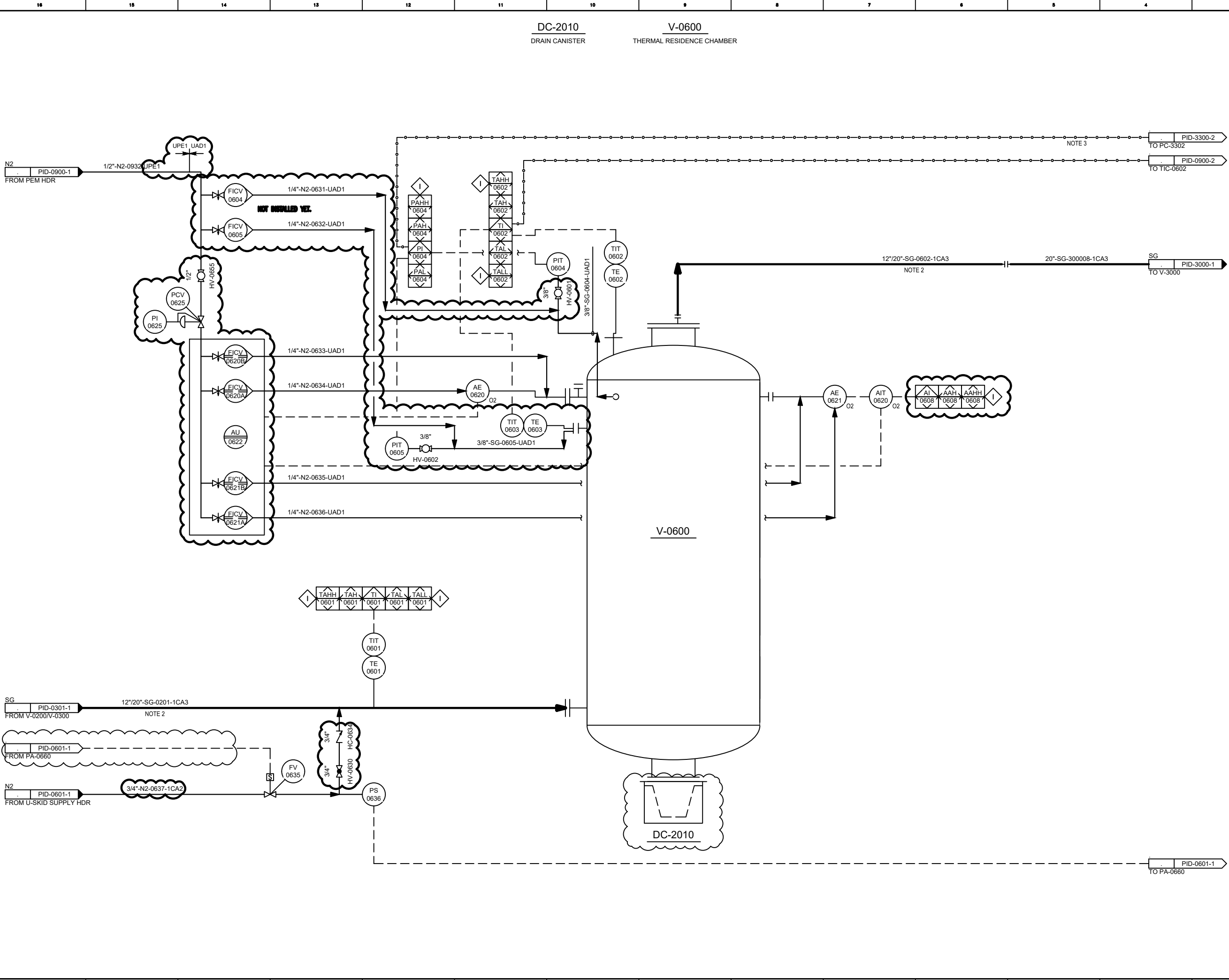
1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
2. THIS LINE ENDS WITHIN 3 FT OF DC RETURN HOLDER AND ENDS IN 3/8" HOSE BARB.

DRAWN: A BENNETT	
DATE:	
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REVISIONS

REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

3041 P&ID DC RETURN ELECTRODE HOLDERS	
SCALE	NONE
DWG NO	3041-PID-0500-3
CADFILE	3041-PID-0500-3.DWG
SHEET NO	1
SHEETS	1
REV	B
DATE	4/20/2023



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- NOTES:
1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
 2. PIPE SIZE SHOWN IS APPROXIMATE INSIDE / OUTSIDE DIAMETER REQUIRED. THE DIFFERENCE INDICATES THE ALLOWANCE FOR REFRACTORY.
 3. PRESSURE CONTROL BY SYNGAS CLEANING TRAIN.

DRAWN: A BENNETT	
DATE:	
CHECKED: P PEREZ	
DATE: 6/29/2020	
ENGR:	
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REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE
B	REVISED AND REDRAWN
C	REVISED AND REDRAWN

3041 P&ID TRC	
SCALE	NONE
DWG NO	3041-PID-0600-1
CADFILE	3041-PID-0600-1.DWG
SHEET NO	1
SHEETS	1
REV	C
3/21/2024	

F-0660A
TRC GAD A

F-0660B
TRC GAD B

F-0660C
TRC GAD C

V-0600
THERMAL RESIDENCE CHAMBER

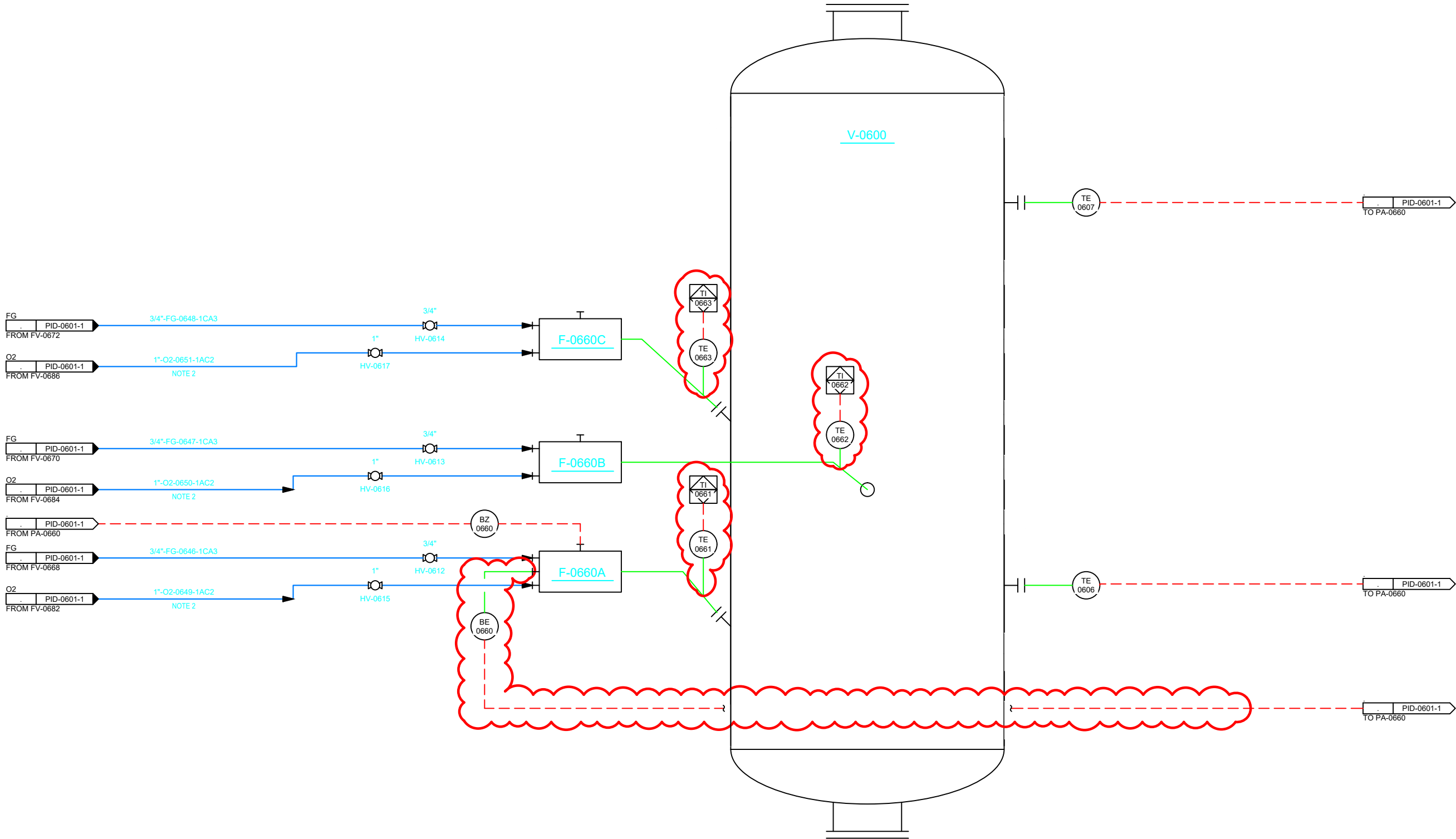
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NOTES:

- REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
- ALL O2 SERVICE VALVES AND INSTRUMENTATION TO BE OXYGEN CLEANED AND COMPATIBLE.



DRAWN: A BENNETT
DATE:
CHECKED: P PEREZ
DATE: 6/29/2020

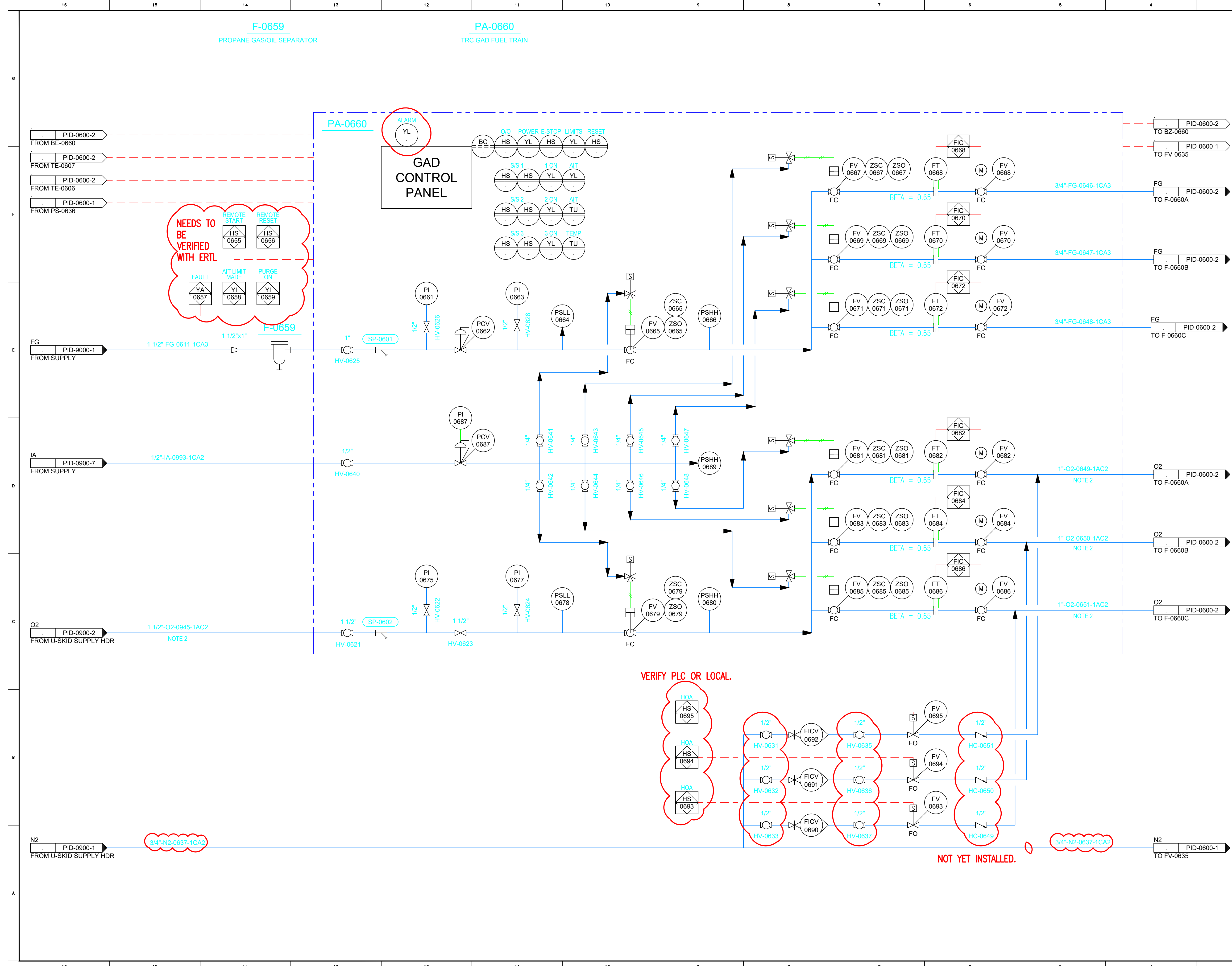
ENGR:
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DATE:

REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE
B	REVISED AND REDRAWN

3041
P&ID
TRC GAD INJECTION

SCALE	NONE	SHEET NO	1	SHEETS	1
DWG NO	3041-PID-0600-2	REV	B		
CADFILE	3041-PID-0600-2.DWG				3/21/2024



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- NOTES:
- REFER TO SHEETS 3041-PID-0001 & 3041-PID-0002 FOR LEGEND.
 - ALL O2 SERVICE VALVES AND INSTRUMENTATION TO BE OXYGEN CLEANED AND COMPATIBLE.

DRAWN: A BENNETT
DATE:
CHECKED: P PEREZ
DATE: 6/29/2020
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REVISIONS

REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

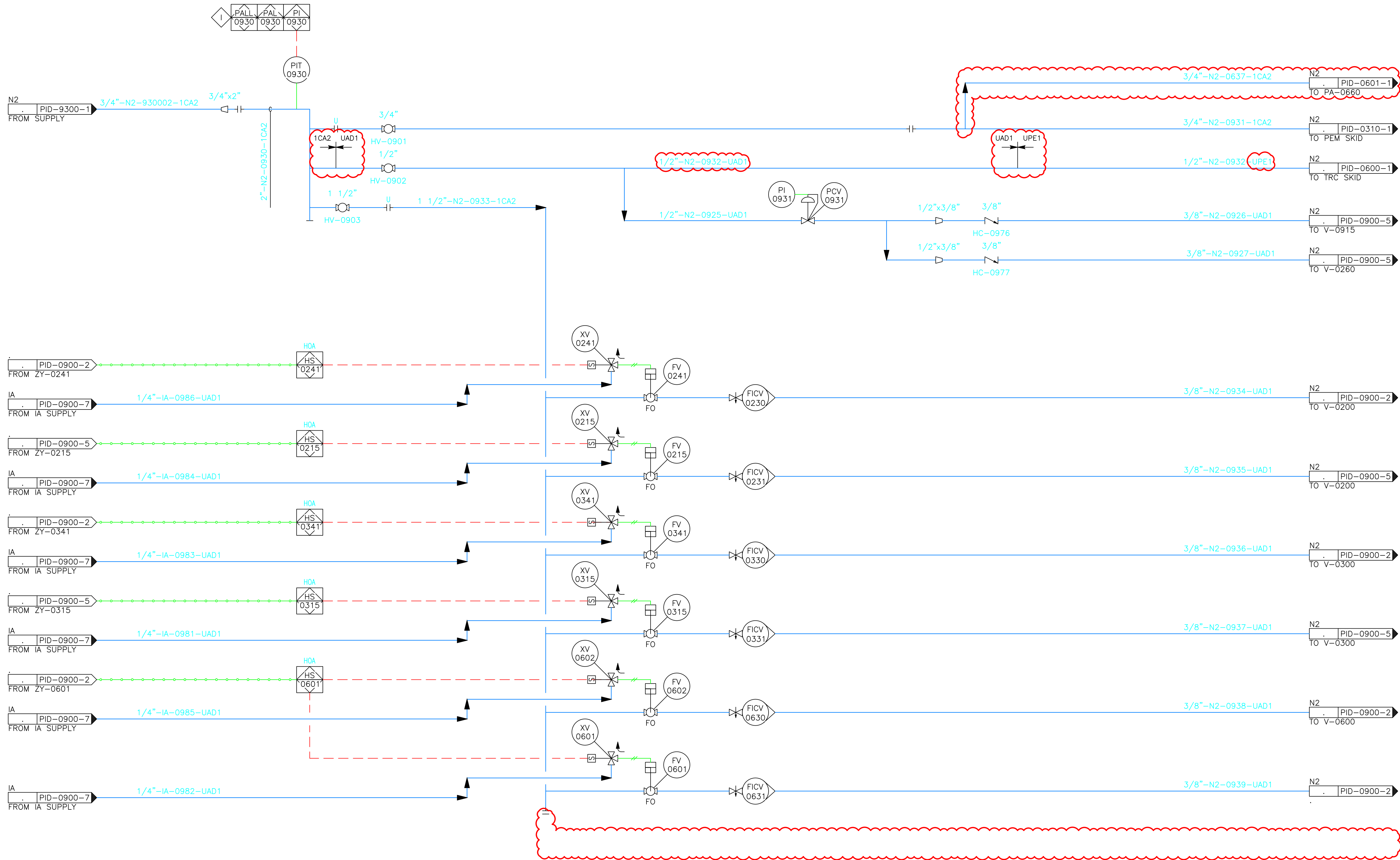
3041 P&ID TRC GAD FUEL TRAIN	
SCALE NONE	SHEET NO 1 SHEETS 1
DWG NO 3041-PID-0601-1	REV A
CADFILE 3041-PID-0601-1.DWG	4/20/2023

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NOTES:

1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.



DRAWN: A BENNETT	
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DATE: 6/29/2020	
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REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

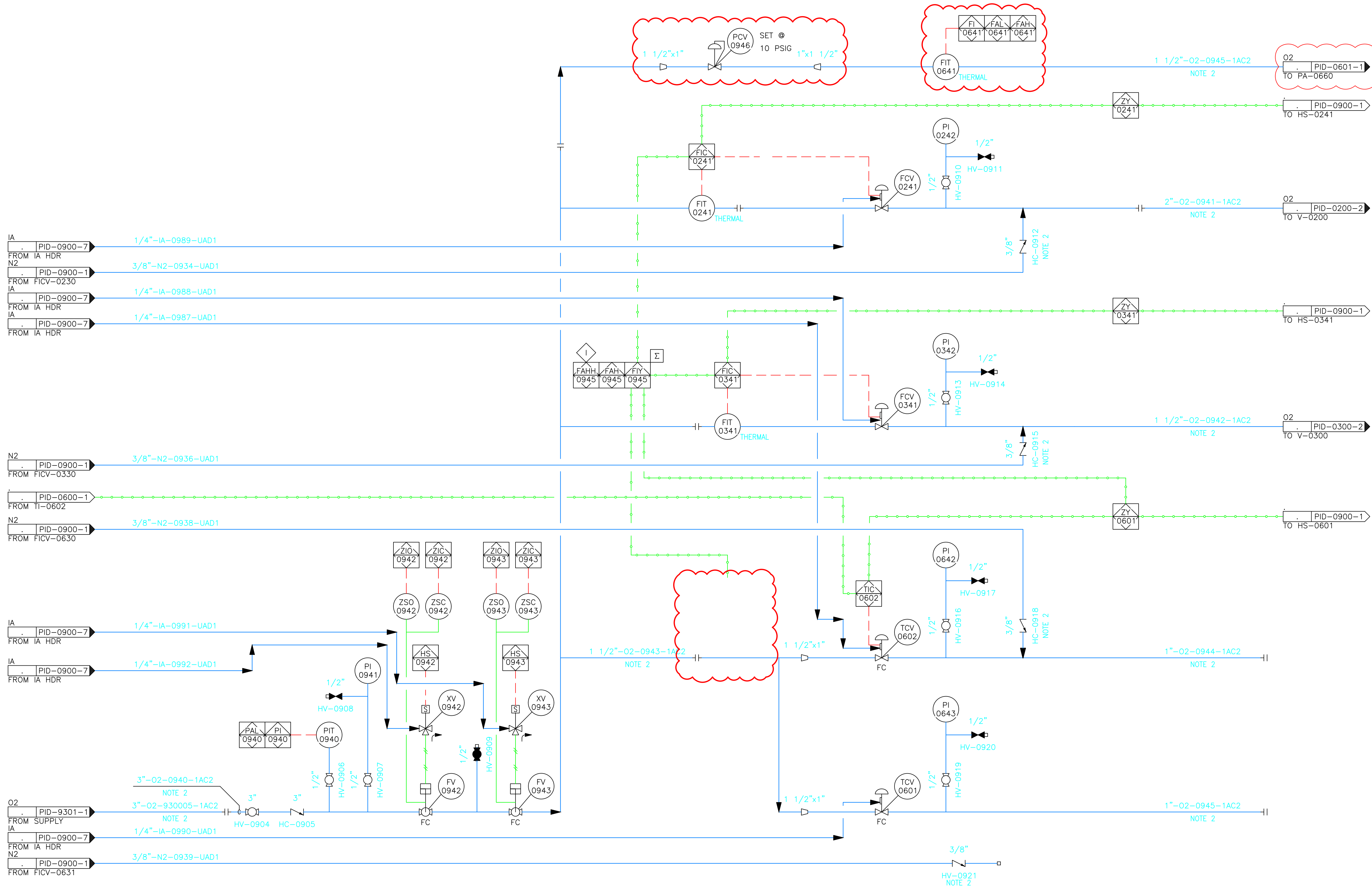
3041 P&ID UTILITY SKID N2 HEADERS	
SCALE	NONE
DWG NO	3041-PID-0900-1
CADFILE	3041-PID-0900-1.DWG
SHEET NO	1
SHEETS	1
REV	A
DATE	4/20/2023

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NOTES:

- REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
- ALL O2 SERVICE VALVES AND INSTRUMENTATION TO BE OXYGEN CLEANED AND COMPATIBLE.



DRAWN: R RAMOS
DATE: 12-4-19

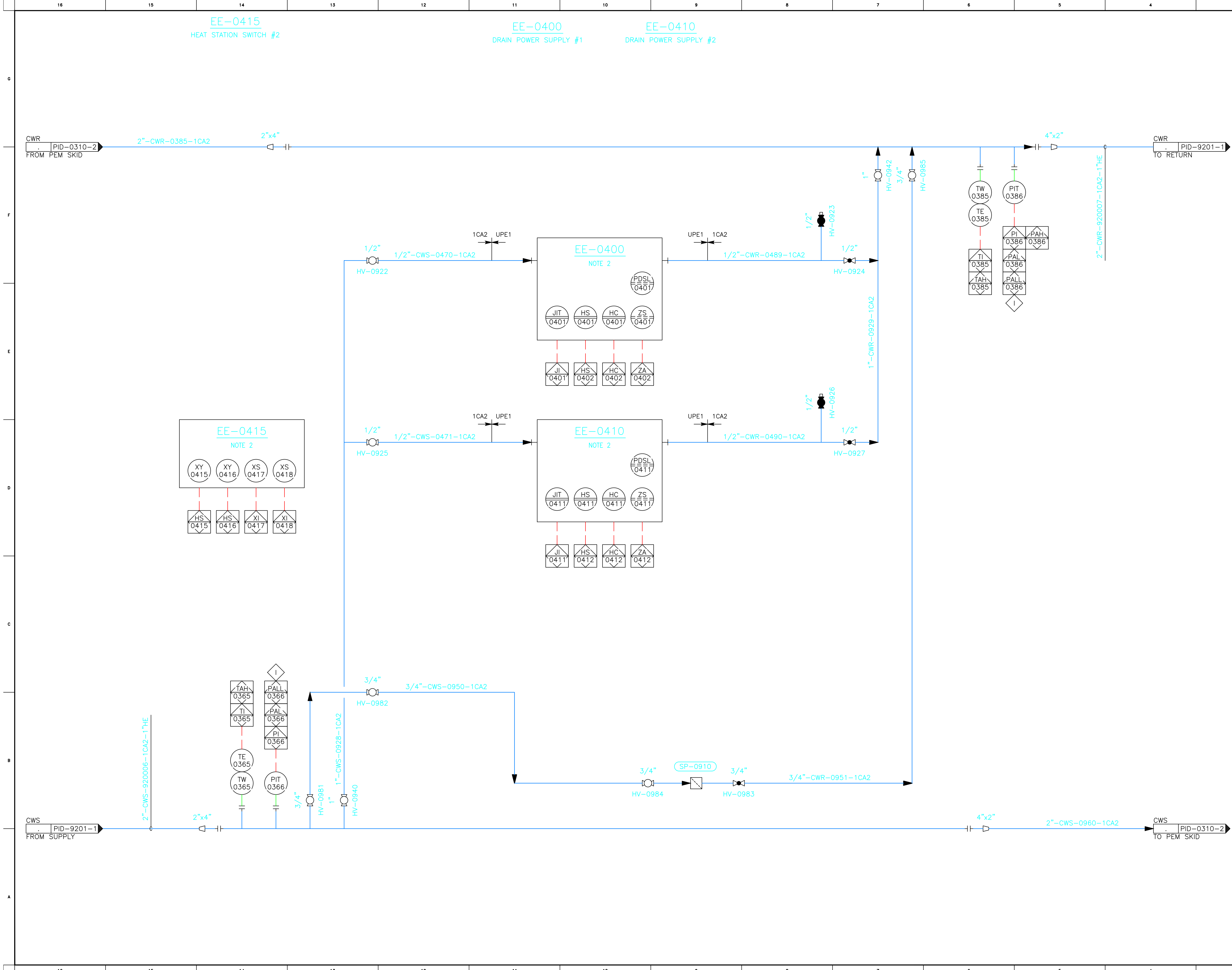
CHECKED: P PEREZ
DATE: 6/29/2020

ENGR: P SALAS
DATE: 12-4-19

APVD:
DATE:

REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

3041 P&ID UTILITY SKID O2 HEADER	
SCALE NONE	SHEET NO 1 SHEETS 1
DWG NO 3041-PID-0900-2	REV B
CADFILE 3041-PID-0900-2.DWG	4/20/2023



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- NOTES:
1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
 2. HEAT STATIONS AND DRAINS LOCATED ON 3041-PID-0400-1.

DRAWN: A BENNETT	
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CHECKED: P PEREZ	
DATE: 6/29/2020	
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REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

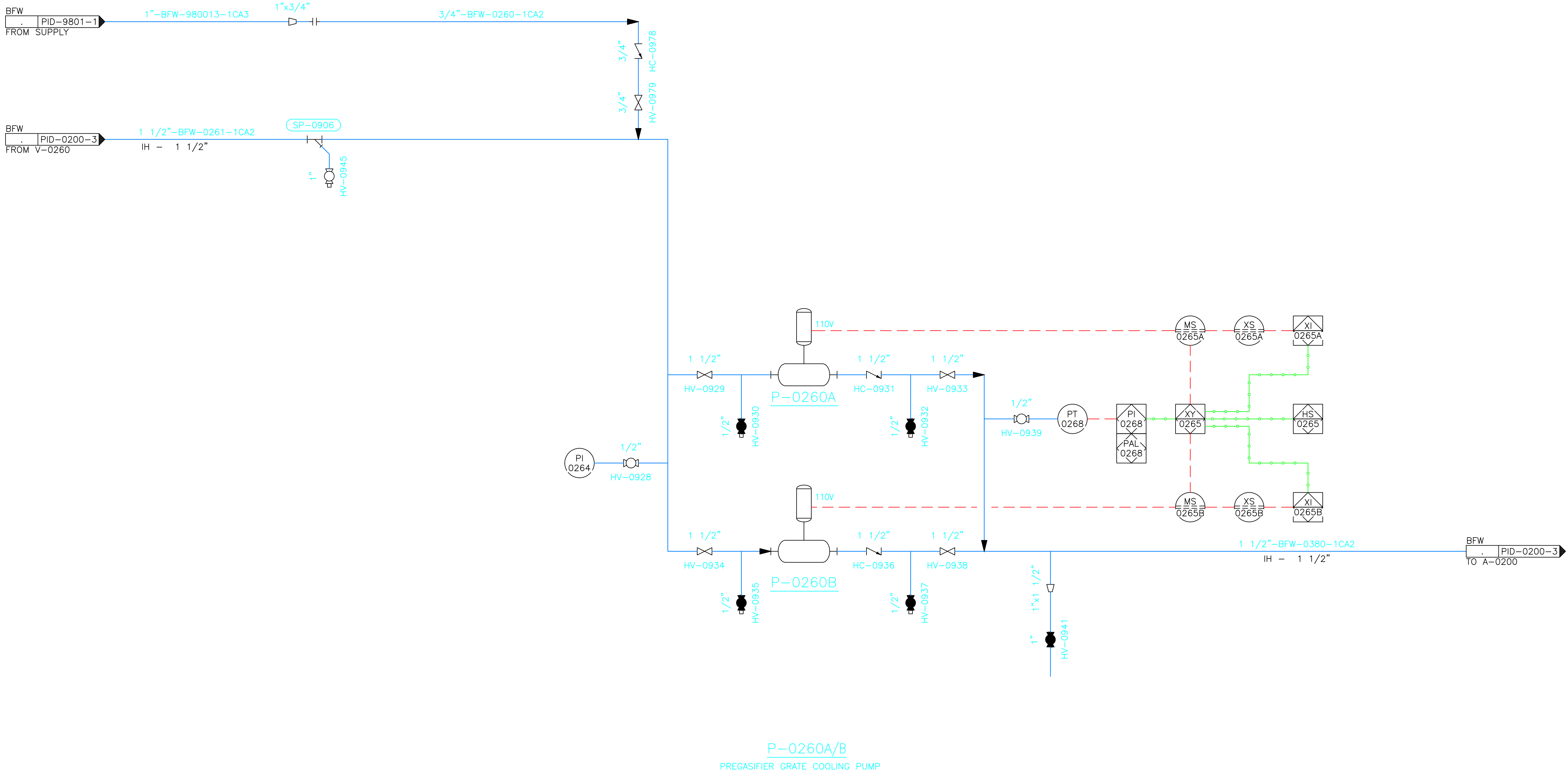
3041	
P&ID	
UTILITY SKID CW HEADERS	
SCALE	NONE
DWG NO	3041-PID-0900-3
CADFILE	3041-PID-0900-3.DWG
SHEET NO	1
SHEETS	1
REV	A
DATE	4/20/2023

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NOTES:

1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.



DRAWN: A BENNETT DATE:	
CHECKED: P PEREZ DATE: 6/29/2020	
ENGR: DATE:	
APVD: DATE:	

REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE
B	REVISED AND REDRAWN

3041 P&ID UTILITY SKID COOLING PUMPS	
SCALE NONE	SHEET NO 1 SHEETS 1
DWG NO 3041-PID-0900-4	REV B
CADFILE 3041-PID-0900-4.DWG	4/20/2023

V-0915
STEAM SUPPLY VESSEL

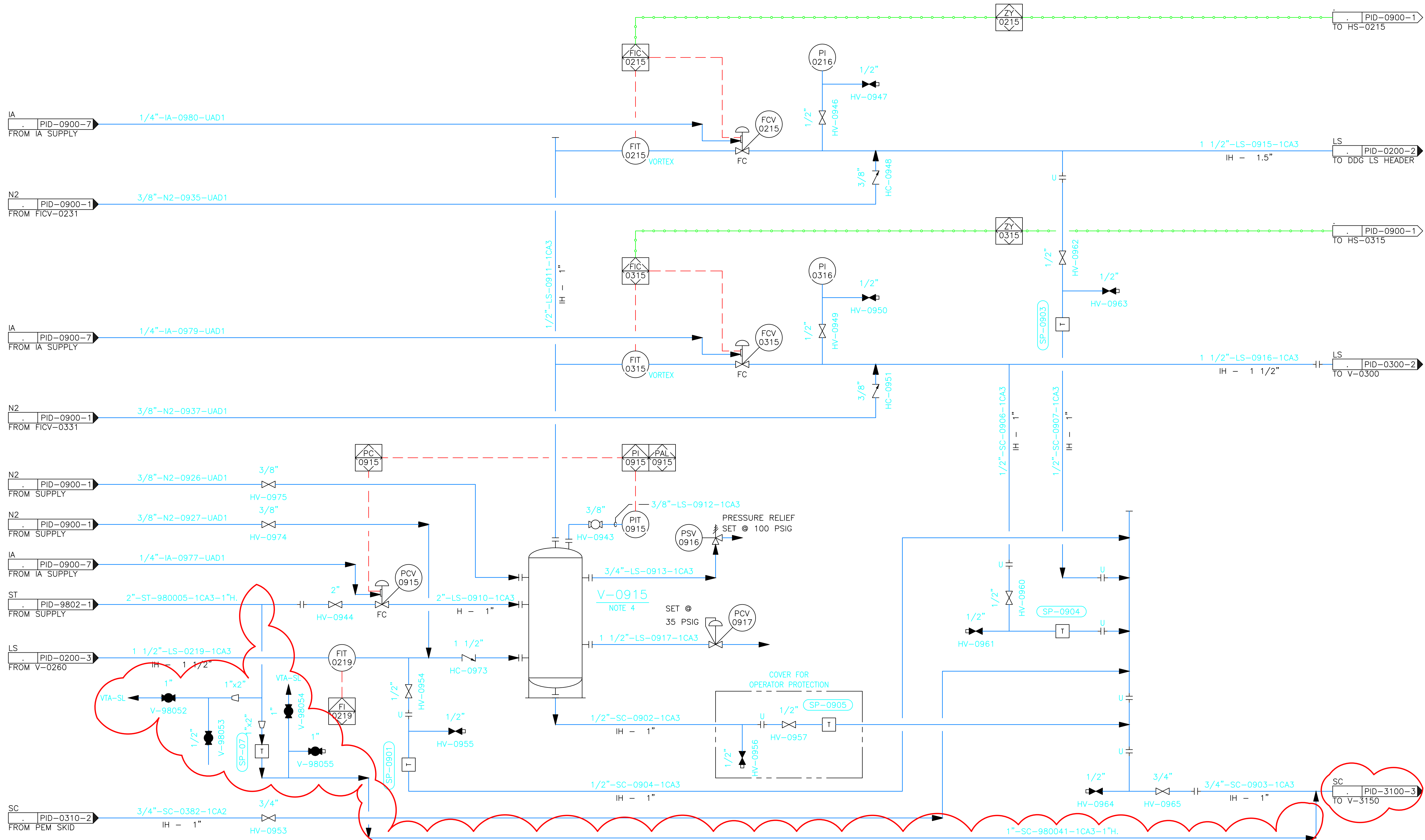
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NOTES:

- REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
- PCV-0915 HAS 3" OF INSULATION.



DRAWN: A BENNETT DATE:	
CHECKED: P PEREZ DATE: 6/29/2020	
ENGR: DATE:	
APVD: DATE:	

REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

3041 P&ID UTILITY SKID LS HEADER	
SCALE NONE	SHEET NO 1 SHEETS 1
DWG NO 3041-PID-0900-5	REV A
CADFILE 3041-PID-0900-5.DWG	4/20/2023

PA-0610
TRC PREHEATER PACKAGE

PA-0210
PREGASIFIER PREHEATER PACKAGE

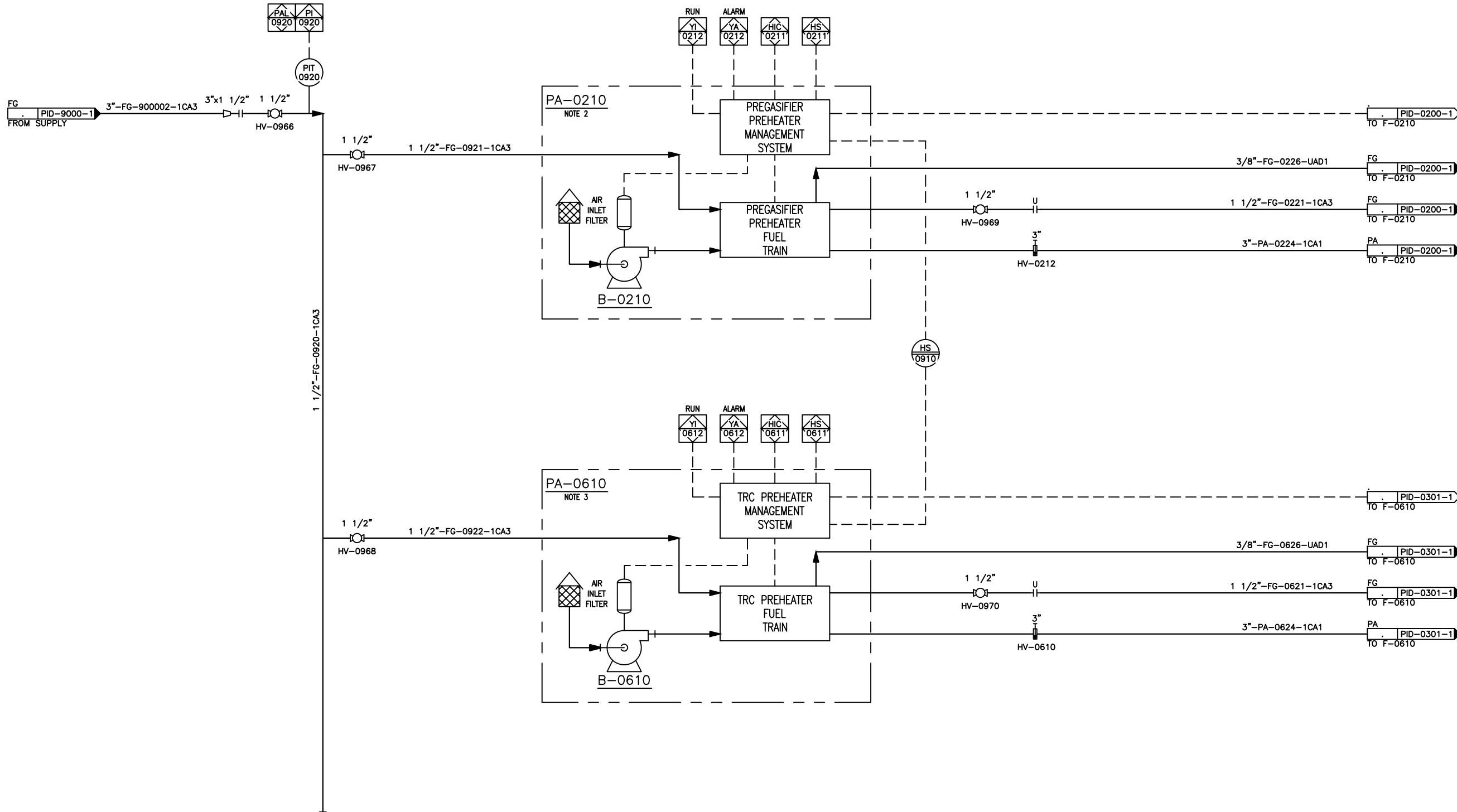
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NOTES:

1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
2. THE PREGASIFIER PREHEATER HEAD IS LOCATED ON 3041-PID-0200-1.
3. THE TRC PREHEATER HEAD IS LOCATED ON 3041-PID-0301-1.



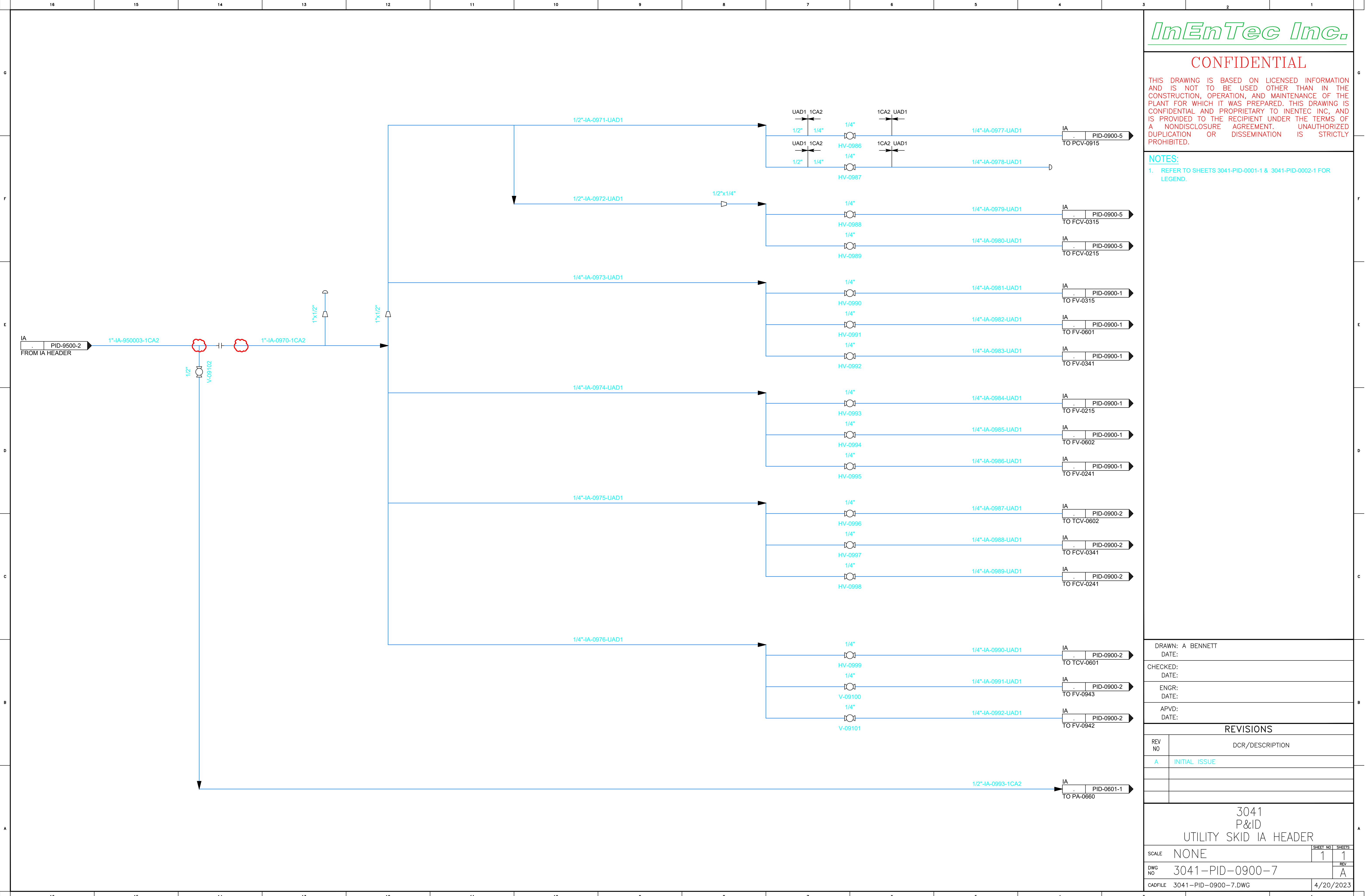
B-0210
PREGASIFIER PREHEATER BLOWER

B-0610
TRC PREHEATER BLOWER

DRAWN: A BENNETT	
DATE:	
CHECKED: P PEREZ	
DATE: 6/29/2020	
ENGR:	
DATE:	
APVD:	
DATE:	

REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

3041 P&ID UTILITY SKID PREHEATERS	
SCALE	NONE
DWG NO	3041-PID-0900-6
CADFILE	3041-PID-0900-6.DWG
SHEET NO	1
SHEETS	1
REV	A



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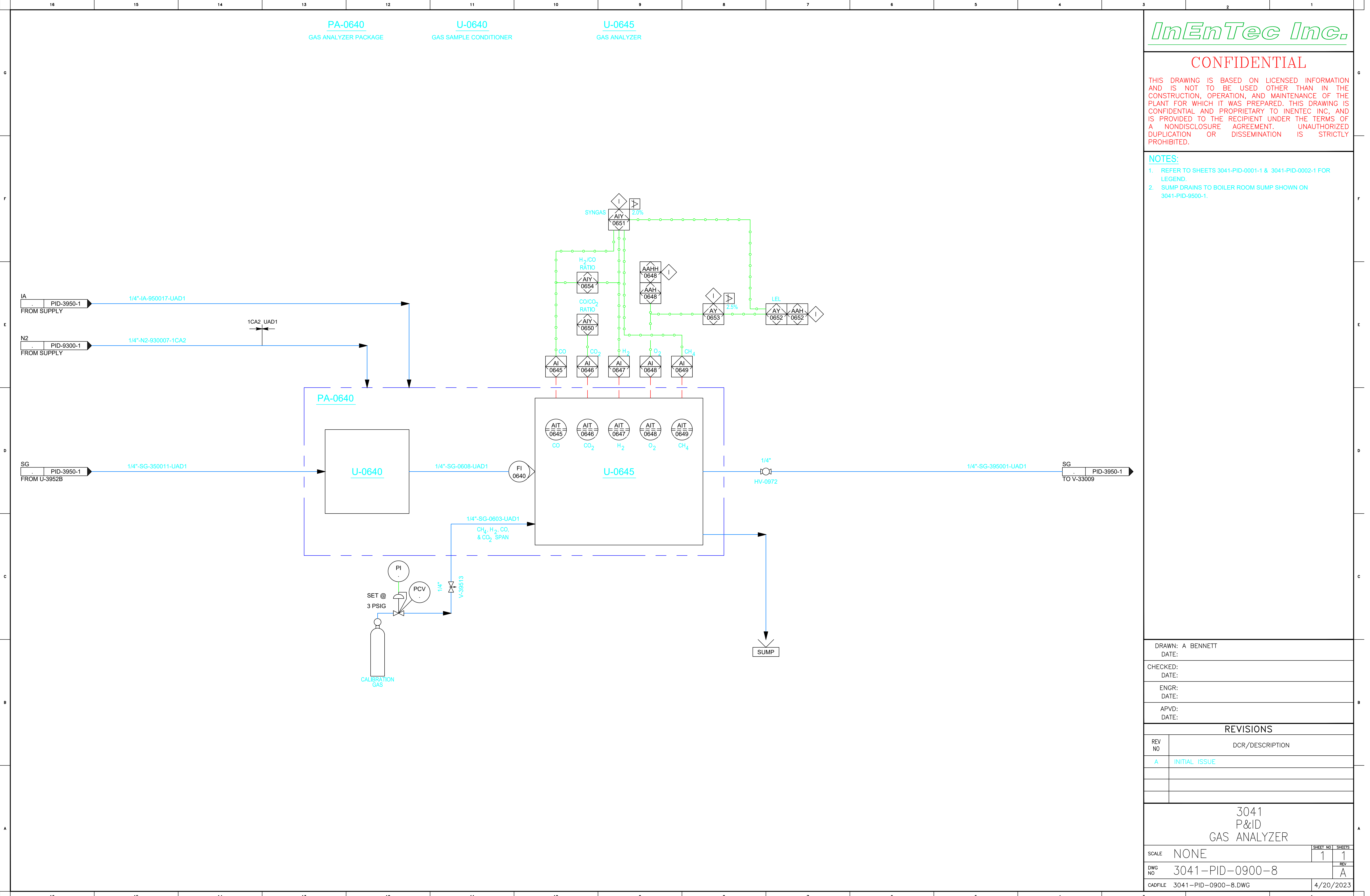
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- NOTES:
- 1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.

DRAWN: A BENNETT	
DATE:	
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ENGR:	
DATE:	
APVD:	
DATE:	
REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

3041 P&ID UTILITY SKID IA HEADER			
SCALE	NONE	SHEET NO	1
DWG NO	3041-PID-0900-7	REV	A
CADFILE	3041-PID-0900-7.DWG	4/20/2023	



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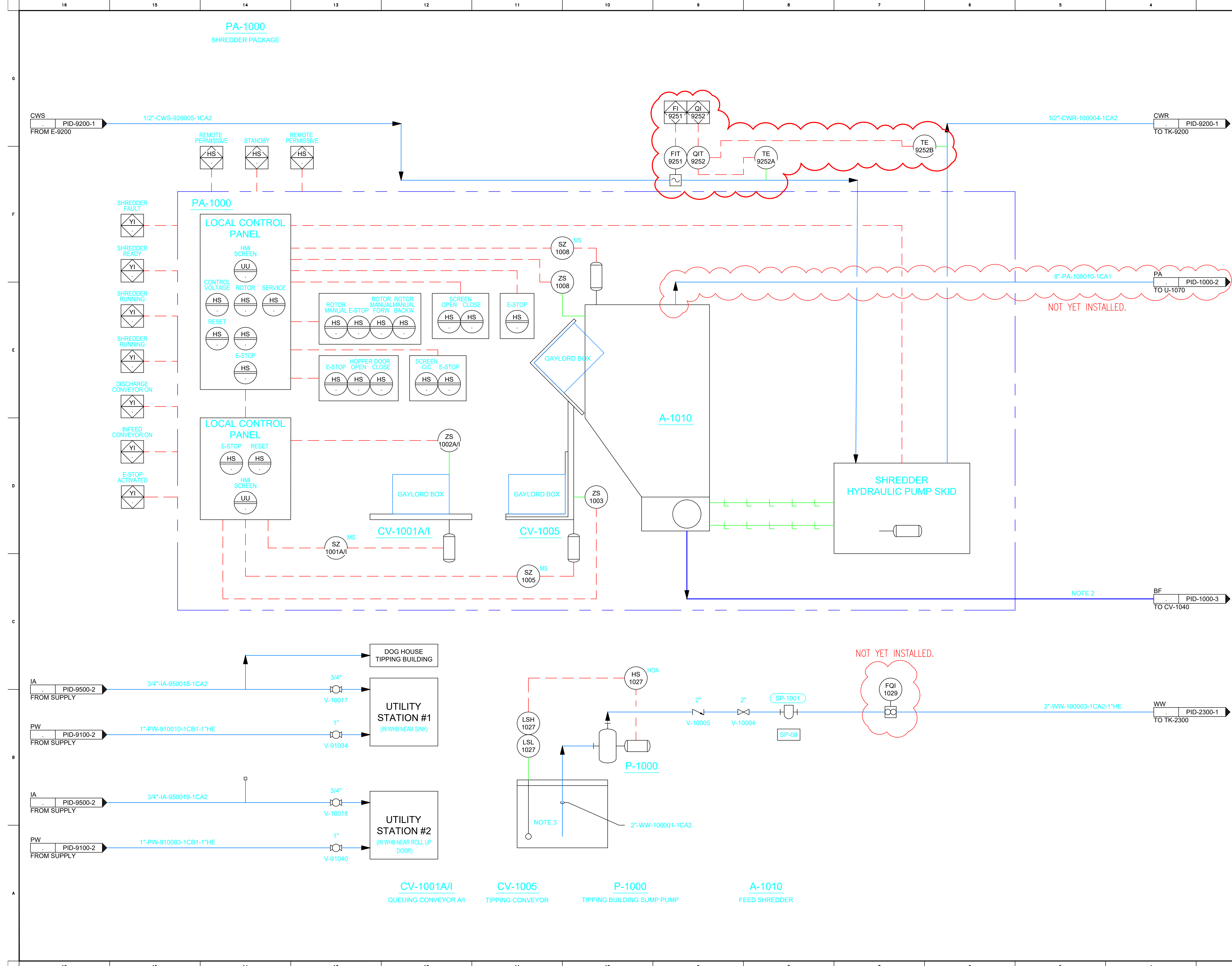
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- NOTES:
- 1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
 - 2. SUMP DRAINS TO BOILER ROOM SUMP SHOWN ON 3041-PID-9500-1.

DRAWN: A BENNETT	
DATE:	
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ENGR:	
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APVD:	
DATE:	

REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

3041 P&ID GAS ANALYZER	
SCALE	NONE
DWG NO	3041-PID-0900-8
CADFILE	3041-PID-0900-8.DWG
SHEET NO	1
SHEETS	1
REV	A
DATE	4/20/2023



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NOTES:

- REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
- THIS LINE IS NOT A PIPELINE. THE SHREDDER A-1010 IS CONNECT TO CV-1040 WITH A FABRICATED SHEET METAL PIECE.
- FEED HANDLING BUILDING SUMP.

DRAWN: P SALAS
DATE:

CHECKED:
DATE:

ENGR:
DATE:

APVD:
DATE:

REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE
B	REVISED AND REDRAWN

3041
P&ID
FEED HANDLING 1 & SHREDDING

SCALE
NONE

DWG NO
3041-PID-1000-1

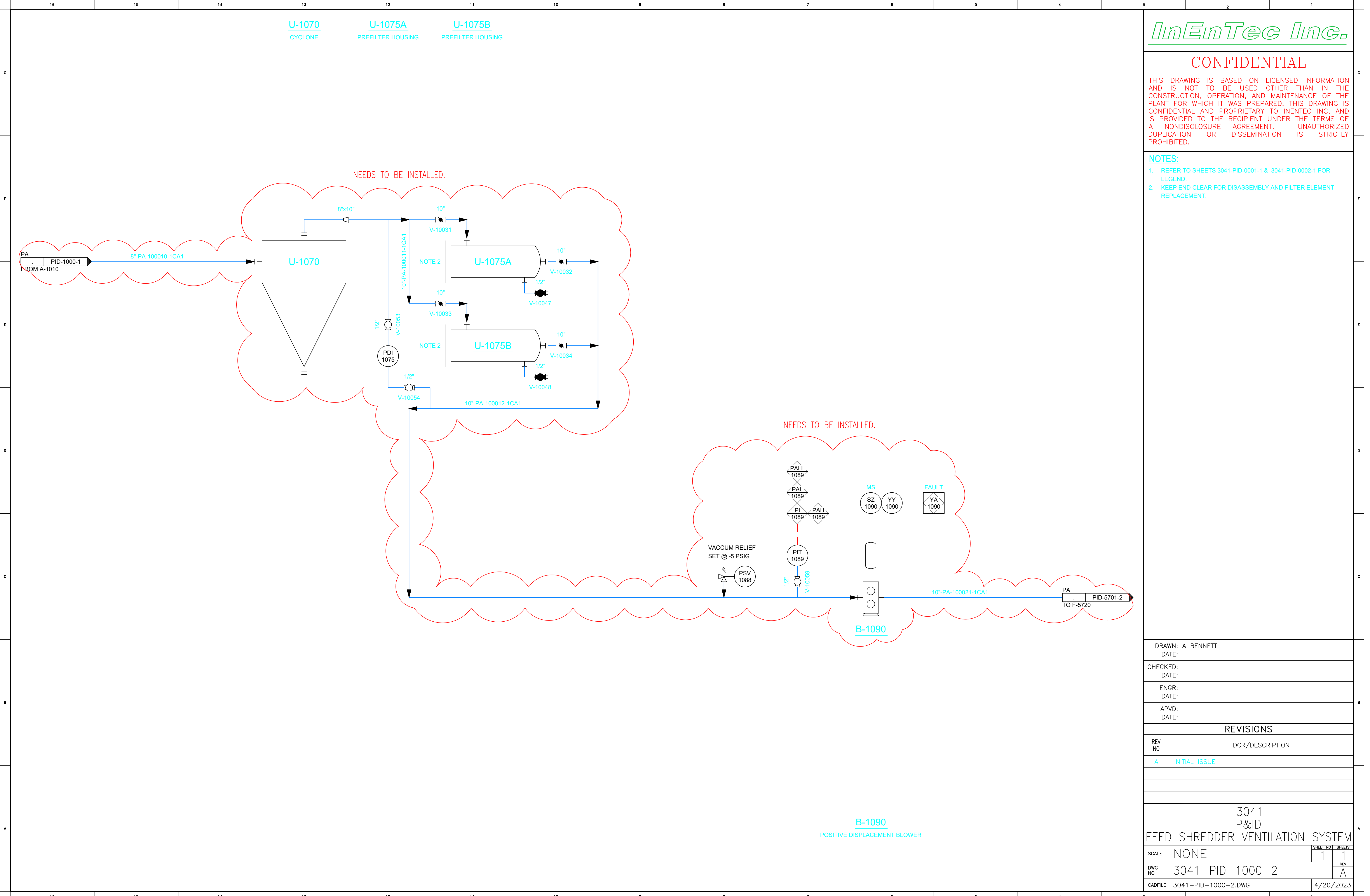
CADFILE
3041-PID-1000-1.DWG

SHEET NO
1

REV
B

SHEETS
1

3/21/2024



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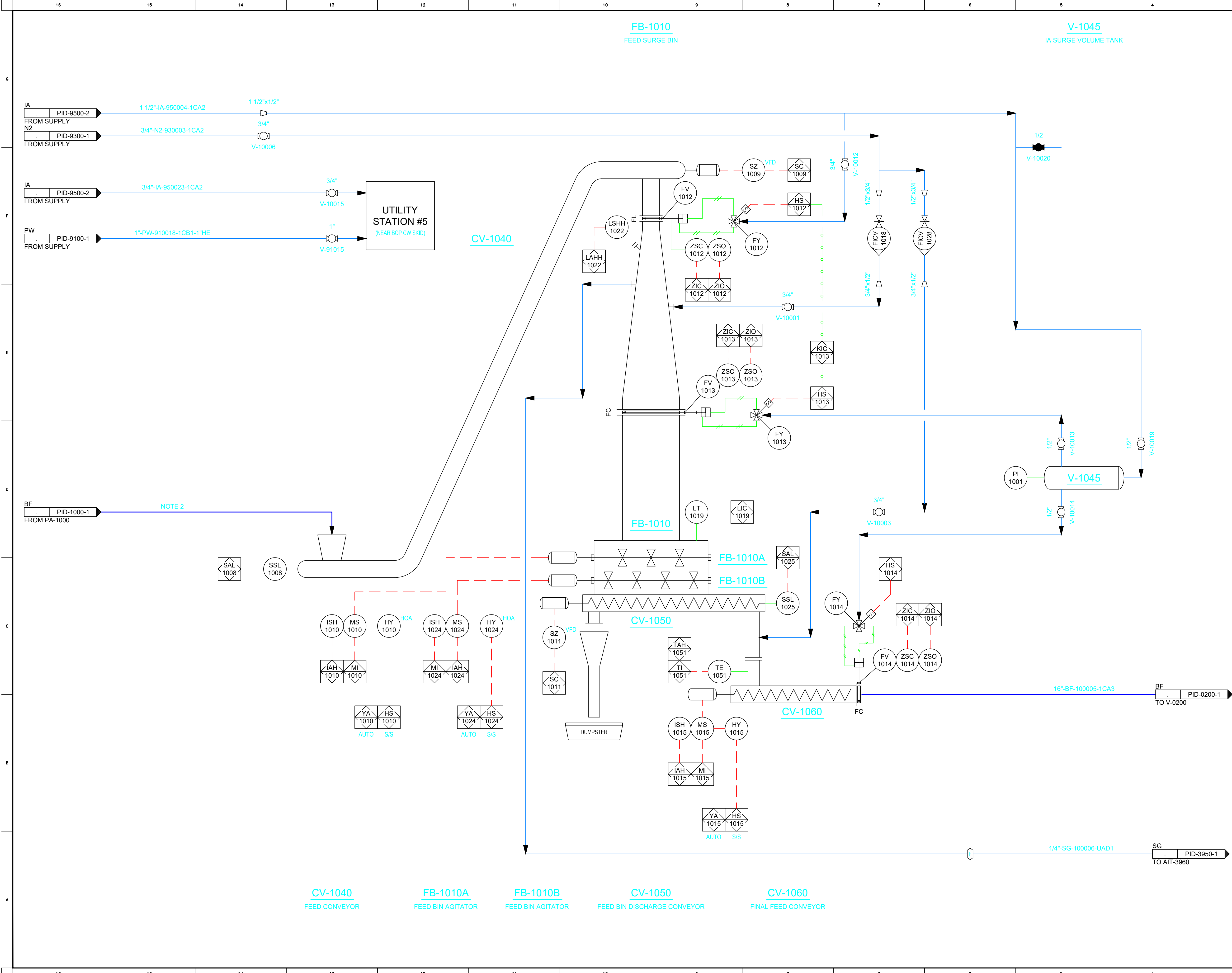
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- NOTES:
- 1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
 - 2. KEEP END CLEAR FOR DISASSEMBLY AND FILTER ELEMENT REPLACEMENT.

DRAWN: A BENNETT	
DATE:	
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DATE:	
ENGR:	
DATE:	
APVD:	
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REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

3041 P&ID FEED SHREDDER VENTILATION SYSTEM	
SCALE	NONE
DWG NO	3041-PID-1000-2
CADFILE	3041-PID-1000-2.DWG
SHEET NO	1
SHEETS	1
REV	A
4/20/2023	



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NOTES:

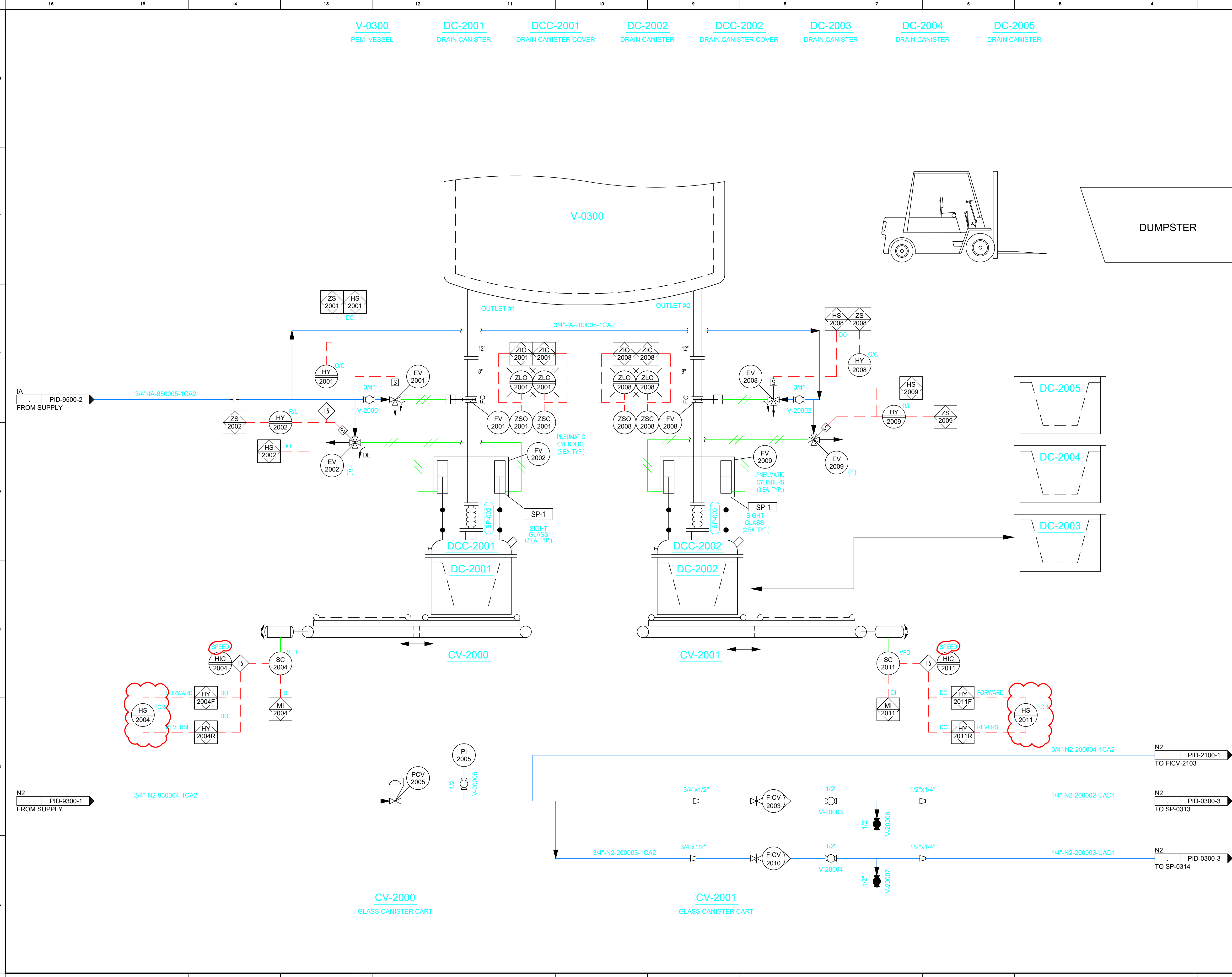
1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
2. THIS LINE IS NOT A PIPELINE. THE SHREDDER A-1010 CONNECTS TO CV-1040 WITH A FABRICATED SHEET METAL TRANSITION PIECE.

DRAWN: A BENNETT	
DATE:	
CHECKED: P PEREZ	
DATE: 6/30/2020	
ENGR:	
DATE:	
APVD:	
DATE:	

REVISIONS

REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

3041 P&ID FEED HANDLING 2	
SCALE NONE	SHEET NO 1 SHEETS 1
DWG NO 3041-PID-1000-3	REV A
CADFILE 3041-PID-1000-3.DWG	3/21/2024



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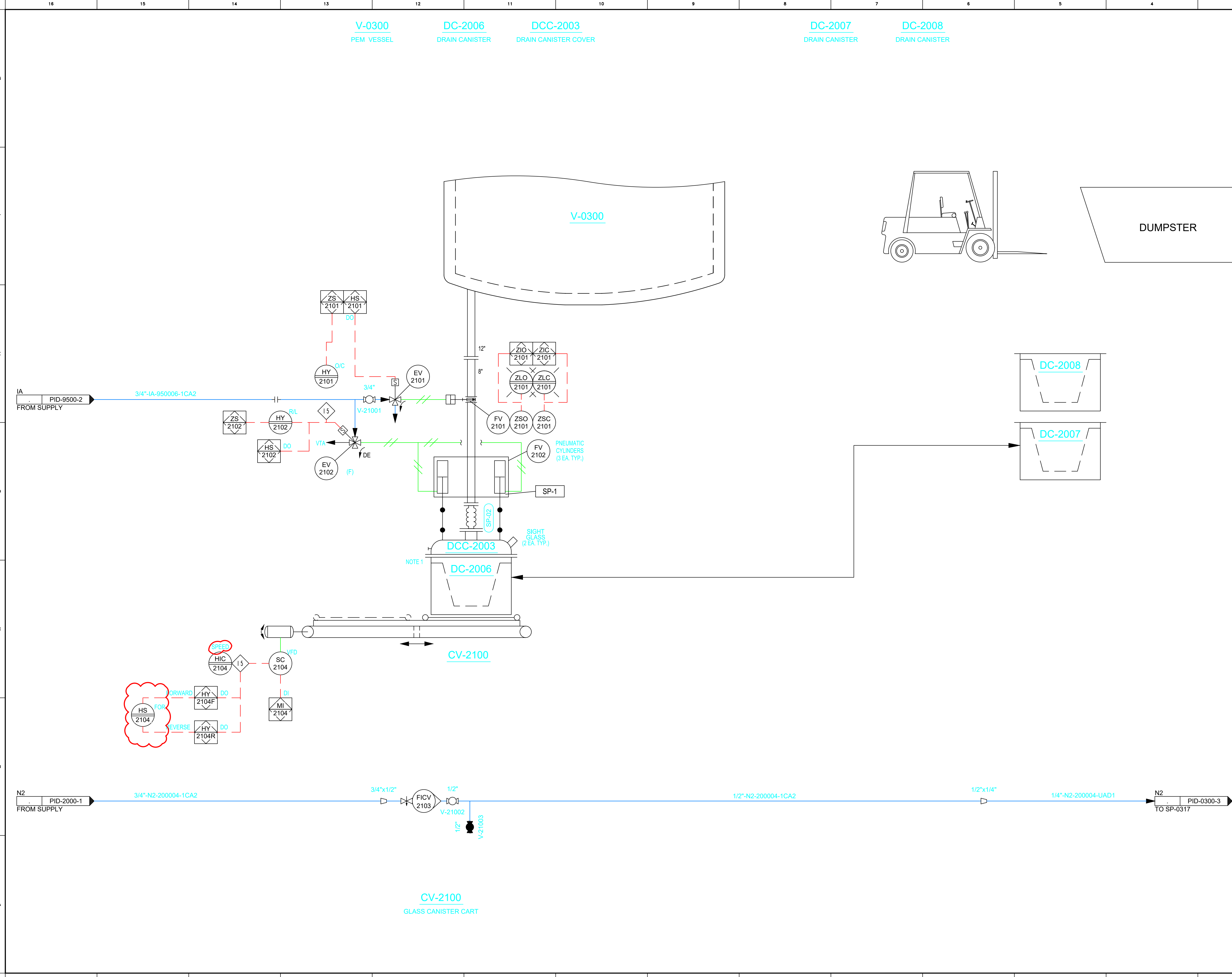
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- NOTES:
- REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.

DRAWN: A BENNETT
DATE:
CHECKED: P PEREZ
DATE: 7/6/2020
ENGR:
DATE:
APVD:
DATE:

REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE
B	REVISED AND REDRAWN

3041 P&ID GLASS HANDLING SYSTEM	
SCALE NONE	SHEET NO 1 SHEETS 1
DWG NO 3041-PID-2000-1	REV B
CADFILE 3041-PID-2000-1.DWG	3/21/2024



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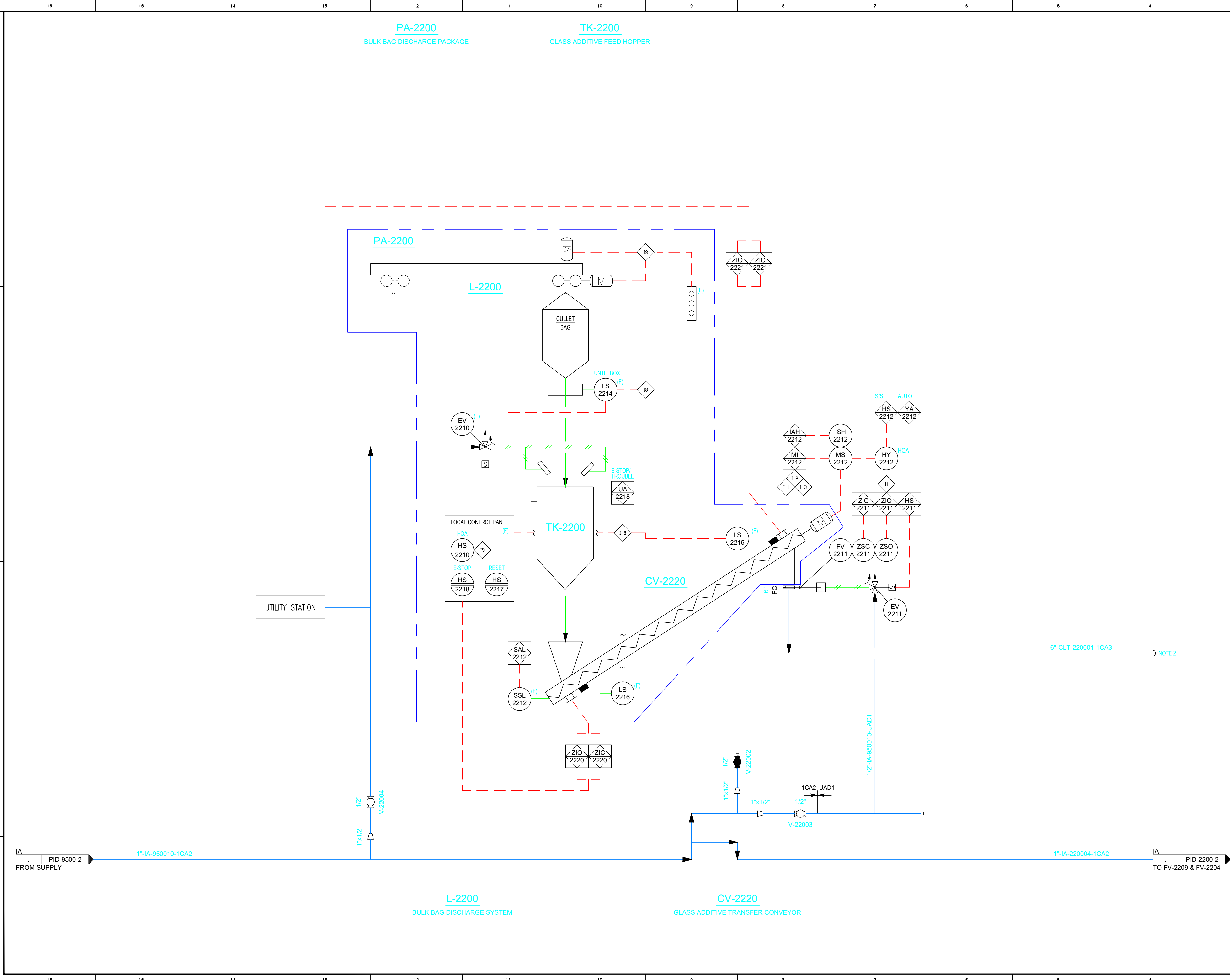
NOTES:

- 1. REFER TO SHEETS 3041-PID-0001 & 3041-PID-0002 FOR LEGEND.

DRAWN: A BENNETT	
DATE:	
CHECKED: P PEREZ	
DATE: 7/6/2020	
ENGR:	
DATE:	
APVD:	
DATE:	

REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE
B	REVISED AND REDRAWN

3041 P&ID METAL HANDLING SYSTEM			
SCALE	NONE	SHEET NO	1
DWG NO	3041-PID-2100-1	REV	B
CADFILE	3041-PID-2100-1.DWG	DATE	8/21/2024



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NOTES:

1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
2. THIS LINE IS CURRENTLY DISCONNECTED DUE TO PLANT MODIFICATIONS.

DRAWN: A BENNETT
DATE:

CHECKED:
DATE:

ENGR:
DATE:

APVD:
DATE:

REVISIONS

REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

3041
P&ID
GLASS ADDITIVES CONVEYOR

SCALE	NONE	SHEET NO	1	SHEETS	2
DWG NO	3041-PID-2200-1	REV	A		
CADFILE	3041-PID-2200-1.DWG	4/20/2023			

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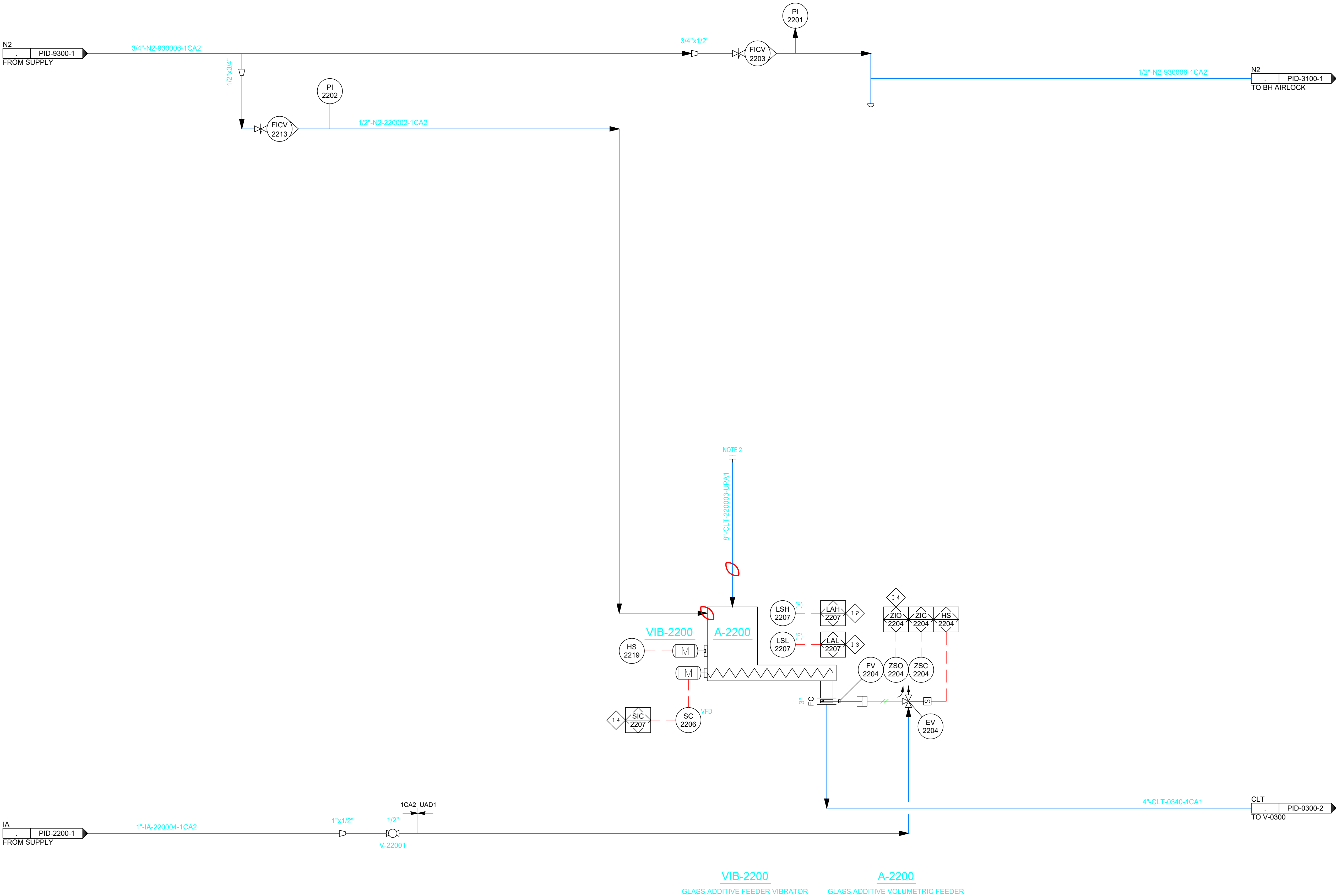
- NOTES:
- REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
 - ADDITIVE FEEDER IS CURRENTLY FILLED MANUALLY THRU FLANGED CONNECTION DUE TO PLANT MODIFICATIONS.

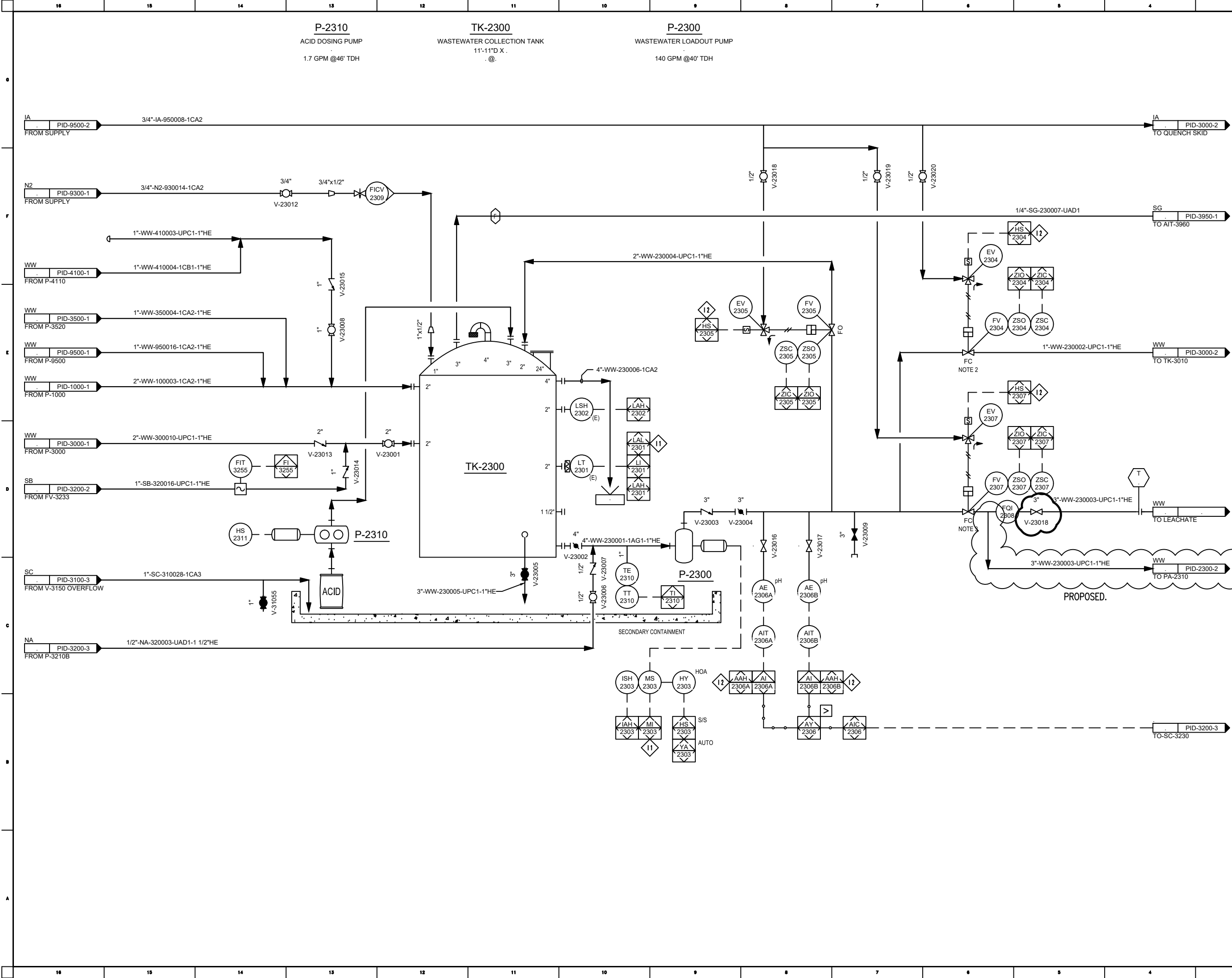
DRAWN: A BENNETT
DATE:
CHECKED: P PEREZ
DATE: 7/6/2020
ENGR:
DATE:
APVD:
DATE:

REVISIONS

REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

3041 P&ID GLASS ADDITIVES FEEDER	
SCALE NONE	SHEET NO 1 SHEETS 1
DWG NO 3041-PID-2200-2	REV A
CADFILE 3041-PID-2200-2.DWG	4/20/2023





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- NOTES:
1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
 2. VALVE FV-2304 IS LOCATED ON THE EDGE OF THE QUENCH SKID.
 3. VALVE FV-2307 IS LOCATED IN THE PIPE RACK BETWEEN THE QUENCH SKID AND SCRUBBER SKID.

DRAWN: A BENNETT	
DATE:	
CHECKED: P PEREZ	
DATE: 7/6/2020	
ENGR:	
DATE:	
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DATE:	
REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE
B	REVISED AND REDRAWN

3041 P&ID WASTEWATER COLLECTION	
SCALE	NONE
DWG NO	3041-PID-2300-1
CADFILE	3041-PID-2300-1.DWG
SHEET NO	1
SHEETS	1
REV	B
DATE	3/21/2024

PA-2310
MVR SYSTEM

TK-2340
MVR CLEANING SOLUTION TANK

TK-2320
MVR DESTILATE TANK
10000 Gallon
0 psig @130 F

TK-2330
MVR CONCENTRATE TANK
2500 Gallon
. @.

P-2340
MVR DESTILATE FEED PUMP
.

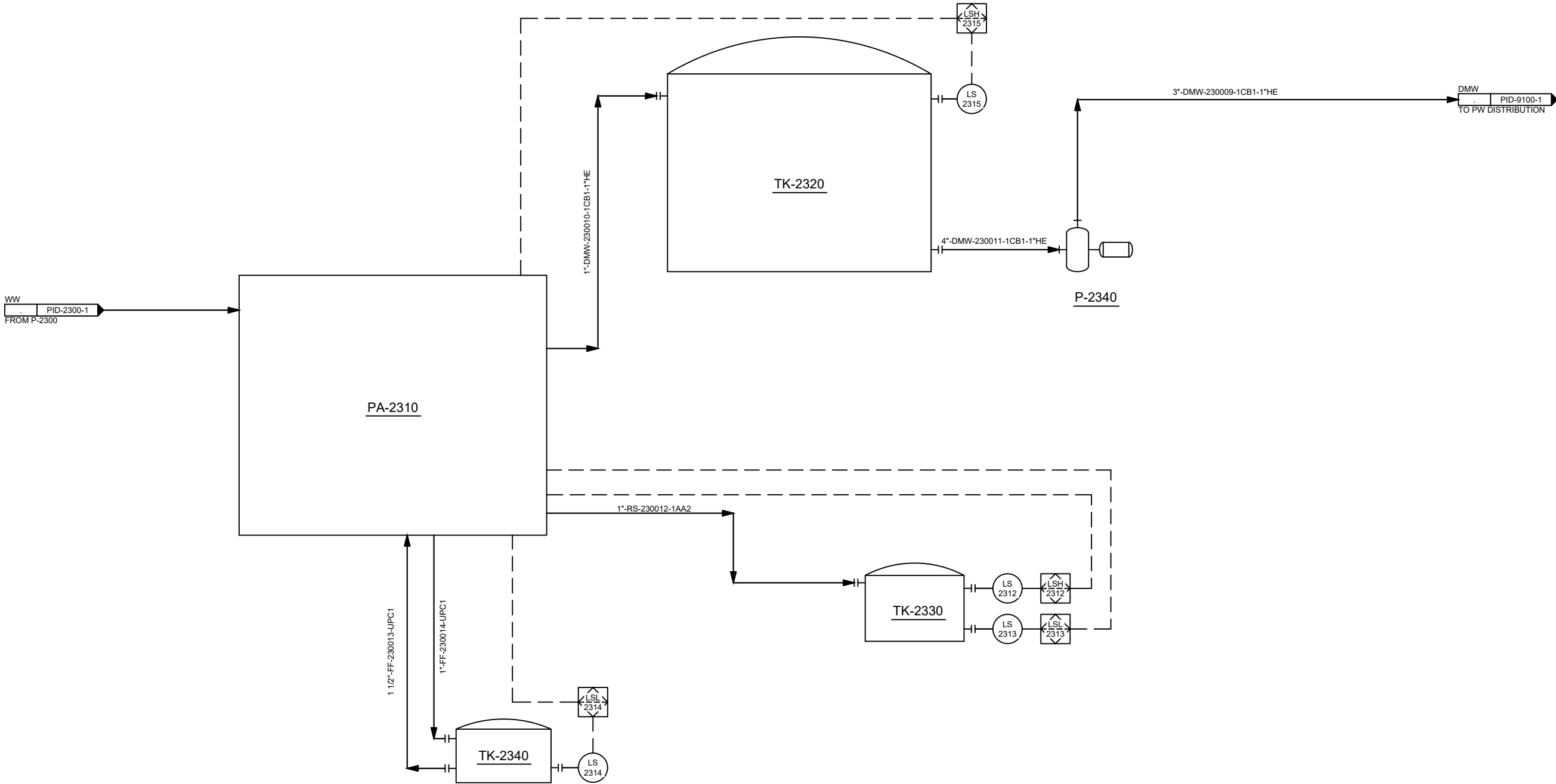
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NOTES:

1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.



DRAWN: P SALAS
DATE: 3/18/2024

CHECKED:
DATE:

ENGR:
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DATE:

REVISIONS

REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

3041
P&ID
WASTEWATER TREATMENT

SCALE NONE

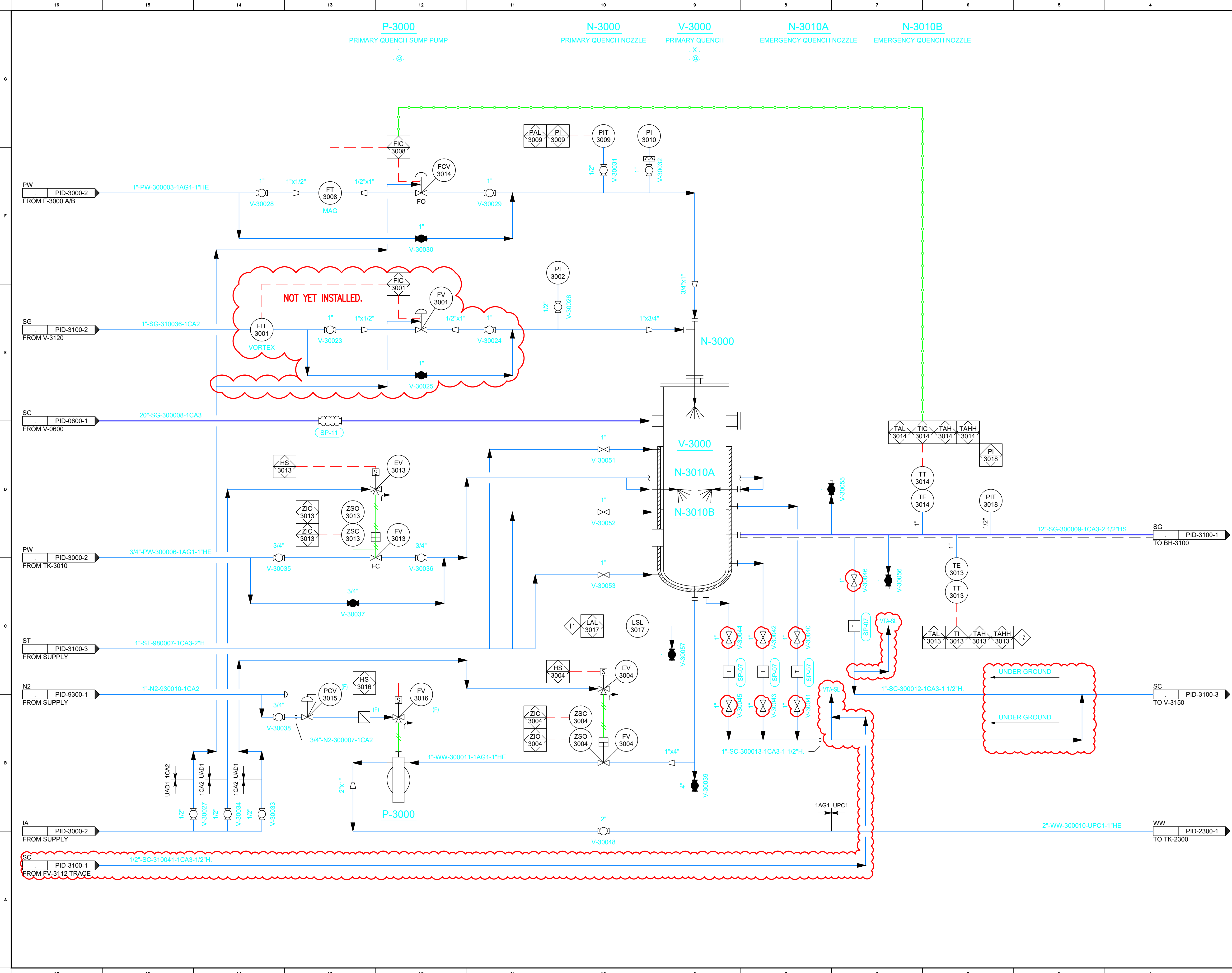
SHEET NO	1	SHEETS	1
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DWG NO 3041-PID-2300-2

REV	A
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CADFILE 3041-PID-2300-2.DWG

3/21/2024



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NOTES:
1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.

DRAWN: A BENNETT
DATE:
CHECKED: P PEREZ
DATE: 7/6/2020
ENGR:
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APVD:
DATE:

REVISIONS

REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE
B	REVISED AND REDRAWN

3041 P&ID PRIMARY QUENCH	
SCALE	NONE
DWG NO	3041-PID-3000-1
CADFILE	3041-PID-3000-1.DWG
SHEET NO	1
SHEETS	1
REV	B
DATE	4/20/2023

TK-3010
PRIAMRY QUENCH TANK
5'-1" D X 8'-6"
ATM @

F-3000A
PRIMARY QUENCH FILTER
F-3000B
PRIMARY QUENCH FILTER

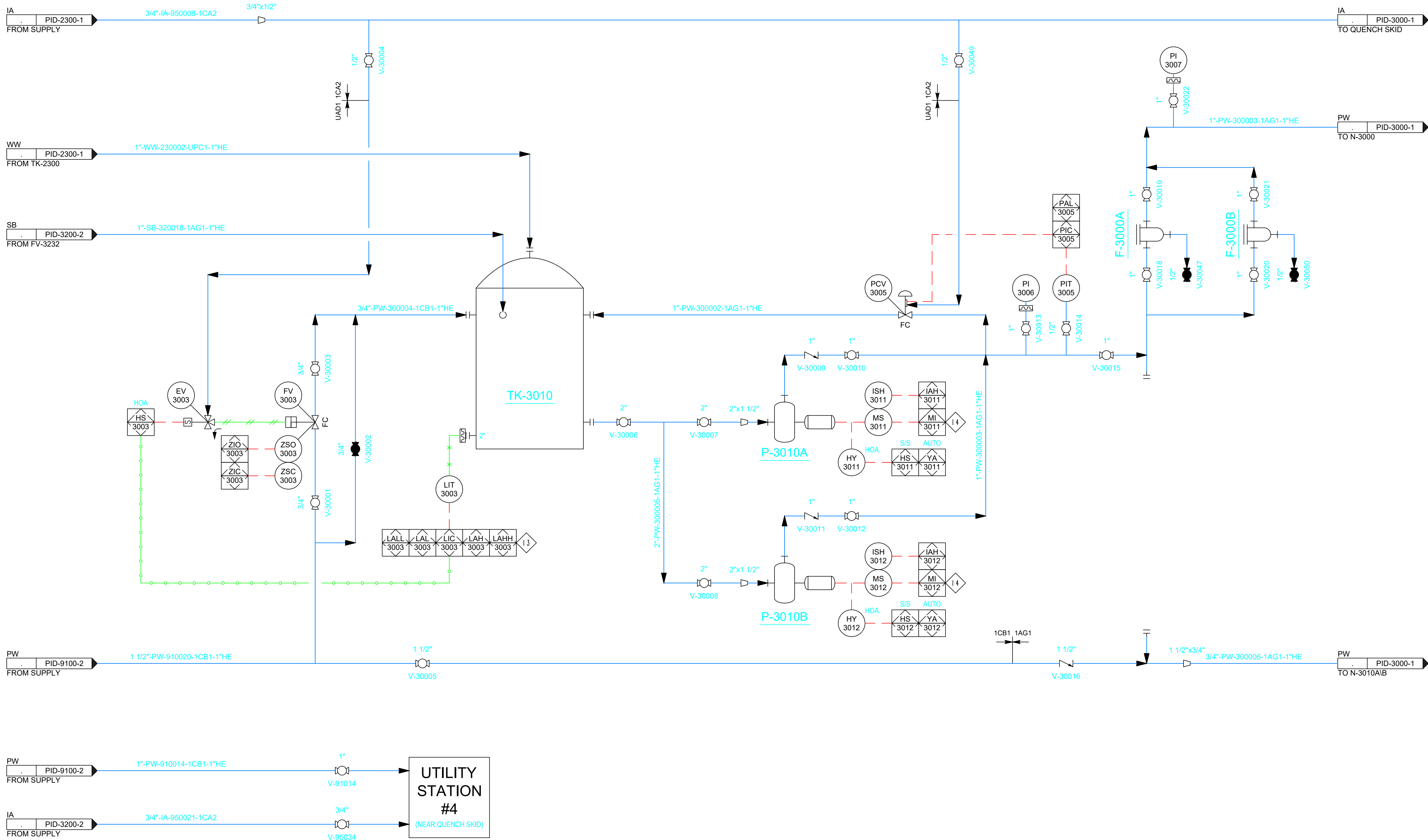
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NOTES:

1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.



DRAWN: A BENNETT
DATE:
CHECKED: P PEREZ
DATE: 7/6/2020
ENGR:
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APVD:
DATE:

REVISIONS

REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

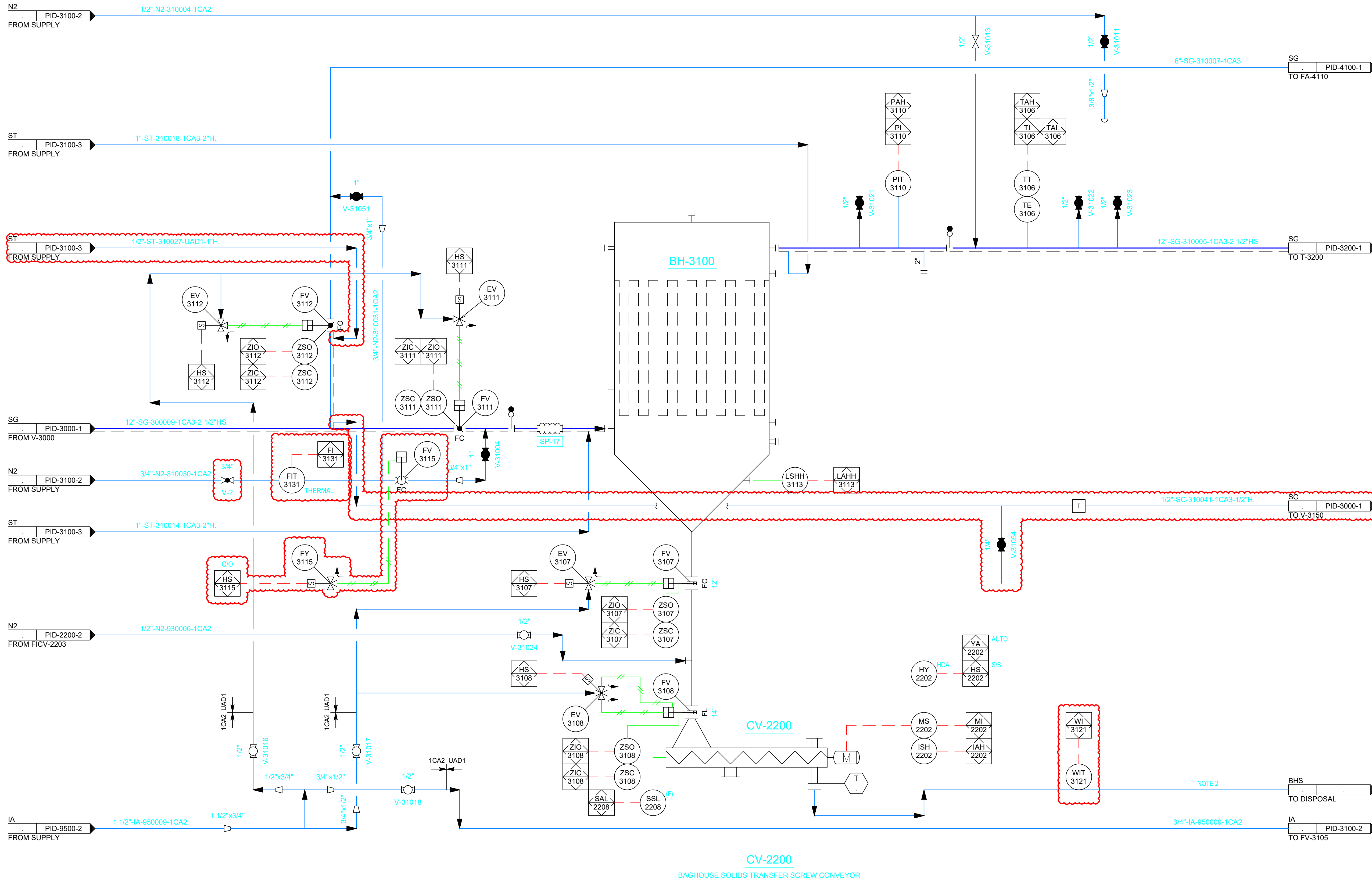
3041 P&ID PRIMARY QUENCH TANK & PUMPS	
SCALE NONE	SHEET NO 1 SHEETS 1
DWG NO 3041-PID-3000-2	REV A
CADFILE 3041-PID-3000-2.DWG	4/20/2023

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NOTES:

1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
2. BAGHOUSE SOLIDS TO BE TRANSFERRED TO SUPER SACKS FOR DISPOSAL.



DRAWN: A BENNETT
DATE:

CHECKED: P PEREZ
DATE: 7/6/2020

ENGR:
DATE:

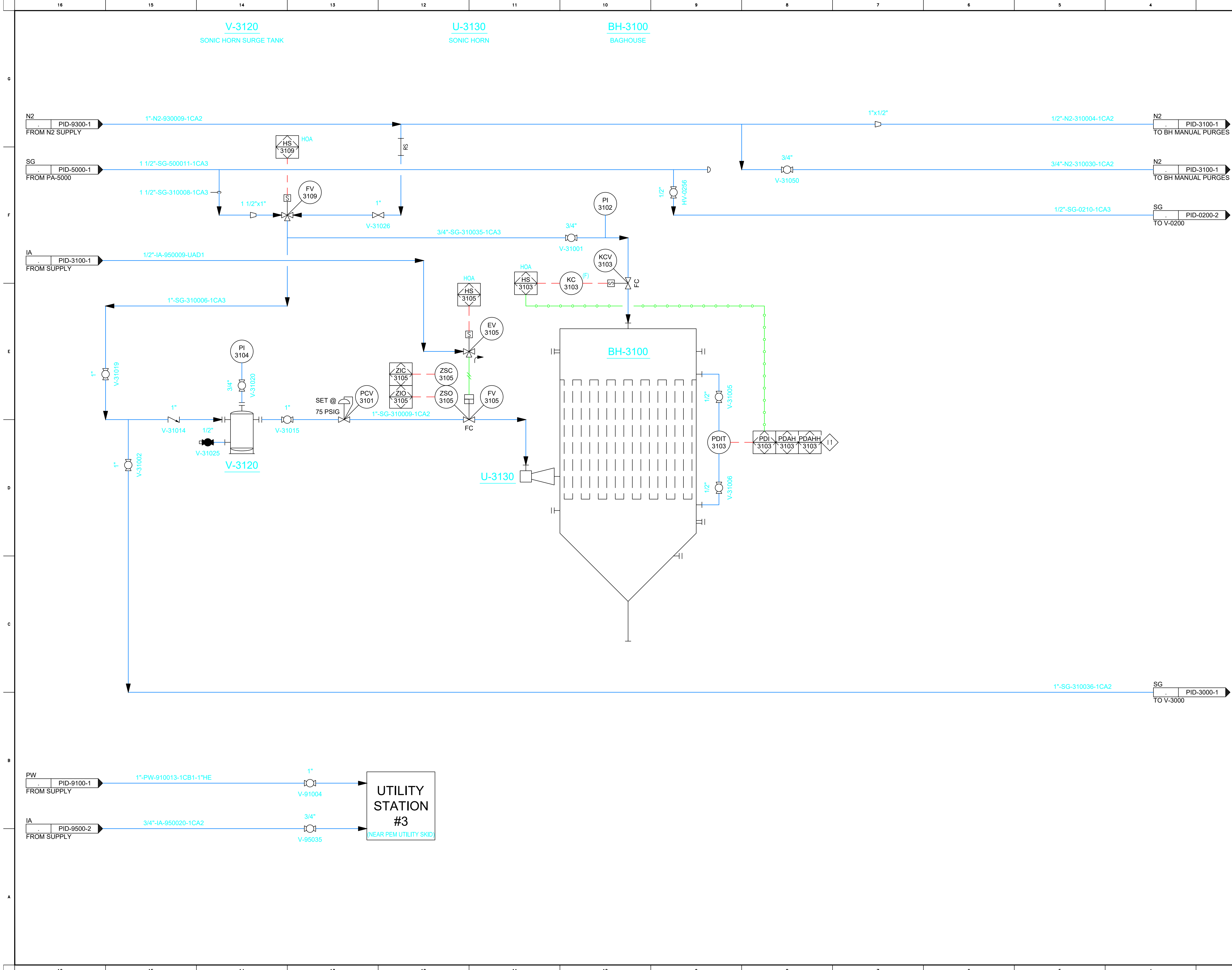
APVD:
DATE:

REVISIONS

REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE
B	REVISED AND REDRAWN

3041
P&ID
BAGHOUSE

SCALE	NONE	SHEET NO	1	SHEETS	1
DWG NO	3041-PID-3100-1	REV	B		
CADFILE	3041-PID-3100-1.DWG	3/21/2024			



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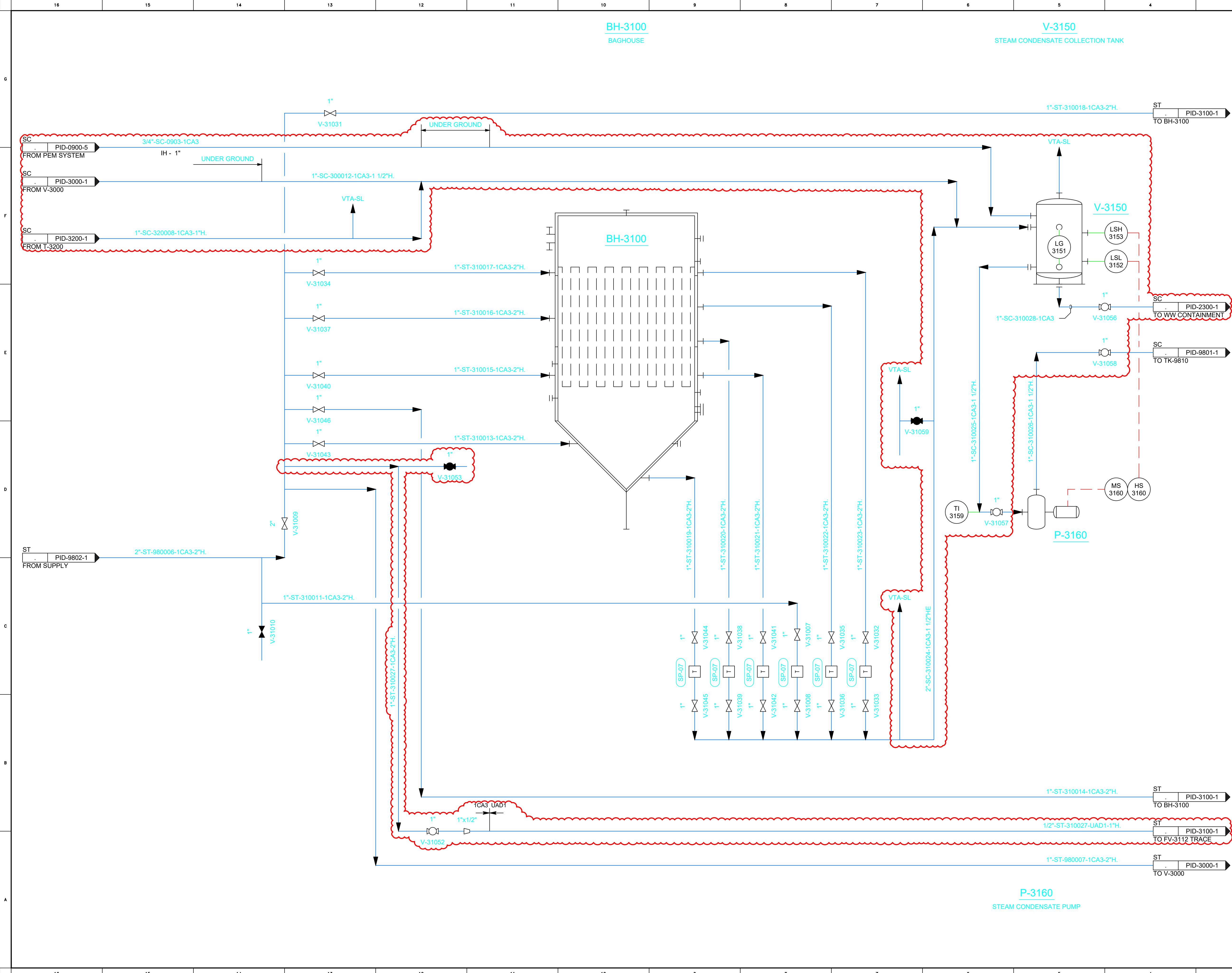
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- NOTES:
- 1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.

DRAWN: A BENNETT	
DATE:	
CHECKED: P PEREZ	
DATE: 7/6/2020	
ENGR:	
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REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE
B	REVISED AND REDRAWN

3041 P&ID BAGHOUSE	
SCALE NONE	SHEET NO 1 SHEETS 1
DWG NO 3041-PID-3100-2	REV B
CADFILE 3041-PID-3100-2.DWG	4/20/2023



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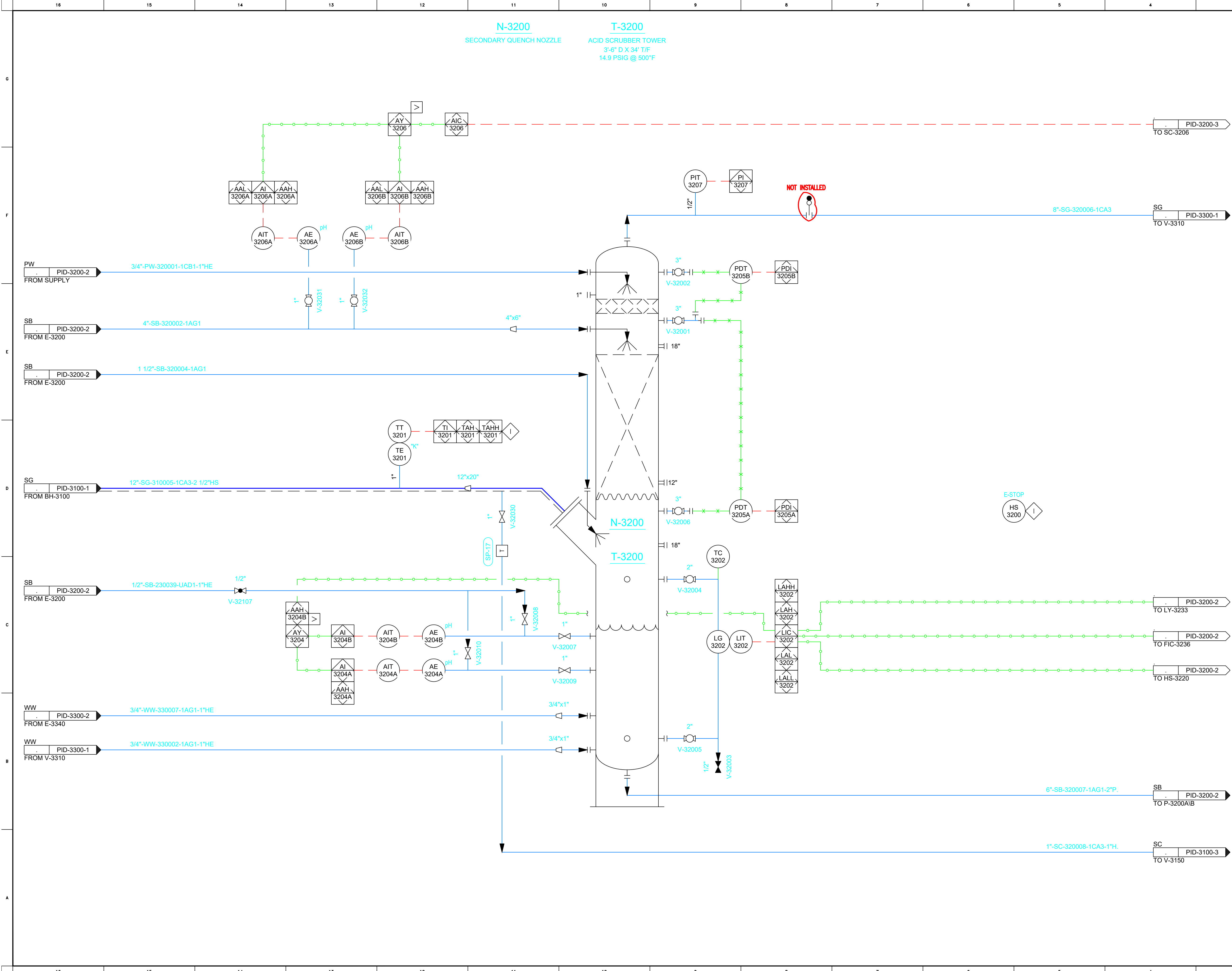
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NOTES:
1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.

DRAWN: A BENNETT DATE:	
CHECKED: P PEREZ DATE: 7/6/2020	
ENGR: DATE:	
APVD: DATE:	
REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

3041 P&ID BAGHOUSE HEAT TRACE	
SCALE NONE	SHEET NO 1 SHEETS 1
DWG NO 3041-PID-3100-3	REV A
CADFILE 3041-PID-3100-3.DWG	4/20/2023



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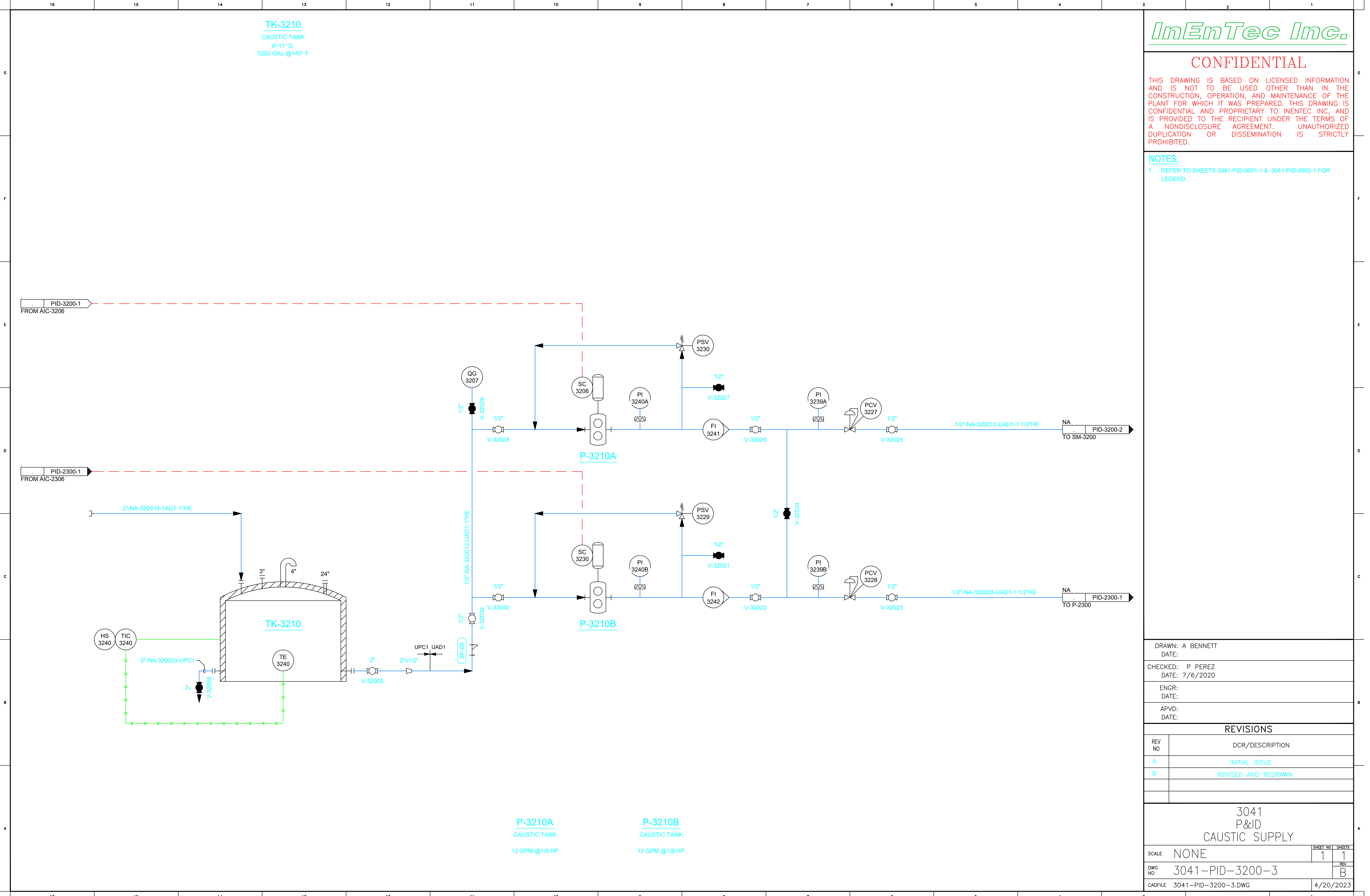
- NOTES:
- 1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.

DRAWN: A BENNETT	
DATE:	
CHECKED: P PEREZ	
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ENGR:	
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APVD:	
DATE:	

REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

3041 P&ID SCRUBBER	
SCALE NONE	SHEET NO 1 SHEETS 1
DWG NO 3041-PID-3200-1	REV A
CADFILE 3041-PID-3200-1.DWG	3/21/2024

3041 P&ID SCRUBBER PUMP AND PIPING		SHEET NO	SHEETS
SCALE	NONE	1	1
DWG NO	3041-PID-3200-2	REV	B
CADFILE	3041-PID-3200-2.DWG	3/21/2018	



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- NOTES:
- 1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.

DRAWN: A BENNETT	
DATE:	
CHECKED: P PEREZ	
DATE: 7/6/2020	
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DATE:	
APVD:	
DATE:	

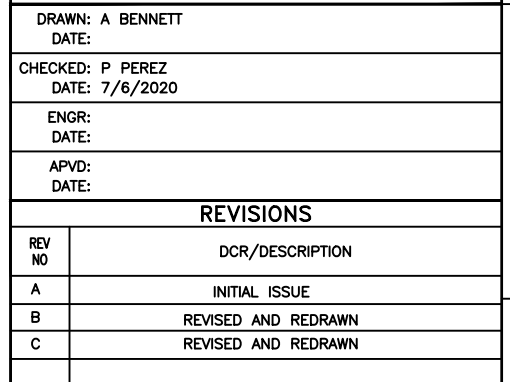
REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE
B	REVISED AND REDRAWN

3041 P&ID CAUSTIC SUPPLY	
SCALE NONE	SHEET NO 1 SHEETS 1
DWG NO 3041-PID-3200-3	REV B
CADFILE 3041-PID-3200-3.DWG	4/20/2023

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1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
2. TRAP MUST SAME HEIGHT OR HIGHER THAN HEIGH LEVEL ALARM ON T-3200.



SCALE	NONE	SHEET NO	1	SHEETS	1
DWG NO	3041-PID-3300-1			REV	B
CADFILE	3041-PID-3300-1.DWG	4/20/2023			

E-3340
SG BLOWER AFTERCOOLER
SHELL: . @
TUBE: . @

U-3952B
HEATED SAMPLE BOX

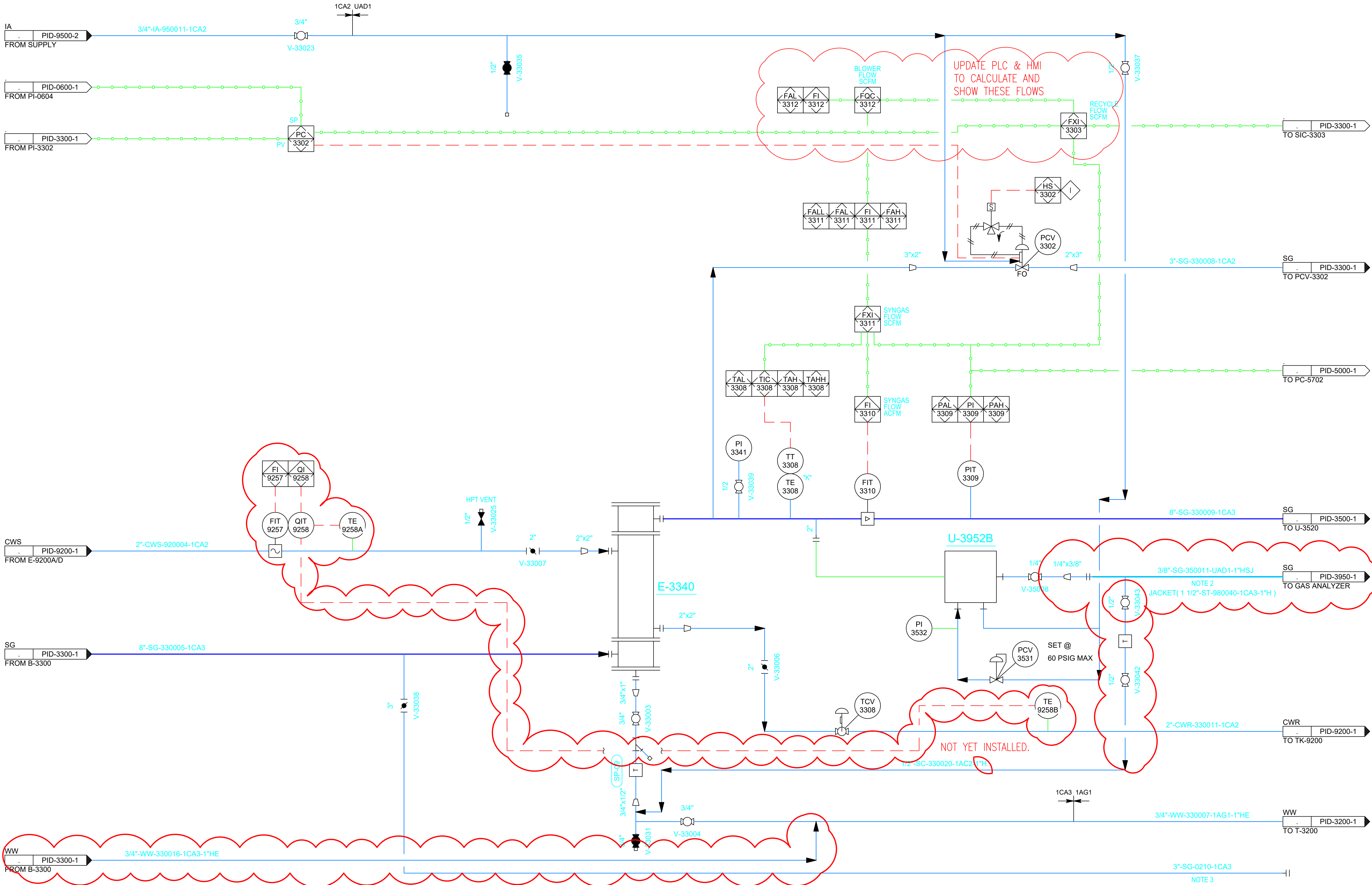
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NOTES:

1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
2. 1-1/2" STEAM JACKETED LINE.
3. CURRENTLY THIS LINE ENDS NEAR THE DDG UPPER DECK.



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DATE:

REVISIONS

REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE
B	REVISED AND REDRAWN

3041 P&ID SYNGAS BLOWER 2	
SCALE NONE	SHEET NO 1 SHEETS 1
DWG NO 3041-PID-3300-2	REV B
CADFILE 3041-PID-3300-2.DWG	4/20/2023

U-3520
SULFUR ADSORBER
10' D X 30'
50 PSIG @ 300°F

P-3520
SULFUR ADSORBER CONDENSATE PUMP
26 GPM @

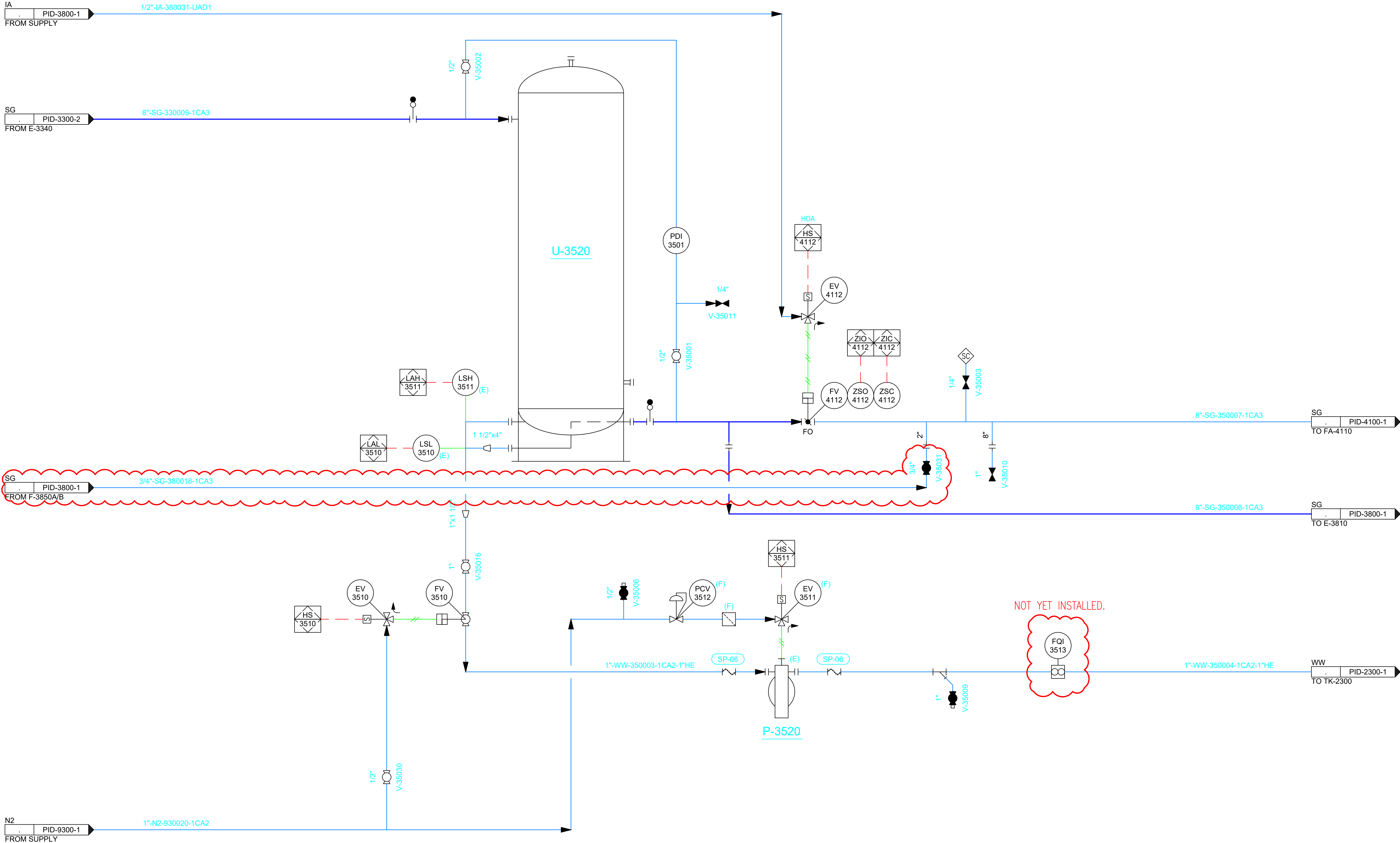
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NOTES:

1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.



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DATE:

CHECKED: P PEREZ
DATE: 7/6/2020

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DATE:

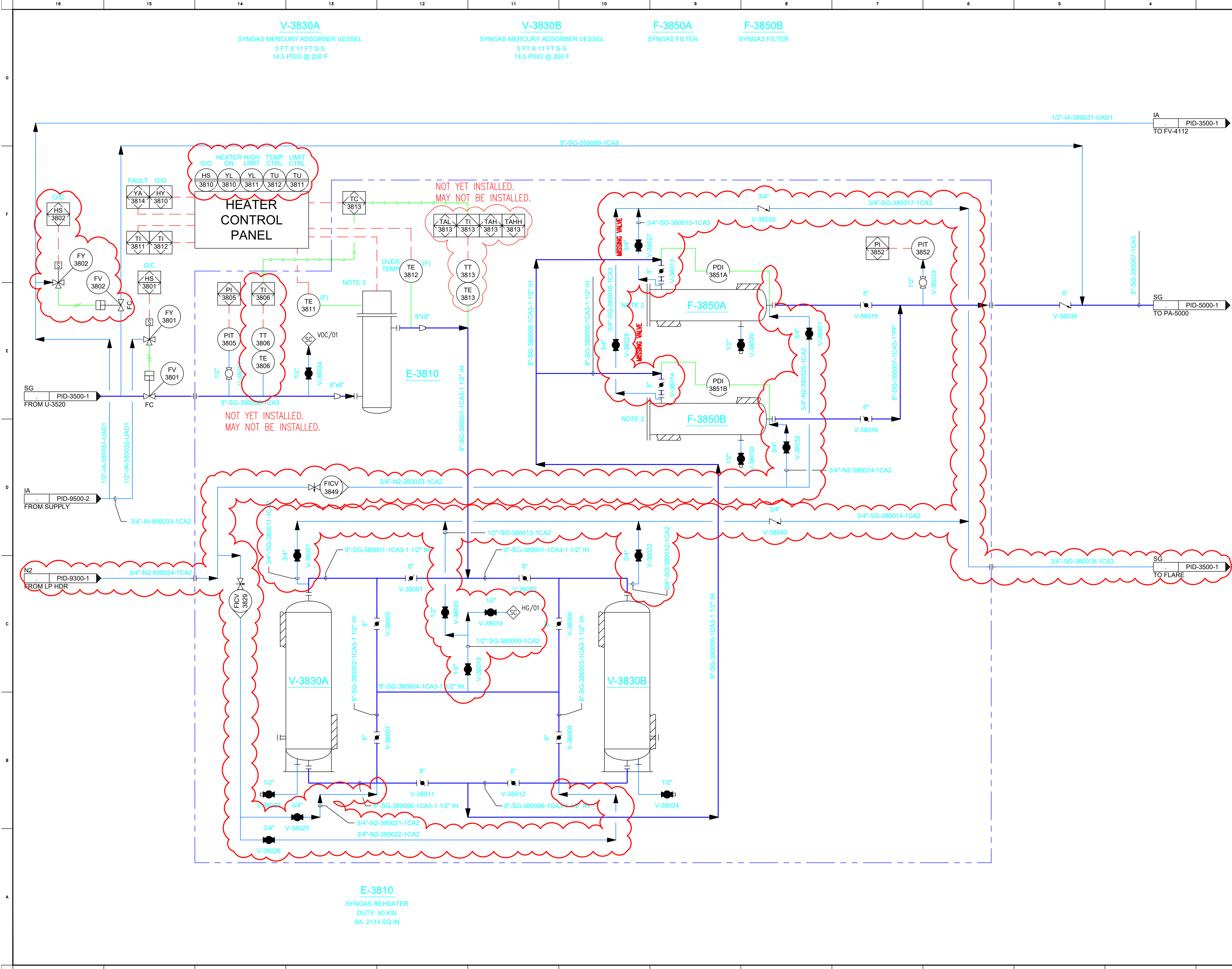
APVD:
DATE:

REVISIONS

REV NO	DCR/DESCRIPTION
0	OFFICIAL RELEASE

3041
P&ID
SULFUR ADSORBER SYSTEM

SCALE	NONE	SHEET NO	1	SHEETS	1
DWG NO	3041-PID-3500-1	REV	0		
CADFILE	3041-PID-3500-1.DWG				4/20/2023



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- NOTES:
- 1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
 - 2. KEEP END CLEAR FOR DISASSEMBLY.

DRAWN: A BENNETT	
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DATE: 7/10/2020	
ENGR:	
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REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

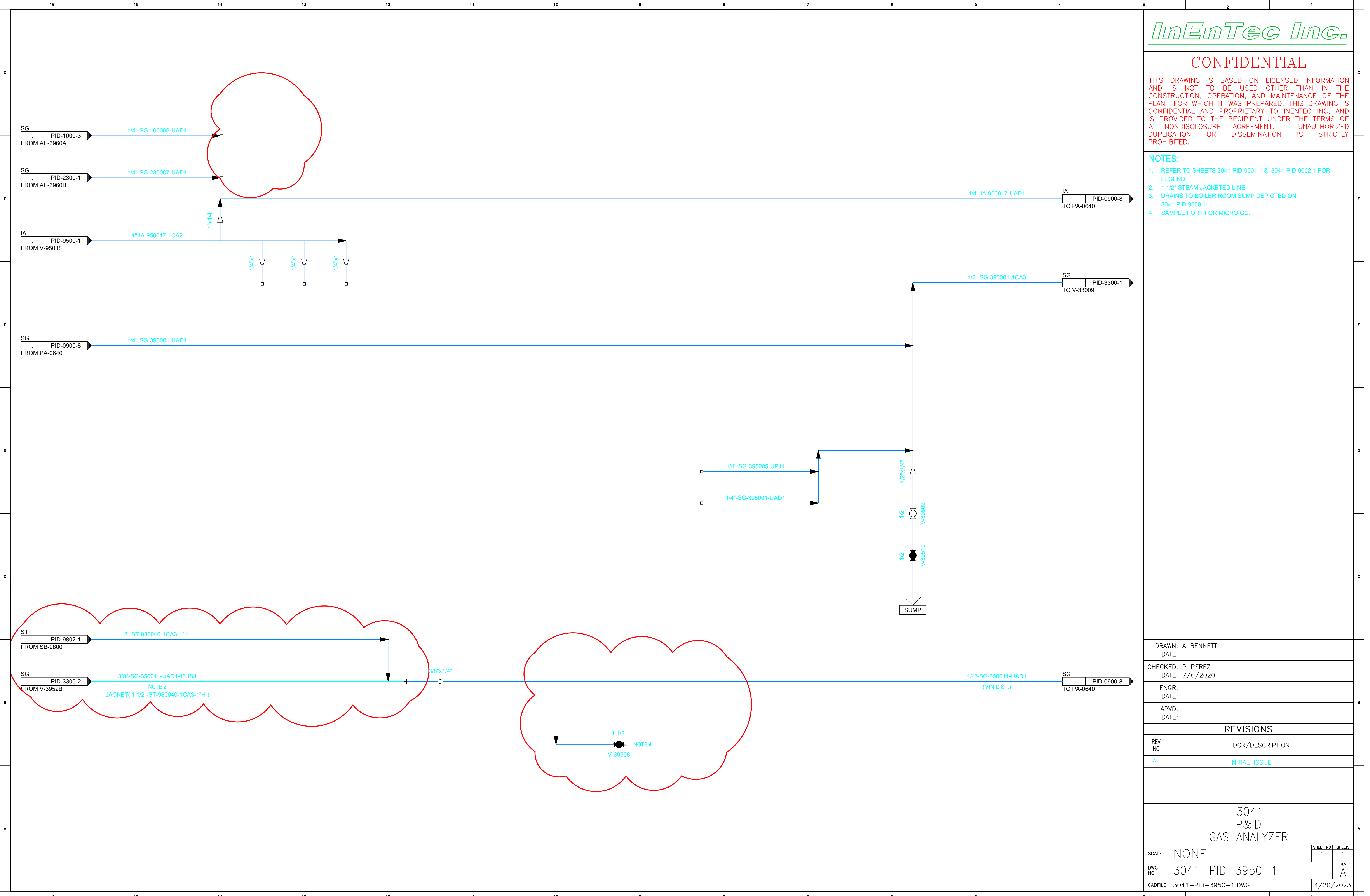
3041 P&ID HG REMOVAL & FILTRATION	
SCALE	NONE
DWG NO	3041-PID-3800-1
CADFILE	3041-PID-3800-1.DWG
SHEET NO	1
SHEETS	1
REV	A
DATE	4/20/2023

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NOTES:

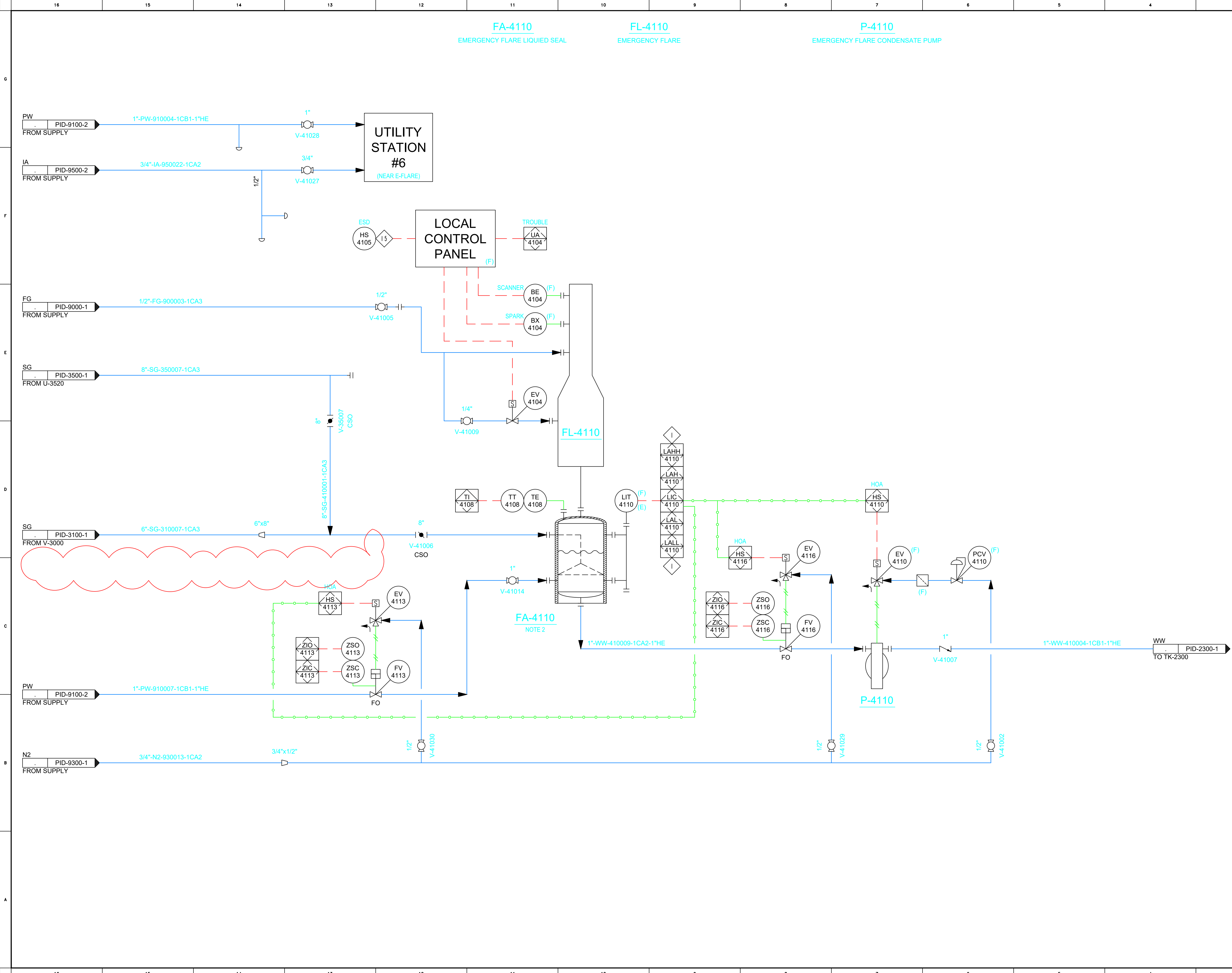
1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
2. 1-1/2" STEAM JACKETED LINE.
3. DRAINS TO BOILER ROOM SUMP DEPICTED ON 3041-PID-9500-1.
4. SAMPLE PORT FOR MICRO GC.



DRAWN: A BENNETT DATE:	
CHECKED: P PEREZ DATE: 7/6/2020	
ENGR: DATE:	
APVD: DATE:	

REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

3041 P&ID GAS ANALYZER	
SCALE NONE	SHEET NO 1 SHEETS 1
DWG NO 3041-PID-3950-1	REV A
CADFILE 3041-PID-3950-1.DWG	4/20/2023



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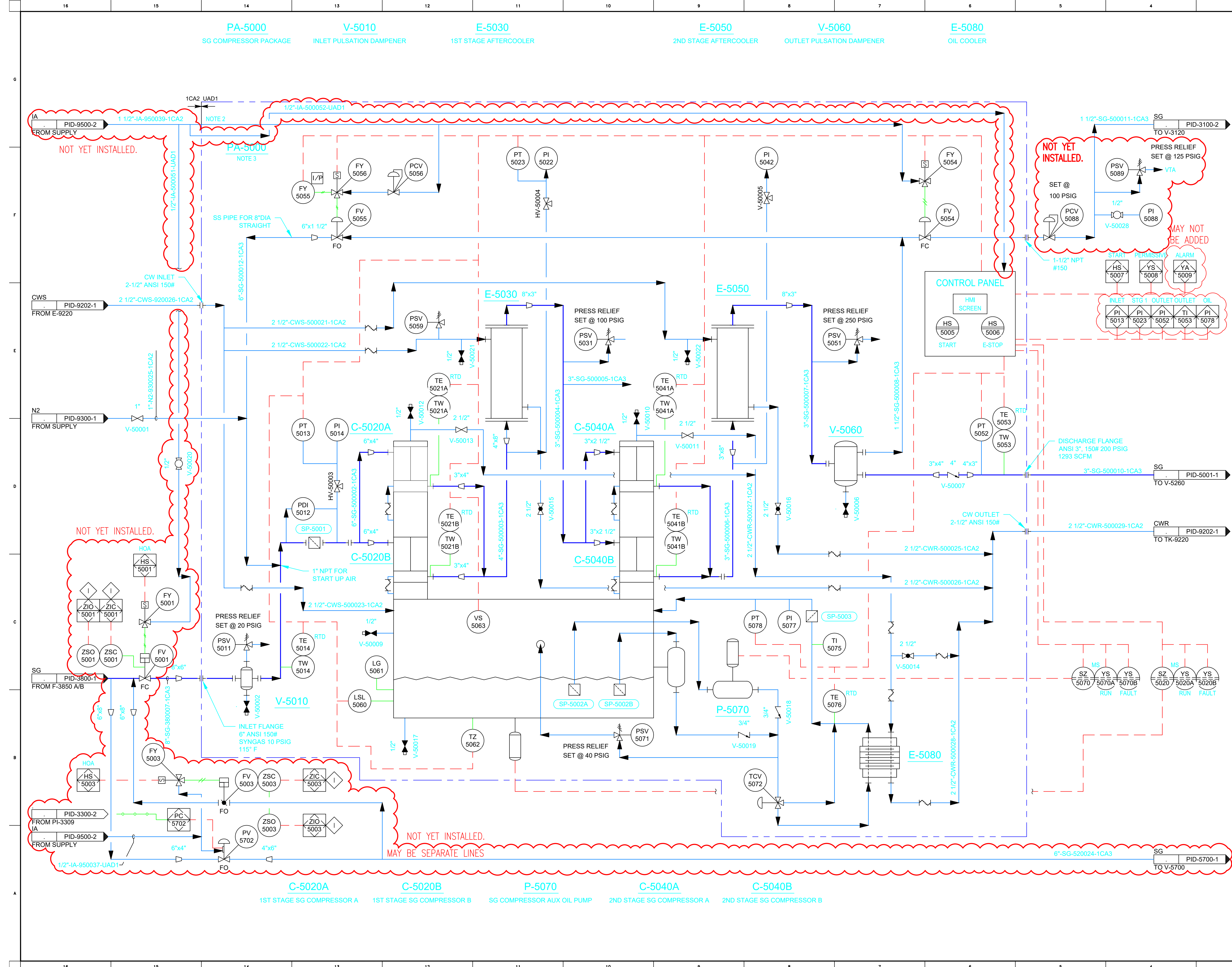
- NOTES:
- 1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
 - 2. LIQUID SEAL POT IS ELECTRIC TRACED FOR FREEZE PROTECTION.

DRAWN: A BENNETT
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DATE: 7/6/2020
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DATE:
APVD:
DATE:

REVISIONS

REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

3041 P&ID FLARE	
SCALE NONE	SHEET NO 1 SHEETS 1
DWG NO 3041-PID-4100-1	REV A
CADFILE 3041-PID-4100-1.DWG	4/20/2023



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NOTES:

- REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
- TUBE TO EDGE OF BEDPLATE WITH 1/4" FEMALE NPT CONNECTION, INSTRUMENT AIR 80-125 PSI, SEE INSTALLATION ARRANGEMENT.
- PA-5000 HAS ITS OWN INDEPENDENT PLC.

DRAWN: P. SALAS DATE:	
CHECKED: P. PEREZ DATE: 6/30/2020	
ENGR: DATE:	
APVD: DATE:	

REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE
B	REVISED AND REDRAWN
C	REVISED AND REDRAWN

3041 P&ID SYNGAS COMPRESSOR	
SCALE: NONE	SHEET NO. 1 OF 1
DWG NO. 3041-PID-5000-1	REV. C
CADFILE 3041-PID-5000-1.DWG	4/20/2023

V-5090
SYNGAS SURGE TANK

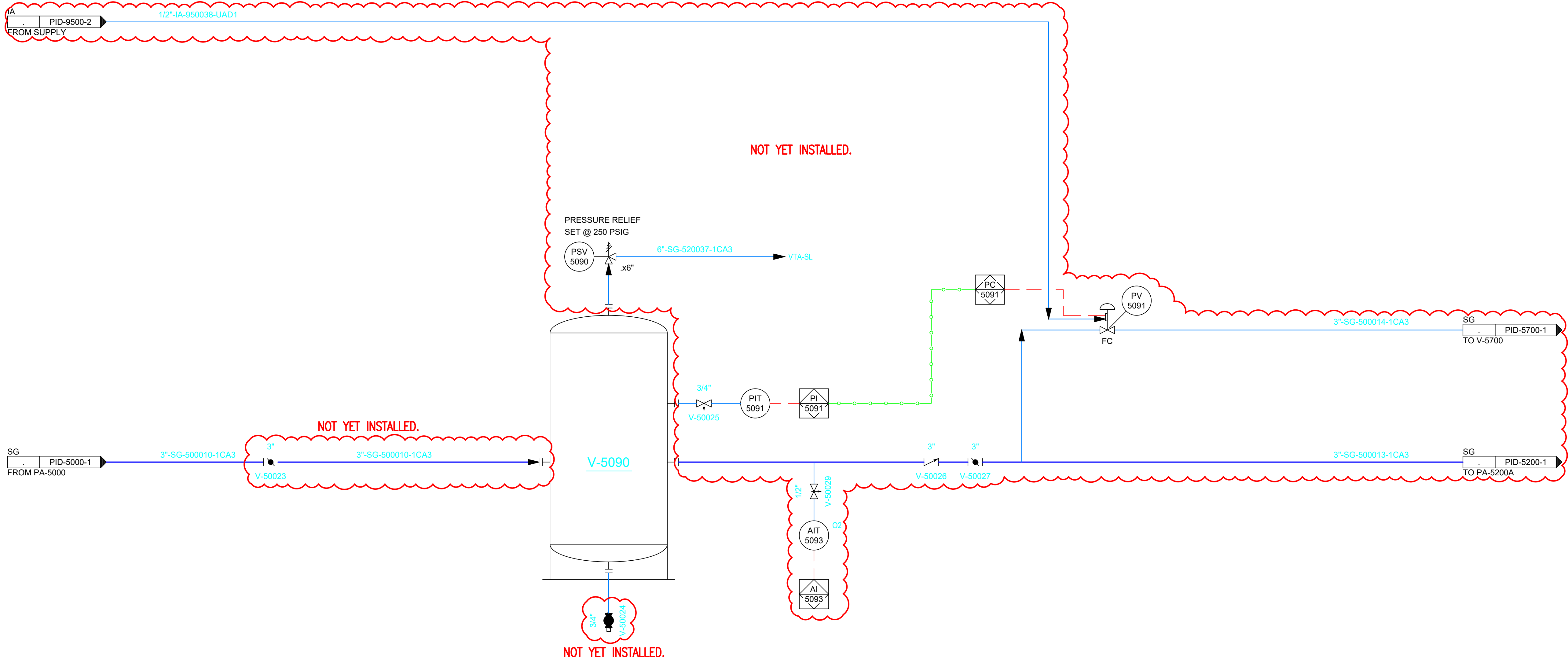
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NOTES:

1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.



DRAWN: A BENNETT
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DATE: 6/30/2020

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APVD:
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REVISIONS

REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

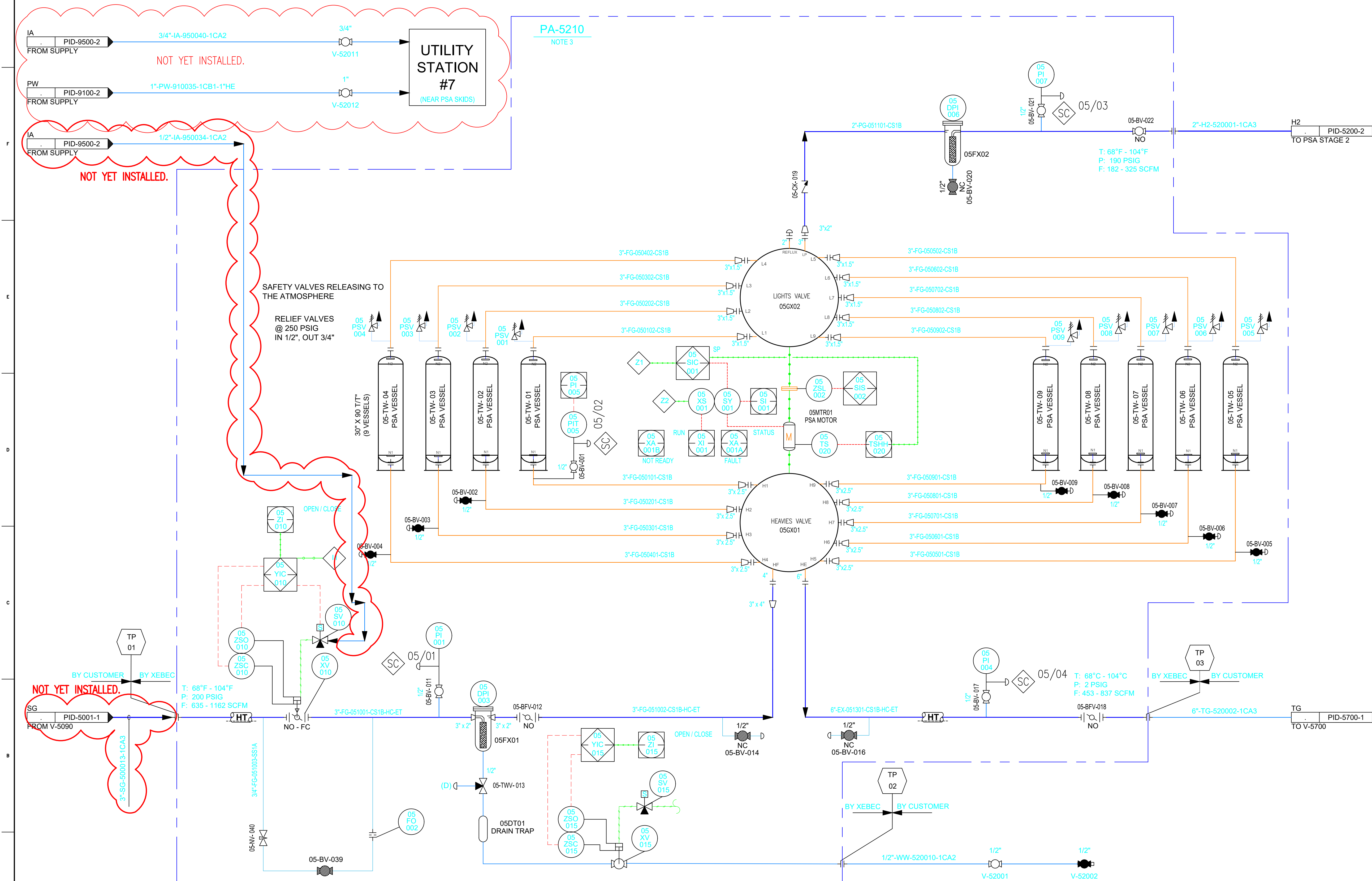
3041
P&ID
SYNGAS SURGE TANK

SCALE	NONE	SHEET NO	1	SHEETS	1
DWG NO	3041-PID-5001-1	REV	A		
CADFILE	3041-PID-5001-1.DWG			4/20/2023	

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IS PROVIDED TO THE RECIPIENT UNDER THE TERMS OF
A NONDISCLOSURE AGREEMENT. UNAUTHORIZED
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NOTES:

1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0001-2 FOR LEGEND.
2. REFER TO XEBEC DOCUMENT 33090-05-PR-PID-001 FOR ORIGINAL EQUIPMENT PACKAGED P&ID AND 33090-05-PR-PID-000 FOR THE LEGEND.
3. PACKAGES PA-5210 AND PA-5220 ARE CONTROLLED BY AN INDEPENDENT PLC.



DRAWN: P SALAS
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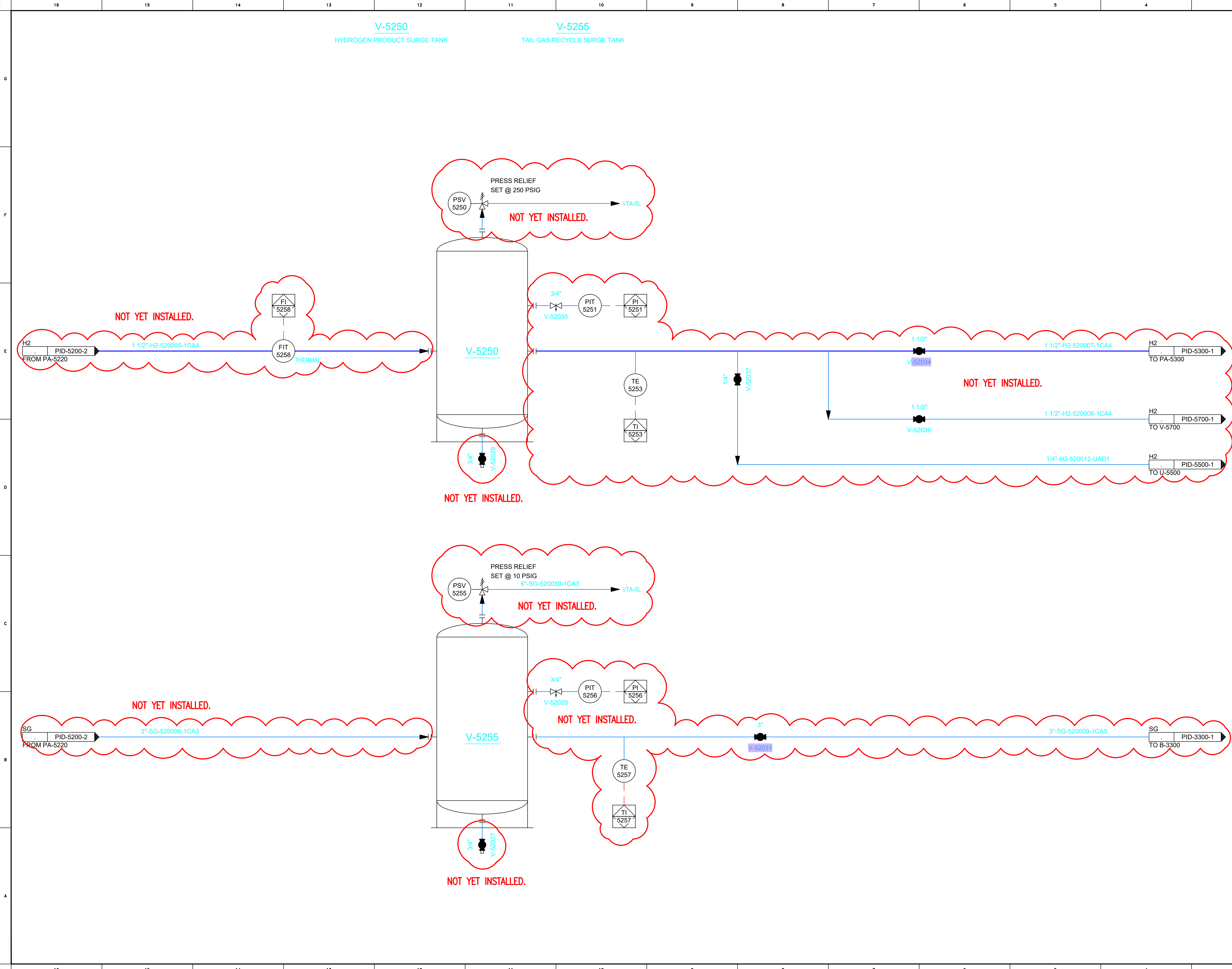
APVD:
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REVISIONS

REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE
B	REVISED AND REDRAWN
C	REVISED AND REDRAWN
D	REVISED AND REDRAWN

3041
P&ID
PRESSURE SWING ADSORPTION

SCALE	NONE	SHEET NO	1	SHEETS	1
DWG NO	3041-PID-5200-1	REV	D		
CADFILE	3041-PID-5200-1.DWG	4/20/2023			



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- NOTES:
- REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
 - ALL INSTRUMENT AIR LINES TO BE RUN IN UAD1.

DRAWN: A BENNETT	
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CHECKED: P PEREZ	
DATE: 6/30/2020	
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REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

3041 P&ID TAIL GAS SURGE TANK	
SCALE NONE	SHEET NO 1 SHEETS 1
DWG NO 3041-PID-5201-1	REV A
CADFILE 3041-PID-5201-1.DWG	4/20/2023

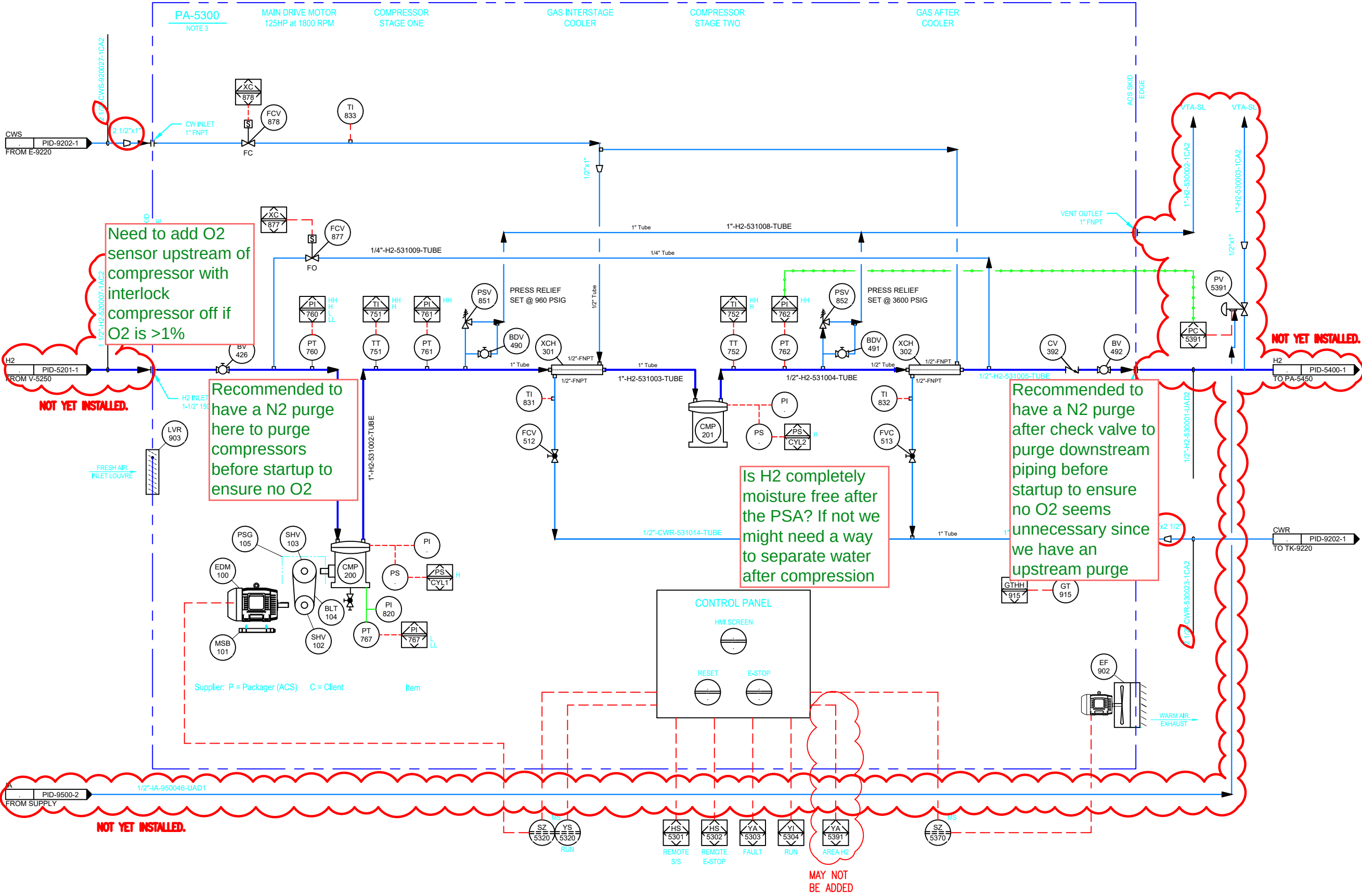
PA-5300
HYDROGEN COMPRESSOR PACKAGE

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- NOTES:
- REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0001-2 FOR LEGEND.
 - REFER TO ACS DOCUMENT PID-10177 FOR PACKAGE'S ORIGINAL P&ID.
 - PACKAGE P-5300 IS CONTROLLED BY AN INDEPENDENT PLC.



DRAWN: P SALAS
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CHECKED: P PEREZ
DATE: 6/30/2020

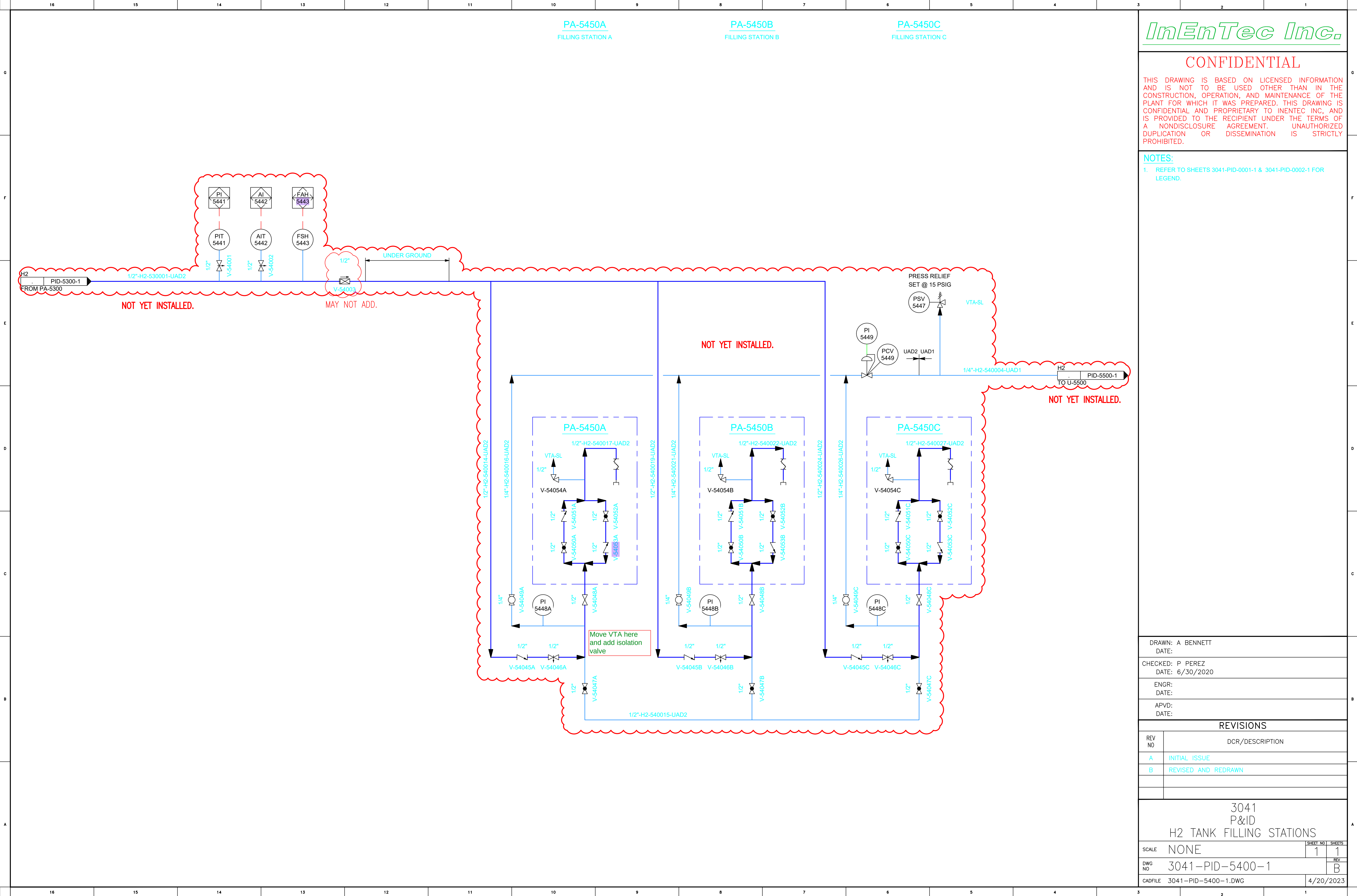
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REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE
B	REVISED AND REDRAWN
C	REVISED AND REDRAWN
D	REVISED AND REDRAWN

3041
P&ID
HYDROGEN COMPRESSOR

SCALE	NONE	SHEET NO	1	SHEETS	1
DWG NO	3041-PID-5300-1	REV	D		
CADFILE	3041-PID-5300-1.DWG				4/20/2023



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- NOTES:
- 1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.

DRAWN: A BENNETT		
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DATE: 6/30/2020		
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REVISIONS		
REV NO	DCR/DESCRIPTION	
A	INITIAL ISSUE	
B	REVISED AND REDRAWN	
3041 P&ID H2 TANK FILLING STATIONS		
SCALE	NONE	SHEET NO 1 SHEETS 1
DWG NO	3041-PID-5400-1	REV B
CADFILE	3041-PID-5400-1.DWG	4/20/2023

U-5500
H2 PRODUCT ANALYZER

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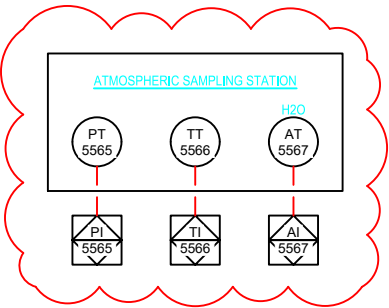
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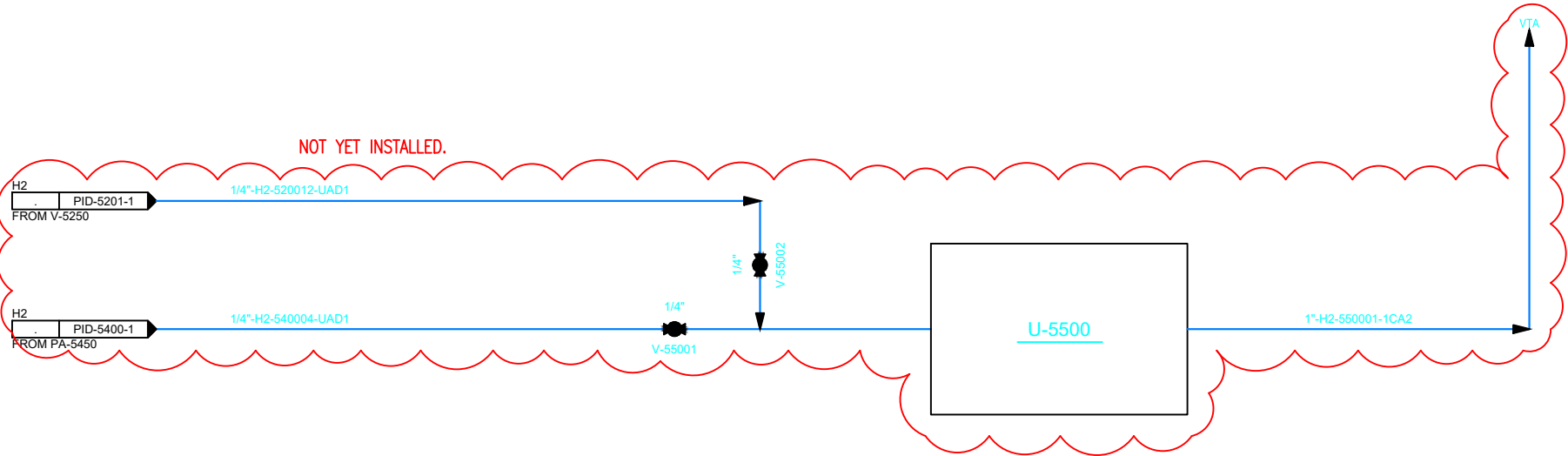
NOTES:

1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.

NOT YET INSTALLED.



NOT YET INSTALLED.



DRAWN: A BENNETT
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REVISIONS

REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE
B	REVISED AND REDRAWN
C	REVISED AND REDRAWN

3041
P&ID
H2 ANALYZER

SCALE NONE

DWG NO 3041-PID-5500-1

CADFILE 3041-PID-5500-1.DWG

SHEET NO 1

REV A

4/20/2023

V-5700
THROX WATER TRAP ARRESTOR

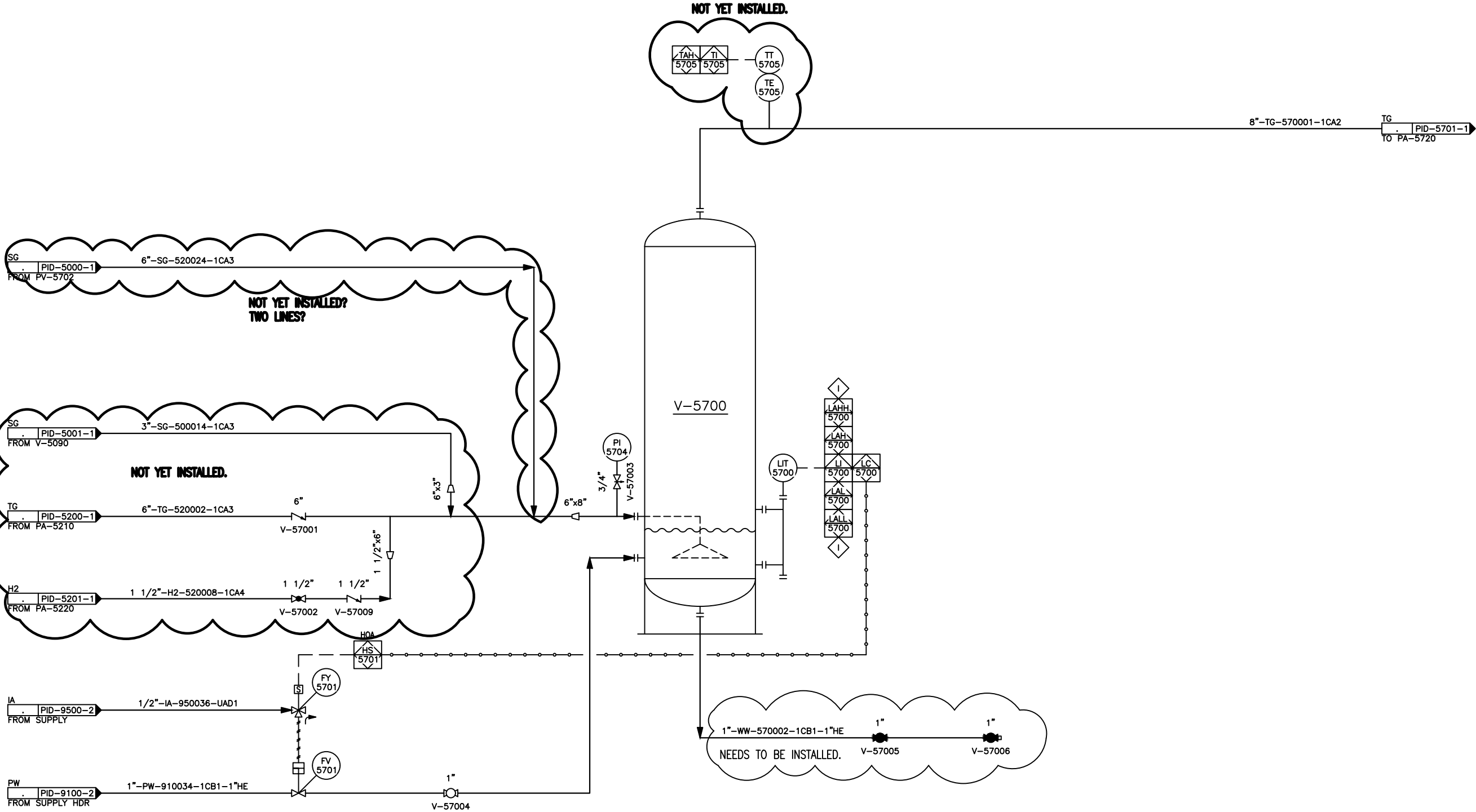
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NOTES:

1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.



DRAWN: A BENNETT
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DATE: 6/30/2020

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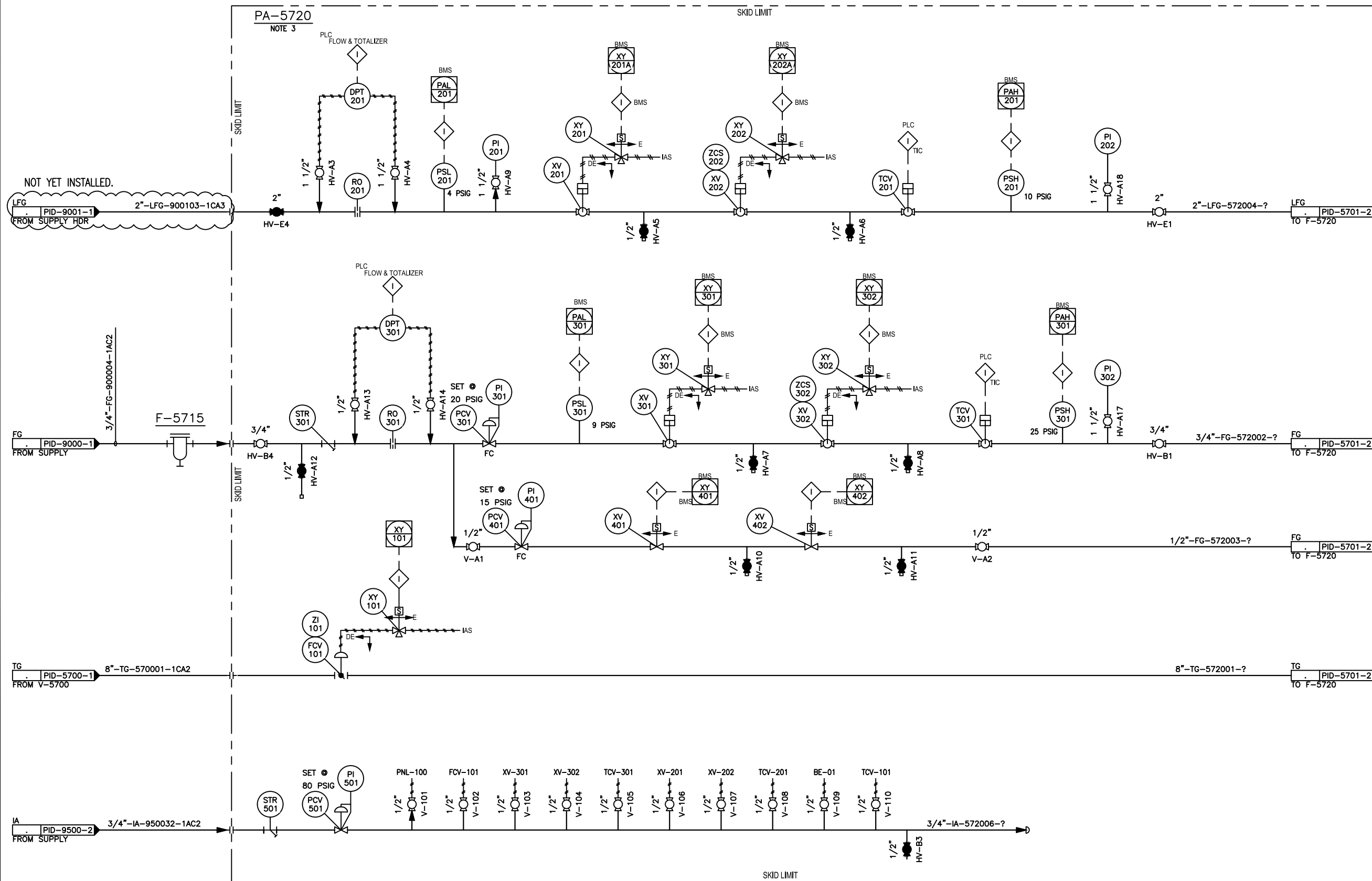
REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE
B	REVISED AND REDRAWN
C	REVISED AND REDRAWN

3041
P&ID
THROX WATER TRAP ARRESTOR

SCALE	NONE	SHEET NO	1	SHEETS	1
DWG NO	3041-PID-5700-1	REV	A		
CADFILE	3041-PID-5700-1.DWG				4/20/2023

NOTES:

1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
2. REFER TO DOCUMENTS TC-16-10-2003-101-Q01 AND TC-16-10-2003-102-Q01 FROM TULSA COMBUSTION LLC FOR ORIGINAL P&IDS FOR PACKAGE.
3. PACKAGE PA-5720 IS CONTROLLED BY AN INDEPENDENT PLC.



DRAWN: A BENNETT
DATE:

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DATE: 6/30/2020

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DATE:

APVD:
DATE:

REVISIONS

REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE
B	REVISED AND REDRAWN
C	REVISED AND REDRAWN

3041
P&ID
THERMAL OXIDIZER HEADERS

SCALE		NONE	SHEET NO.		SHEETS
			1	1	
DWG NO		3041-PID-5701-1			REV
					A
CADFILE		3041-PID-5701-1.DWG			4/20/2023

PA-5720
THERMAL OXIDIZER PACKAGE

F-5720
THERMAL OXIDIZER

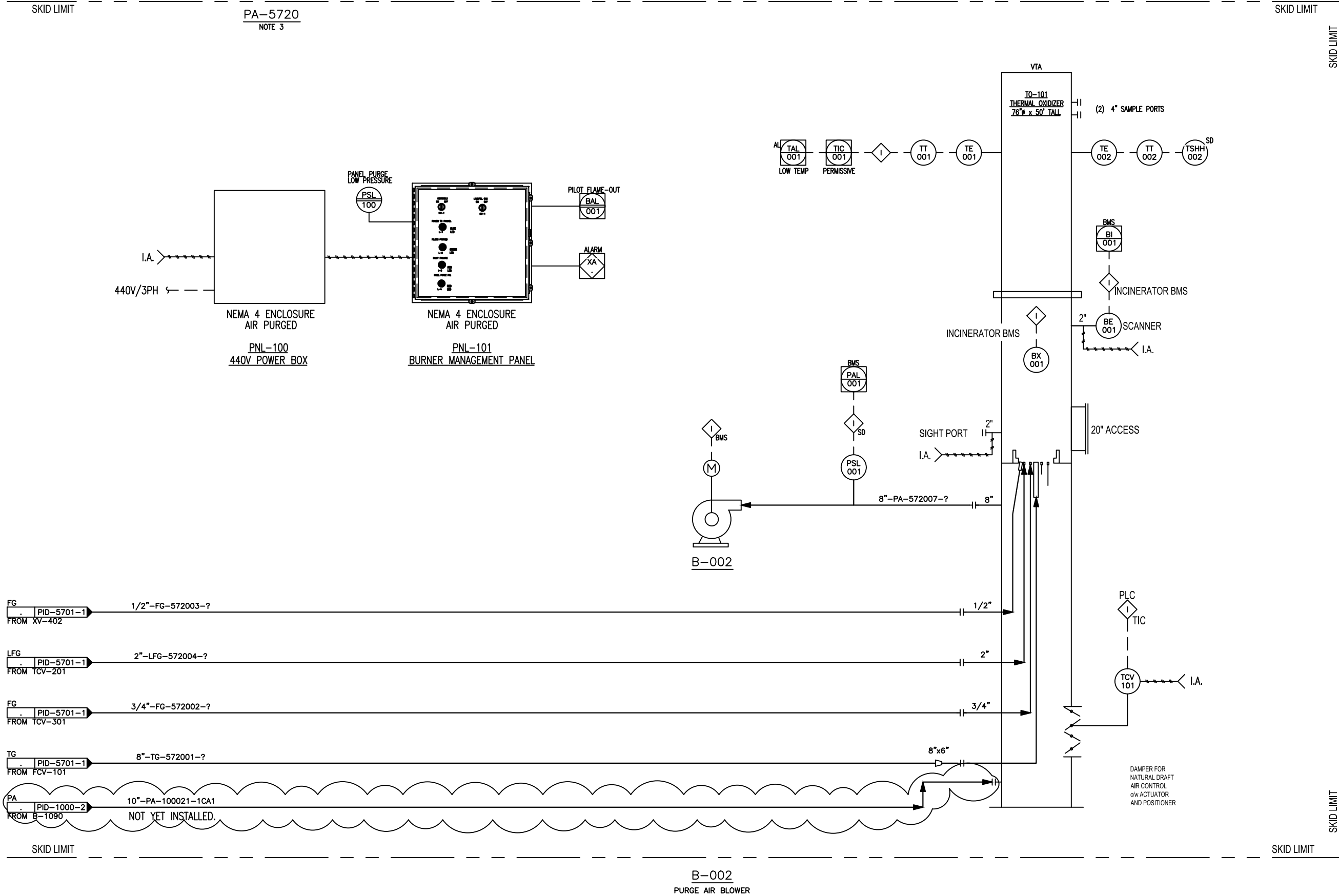
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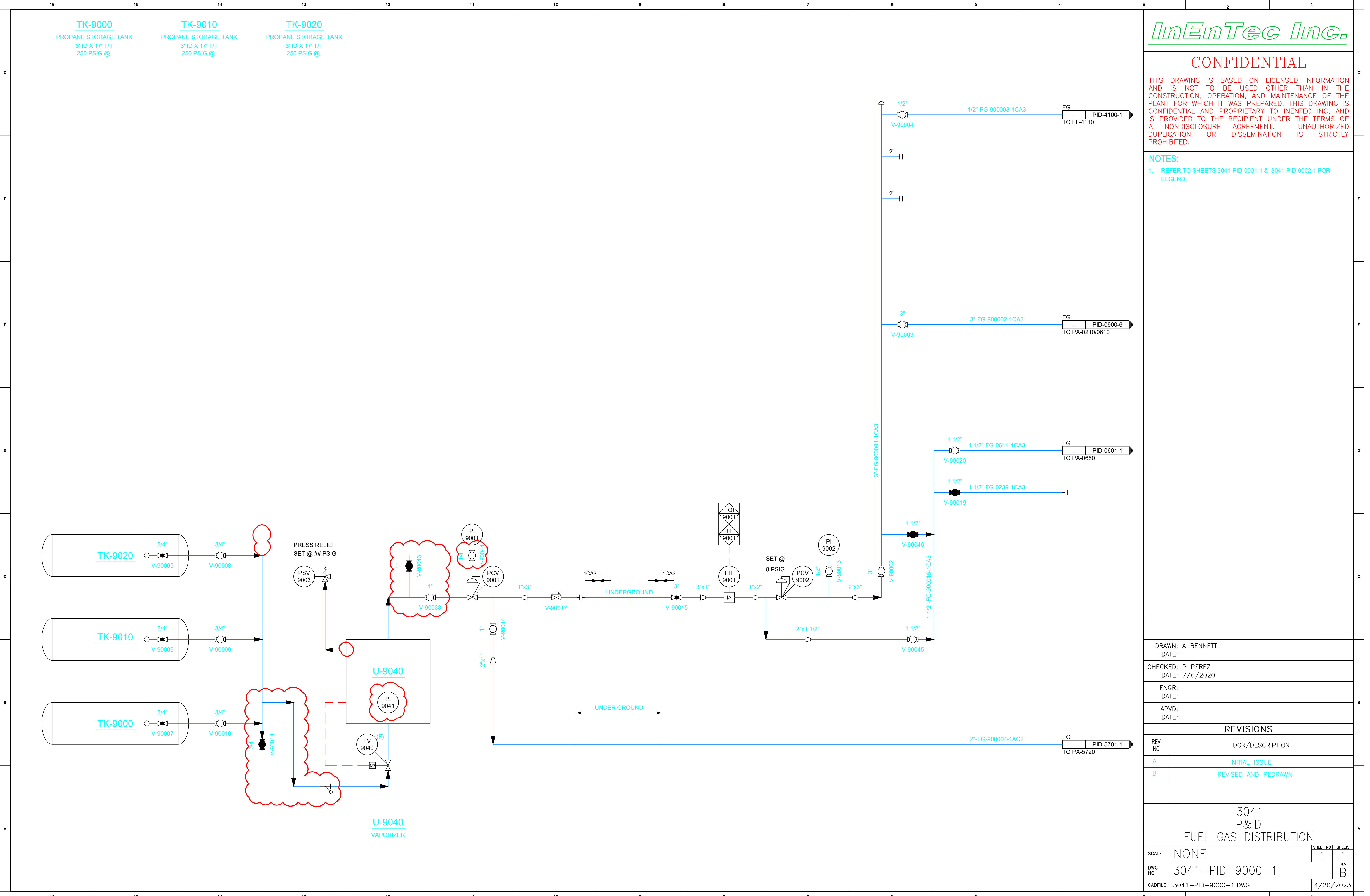
NOTES:

- REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
- REFER TO DOCUMENTS TC-16-10-2003-101-Q01 AND TC-16-10-2003-102-Q01 FROM TULSA COMBUSTION LLC FOR ORIGINAL P&IDS FOR PACKAGE.
- PACKAGE PA-5720 IS CONTROLLED BY AN INDEPENDENT PLC.



DRAWN: A BENNETT DATE:	
CHECKED: P PEREZ DATE: 6/30/2020	
ENGR: DATE:	
APVD: DATE:	
REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

3041 P&ID THERMAL OXIDIZER STACK	
SCALE	NONE
DWG NO	3041-PID-5701-2
CADFILE	3041-PID-5701-2.DWG
SHEET NO	1
SHEETS	1
REV	A
DATE	4/20/2023



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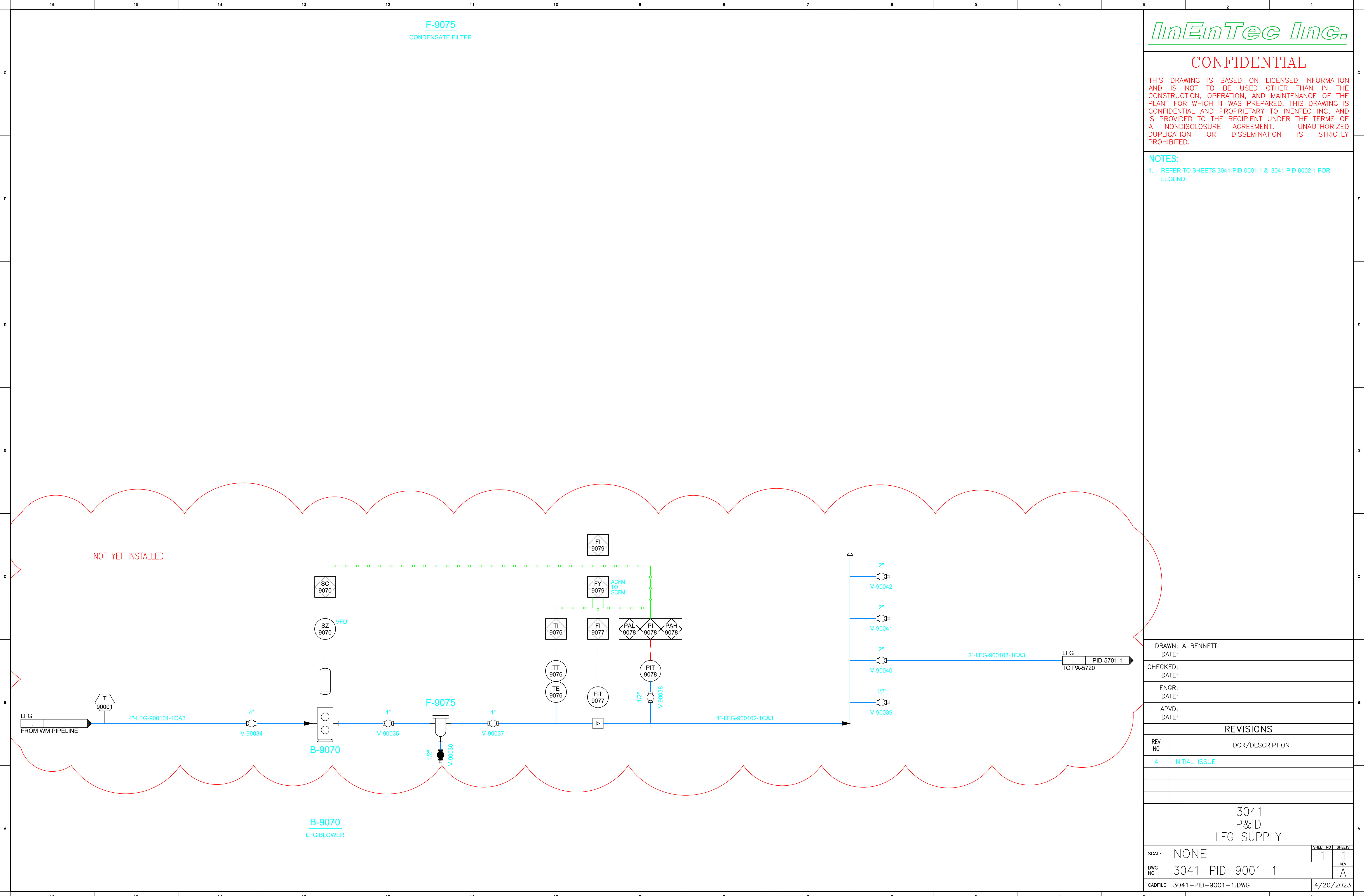
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NOTES:
1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.

DRAWN: A BENNETT
DATE:
CHECKED: P PEREZ
DATE: 7/6/2020
ENGR:
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APVD:
DATE:

REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE
B	REVISED AND REDRAWN

3041 P&ID FUEL GAS DISTRIBUTION	
SCALE NONE	SHEET NO 1 SHEETS 1
DWG NO 3041-PID-9000-1	REV B
CADFILE 3041-PID-9000-1.DWG	4/20/2023



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NOTES:

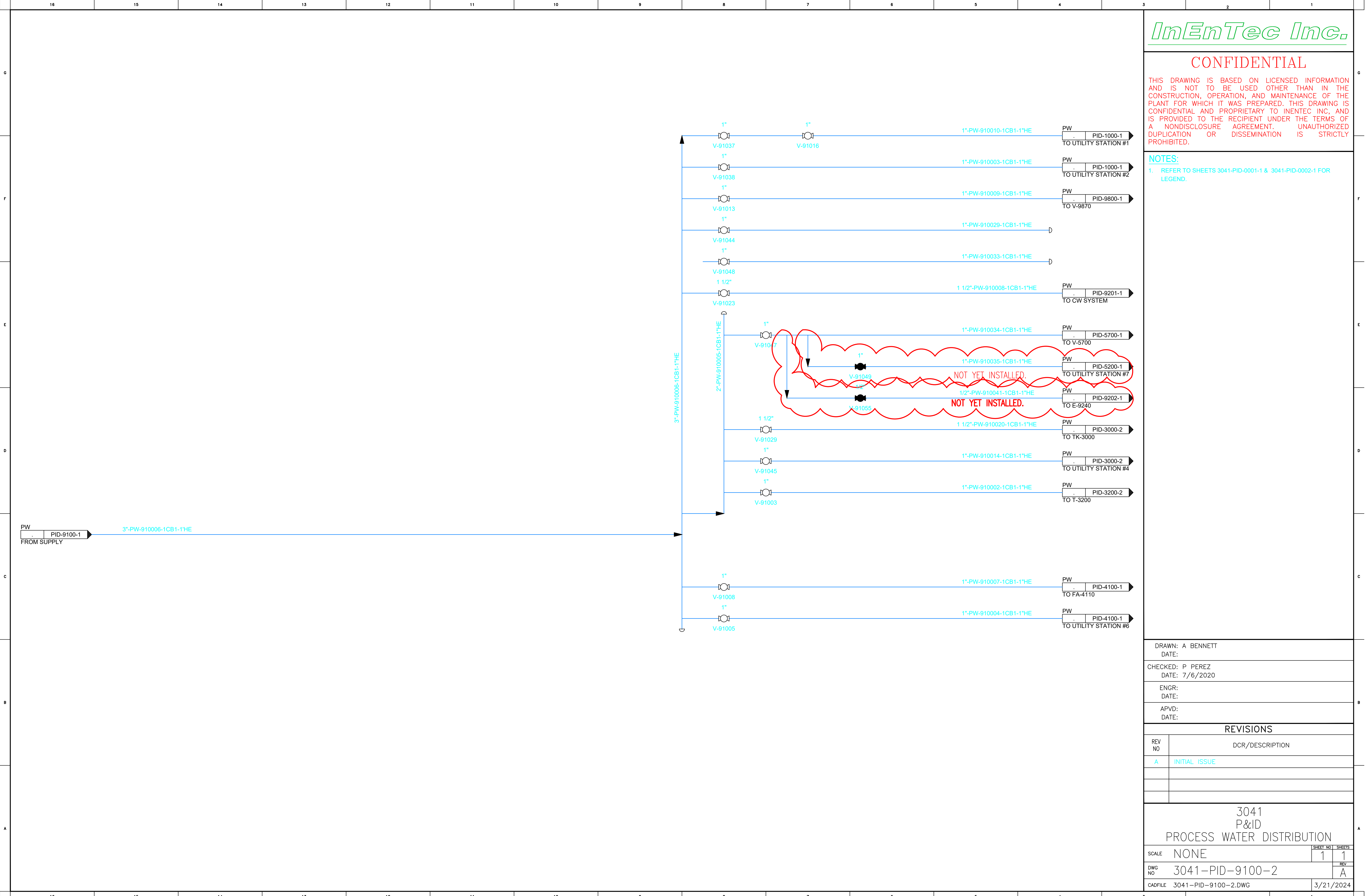
- 1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.

DRAWN: A BENNETT
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REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

3041 P&ID LFG SUPPLY	
SCALE NONE	SHEET NO 1 SHEETS 1
DWG NO 3041-PID-9001-1	REV A
CADFILE 3041-PID-9001-1.DWG	4/20/2023

3041 P&ID PROCESS WATER SYSTEM		SHEET NO	SHEETS
SCALE	NONE	1	1
DWG NO	3041-PID-9100-1	REV	A
CADFILE	3041-PID-9100-1.DWG	3/21/2024	



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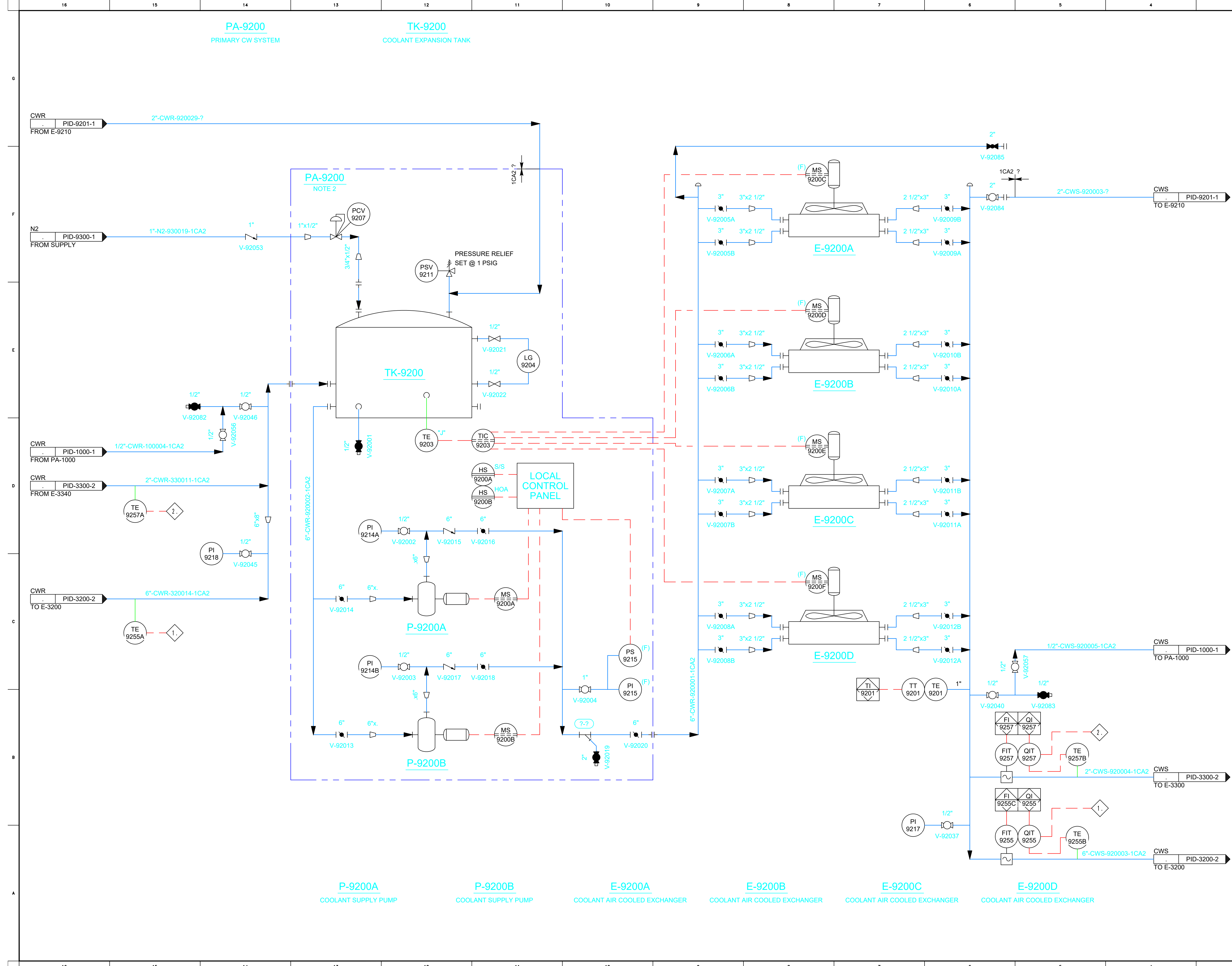
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- NOTES:
- 1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.

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REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

3041	
P&ID	
PROCESS WATER DISTRIBUTION	
SCALE	NONE
DWG NO	3041-PID-9100-2
CADFILE	3041-PID-9100-2.DWG
SHEET NO	1
SHEETS	1
REV	A
DATE	3/21/2024



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NOTES:

- REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
- PA-9200 COOLING WATER SYSTEM USES GLYCOL WATER MIXTURE THAT IS TO BE FILLED MANUALLY THRU A HIGH POINT VENT.

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REVISIONS

REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

3041
P&ID
COOLING WATER SYSTEM

SCALE: NONE

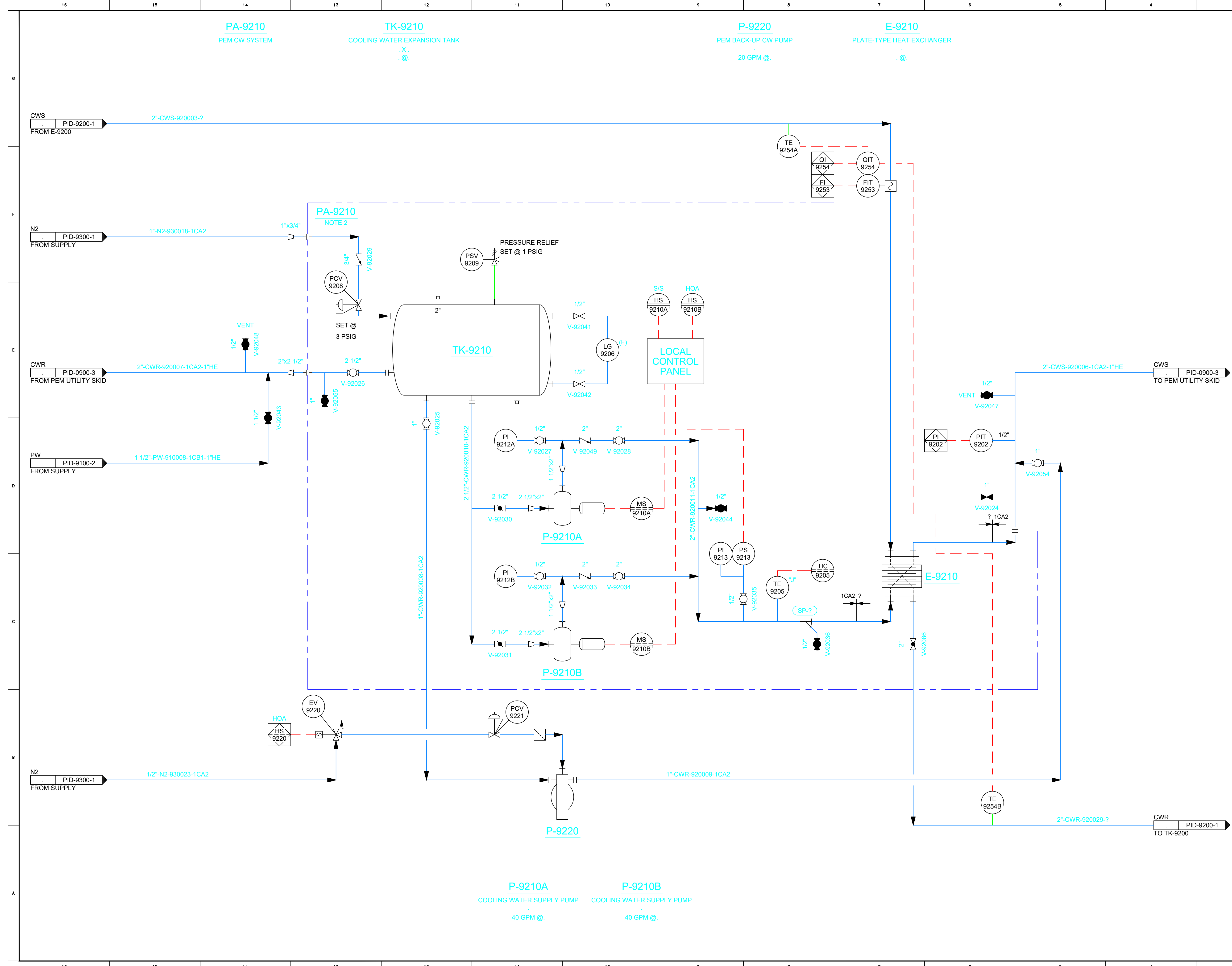
DWG NO: 3041-PID-9200-1

CADFILE: 3041-PID-9200-1.DWG

SHEET NO: 1

REV: A

3/21/2024



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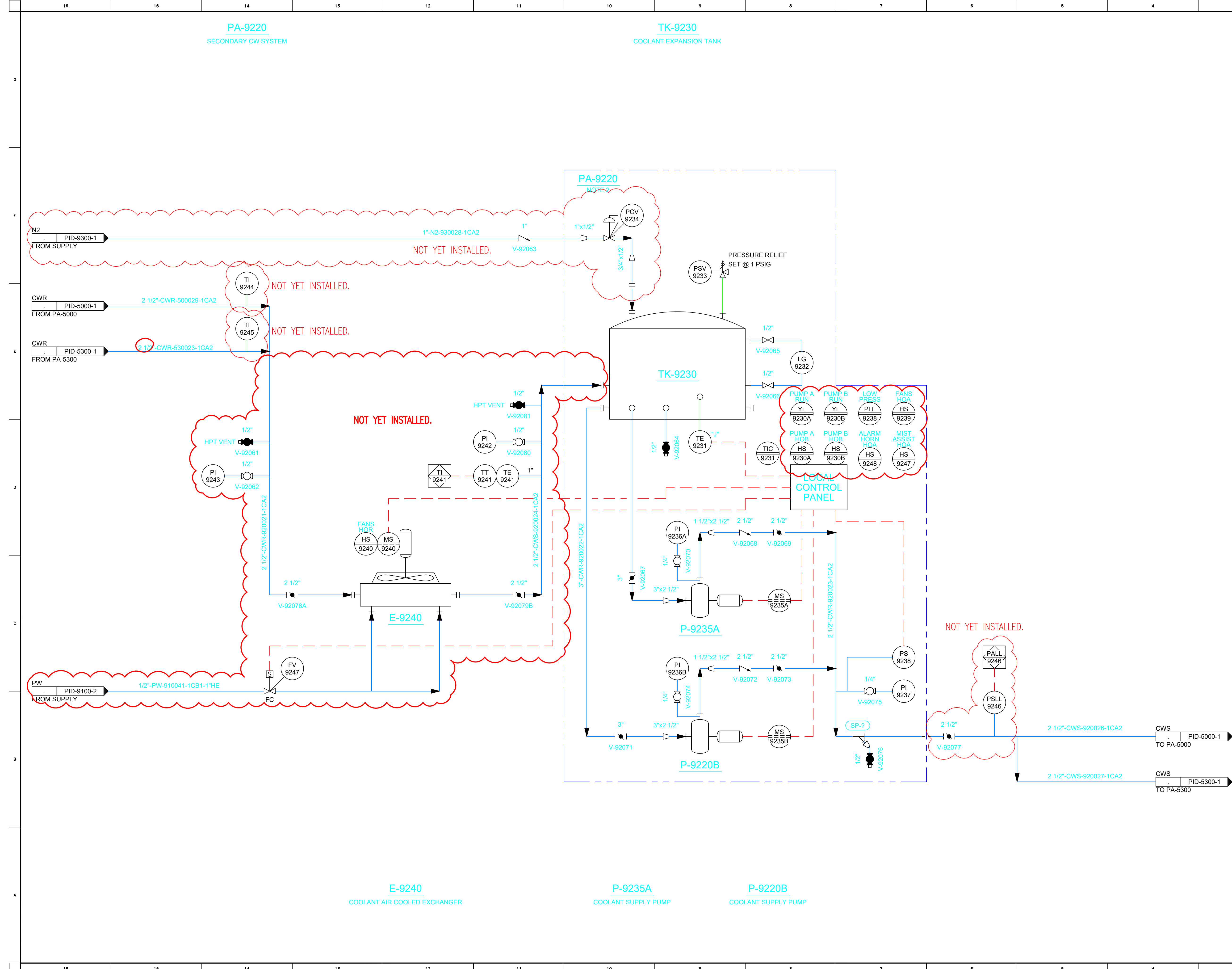
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- NOTES:
- 1. REFER TO SHEETS 3041-PID-0001 & 3041-PID-0002 FOR LEGEND.
 - 2. PA-9210 COOLING WATER SYSTEM USES HIGH PURITY WATER. FILL SYSTEM USING RO WATER MANUALLY FROM HIGH POINT VENT.

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REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

3041 P&ID PEM COOLING WATER SYSTEM			
SCALE	NONE	SHEET NO	1
DWG NO	3041-PID-9201-1	REV	A
CADFILE	3041-PID-9201-1.DWG	3/21/2024	



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NOTES:

- REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
- PA-9220 COOLING WATER SYSTEM USES GLYCOL WATER MIXTURE THAT IS ADDED MANUALLY THRU A HIGH POINT VENT.

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A	INITIAL ISSUE

3041 P&ID H2 COOLING WATER SYSTEM	
SCALE NONE	SHEET NO 1 SHEETS 1
DWG NO 3041-PID-9202-1	REV A
CADFILE 3041-PID-9202-1.DWG	4/20/2023

PA-9300
NITROGEN SUPPLY SYSTEM

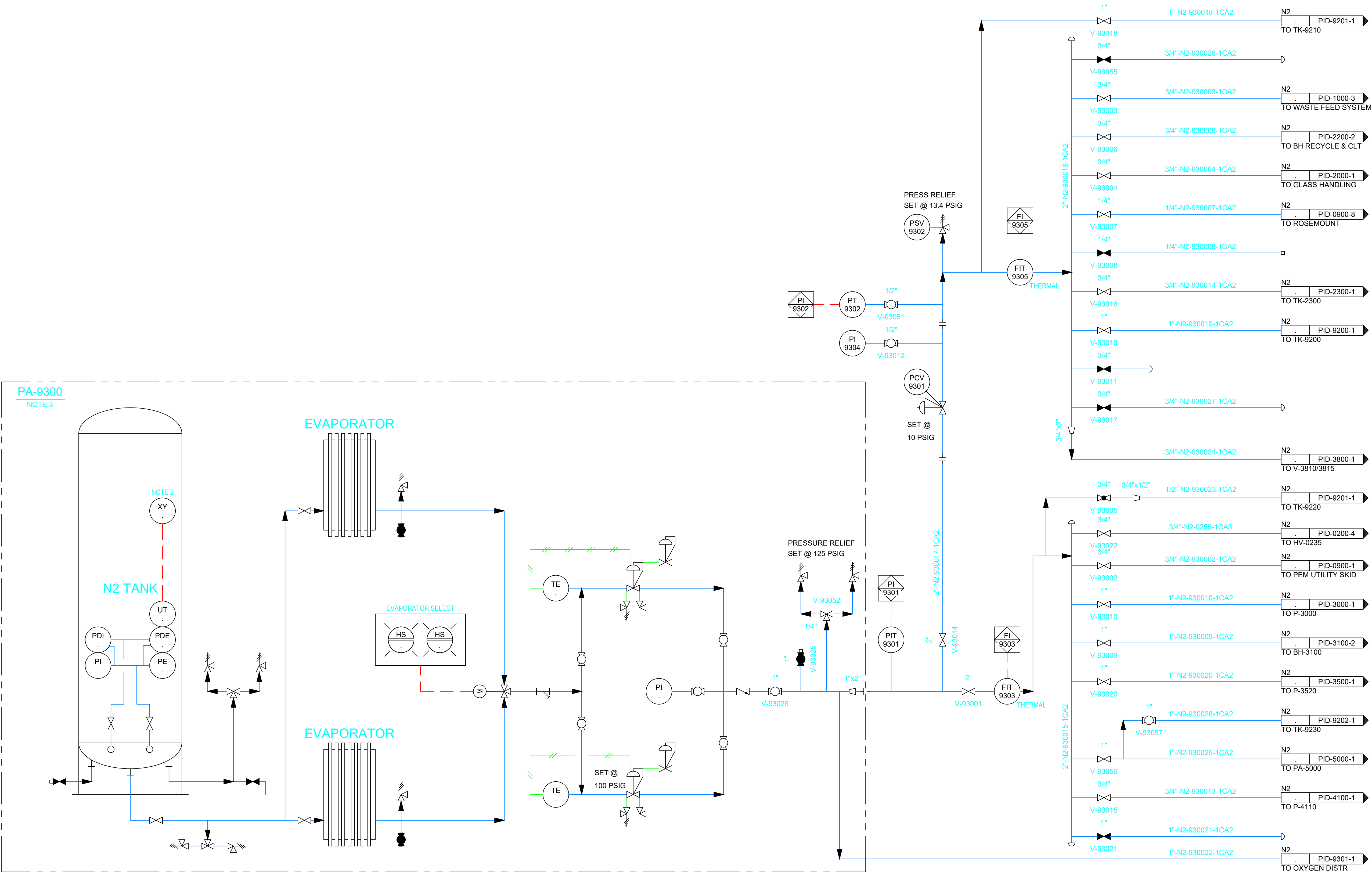
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NOTES:

- REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
- ANTENNA TO TRANSMIT DATA TO NITROGEN VENDOR.
- PA-9300 IS SUPPLIED AND OWNED BY NITROGEN VENDOR.



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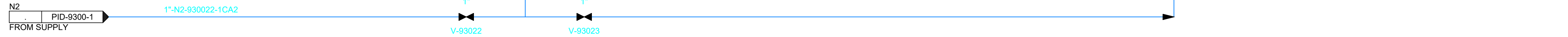
REVISIONS

REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

3041 P&ID NITROGEN DISTRIBUTION	
SCALE NONE	SHEET NO 1 REV 1
DWG NO 3041-PID-9300-1	REV A
CADFILE 3041-PID-9300-1.DWG	3/21/2025

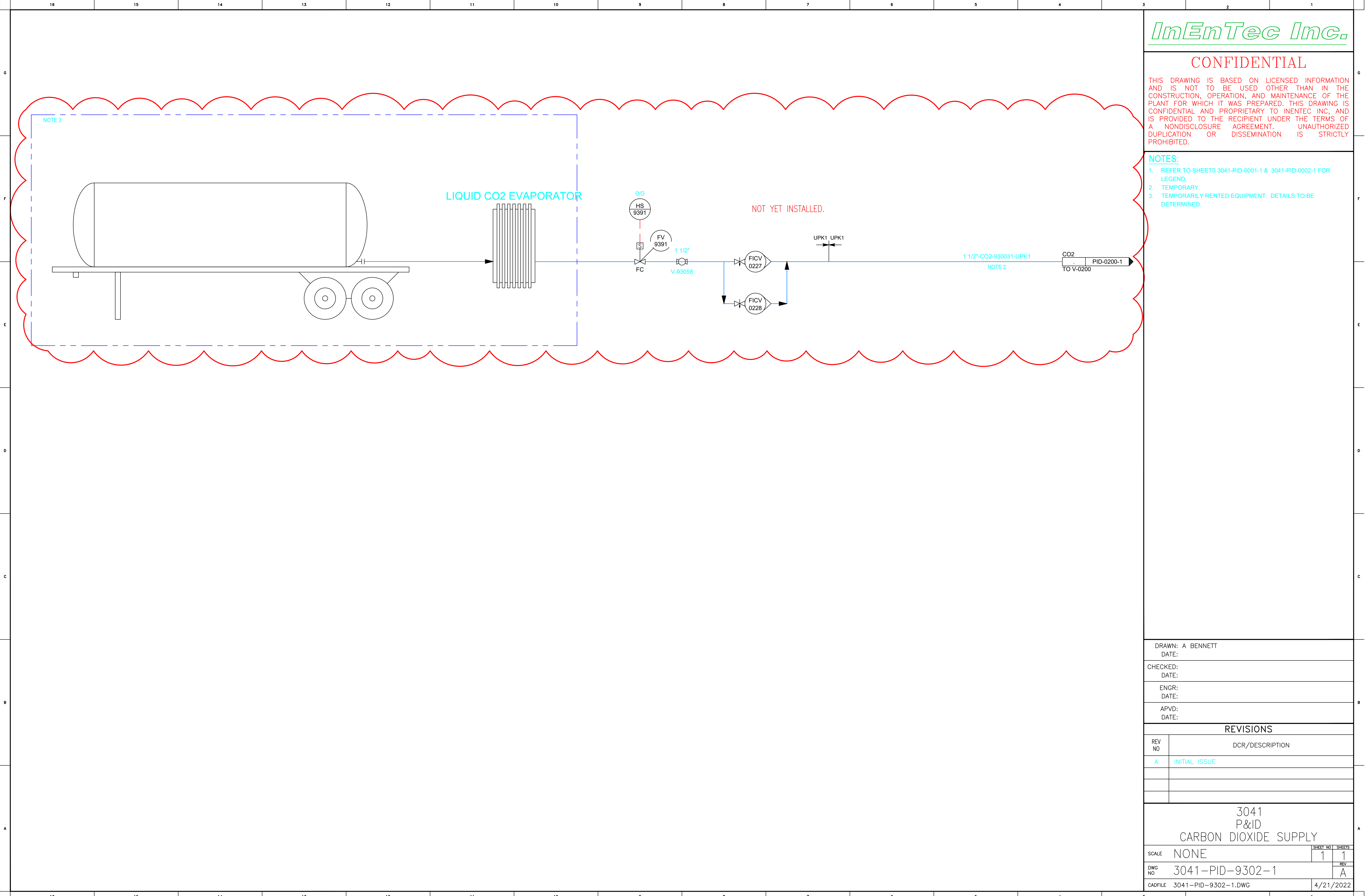
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1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
2. ANTENNA TO TRANSMIT DATA TO OXYGEN VENDOR.
3. PA-9301 IS SUPPLIED AND OWNED BY OXYGEN VENDOR.



APVD:
DATE:

3041 P&ID OXYGEN DISTRIBUTION			SHEET NO. SHEETS	
SCALE	NONE	1		1
DWG NO	3041-PID-9301-1			REV
				A
CADFILE	3041-PID-9301-1.DWG	9/27/2023		



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NOTES:

- 1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
- 2. TEMPORARY.
- 3. TEMPORARILY RENTED EQUIPMENT. DETAILS TO BE DETERMINED.

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A	INITIAL ISSUE

3041 P&ID CARBON DIOXIDE SUPPLY			
SCALE	NONE	SHEET NO	SHEETS
		1	1
DWG NO	3041-PID-9302-1	REV	A
CADFILE	3041-PID-9302-1.DWG	4/21/2022	

V-9510
AIR RECEIVER

S-9510
OIL WATER SEPARATOR

D-9510
DRYER

P-9500
UTILITY BUILDING SUMP PUMP

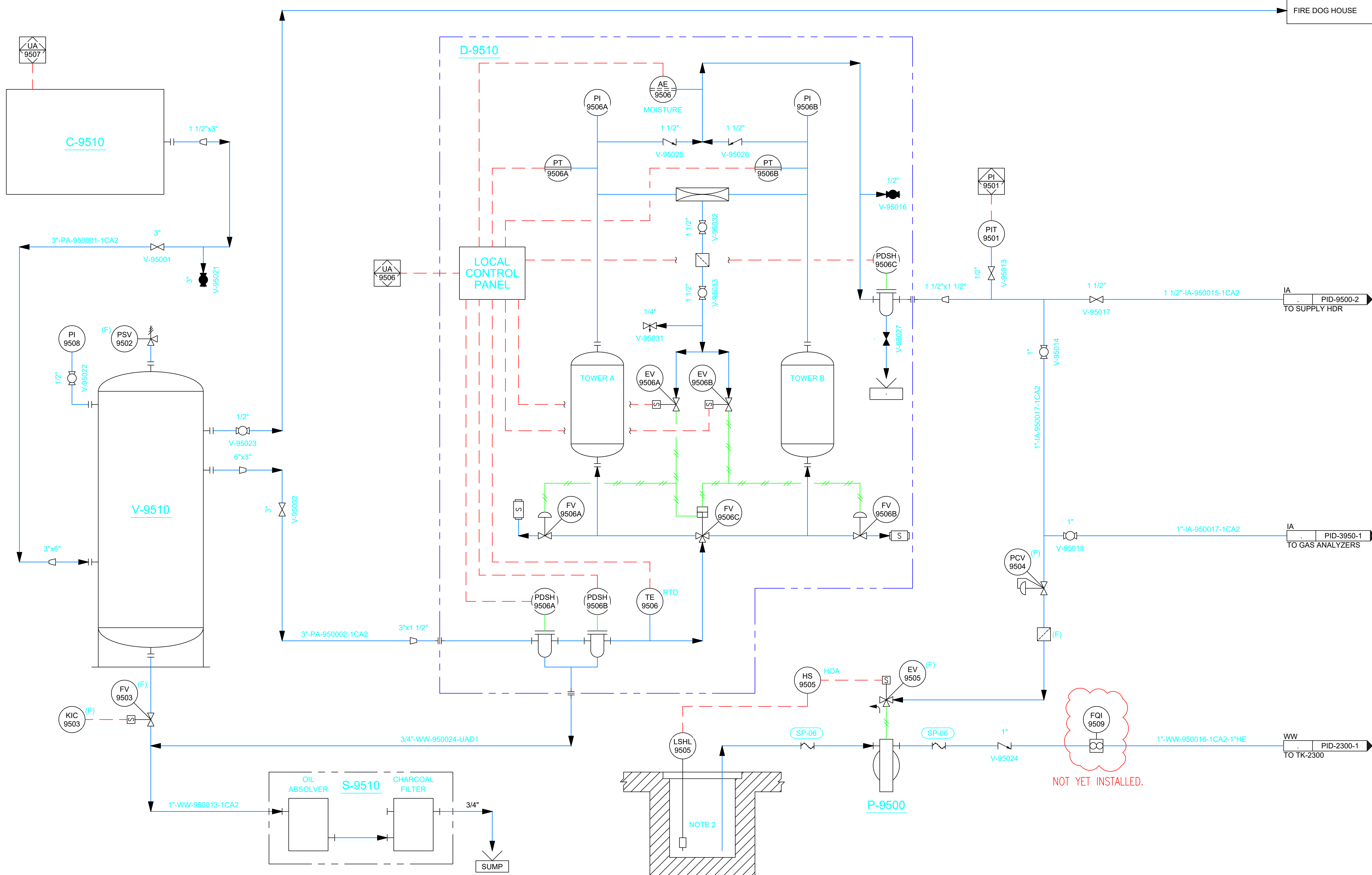
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NOTES:

- REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
- BOILER ROOM SUMP.



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REVISIONS

REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

3041 P&ID INSTRUMENT AIR SYSTEM	
SCALE NONE	SHEET NO 1 REV 1
DWG NO 3041-PID-9500-1	REV A
CADFILE 3041-PID-9500-1.DWG	4/20/2023

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NOTES:

1. REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.



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REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

3041	
P&ID	
INSTRUMENT AIR DISTRIBUTION	
SCALE	NONE
DWG NO	3041-PID-9500-2
CADFILE	3041-PID-9500-2.DWG
SHEET NO	1
SHEETS	1
REV	A
DATE	3/19/2024

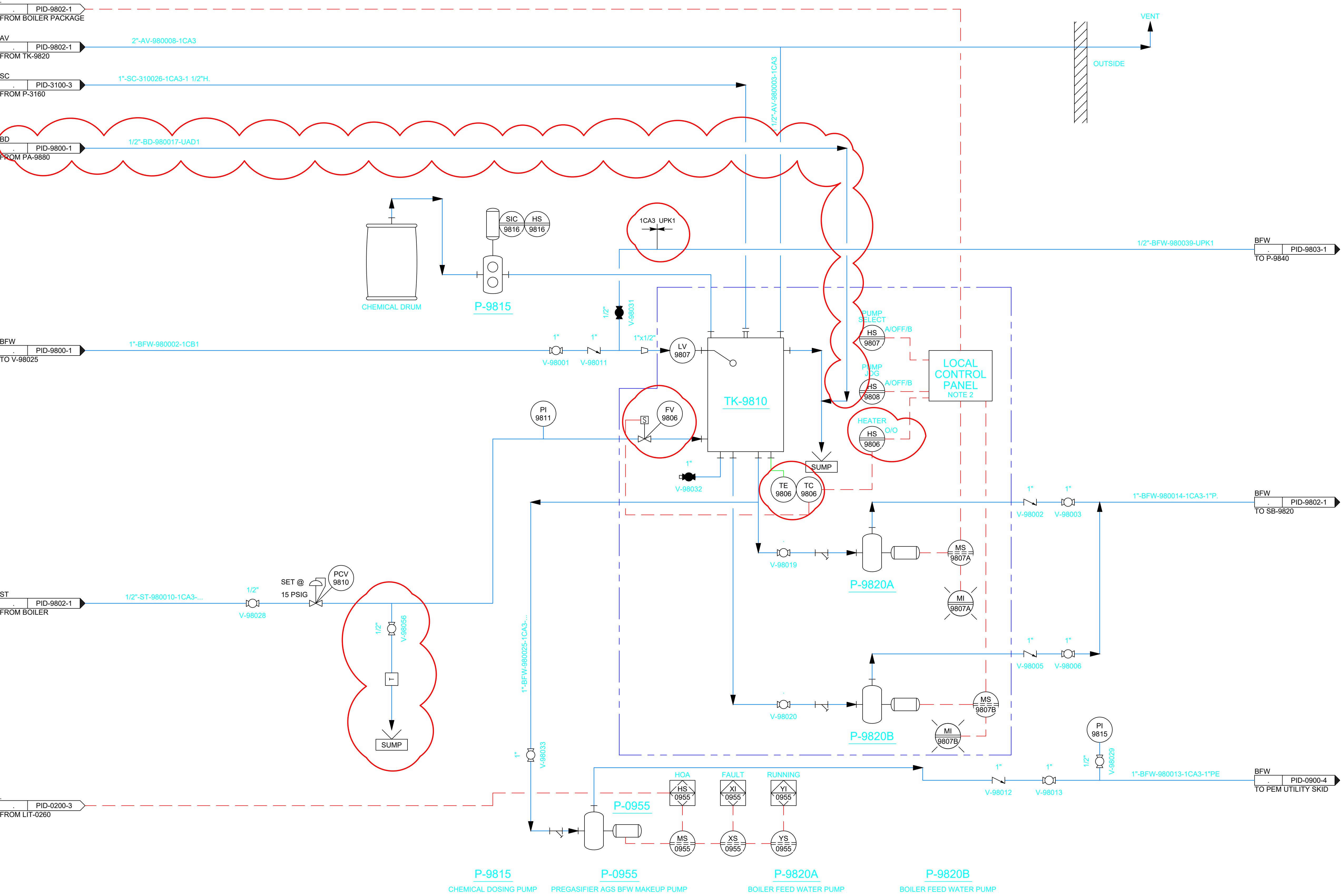
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NOTES:

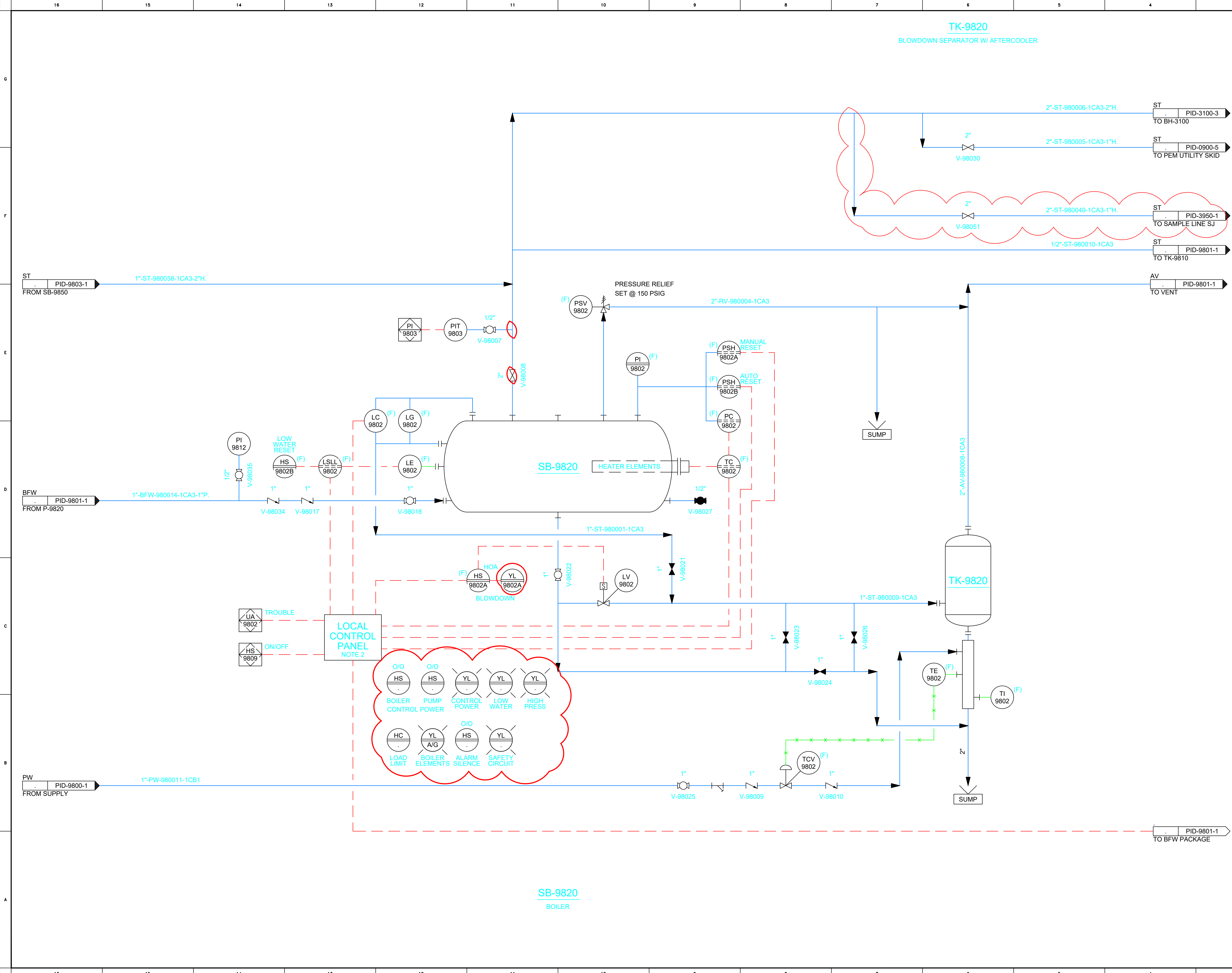
- REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
- FOR CONTROL SYSTEM DIAGRAM SEE LATTNER DWG. WD-10-4998-J(2 OF 2).
- SUMP DRAINS TO BOILER ROOM SUMP SHOWN ON 3041-PID-9500-1.



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REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE

3041 P&ID BFW PUMPS	
SCALE NONE	SHEET NO 1 SHEETS 1
DWG NO 3041-PID-9801-1	REV A
CADFILE 3041-PID-9801-1.DWG	3/19/2024



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NOTES:

- REFER TO SHEETS 3041-PID-0001-1 & 3041-PID-0002-1 FOR LEGEND.
- SEVEN STEP SEQUENCER CONTROLS 7 BANKS OF 30K HEATERS. SEE LATTNER DWG. WD-10-4988-J(1 OF 2).
- SUMP DRAINS TO BOILER ROOM SUMP SHOWN ON 3041-PID-9500-1.

DRAWN: A BENNETT DATE:	
CHECKED: P PEREZ DATE: 7/6/2020	
ENGR: DATE:	
APVD: DATE:	
REVISIONS	
REV NO	DCR/DESCRIPTION
A	INITIAL ISSUE
3041 P&ID STEAM SYSTEM	
SCALE	NONE
DWG NO	3041-PID-9802-1
CADFILE	3041-PID-9802-1.DWG
SHEET NO	1
SHEETS	1
REV	A
DATE	4/20/2023

